Loving or Fighting Inflation: Optimal Policy in High- and Low-Debt Economies*

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Abstract

Does high public debt force governments to tolerate inflation? In a heterogeneous-agent Ramsey framework calibrated to the United Kingdom at three debt levels, the optimal planner keeps inflation at zero regardless of the debt ratio, absorbing shocks entirely through the progressive tax structure — both the level and the progressivity of labor taxation. Progressive taxation dominates inflation as a redistributive tool because it targets income directly, whereas inflation erodes savings regressively. When the tax structure is constrained, inflation re-emerges and scales with debt. The conventional wisdom linking high debt to inflation is therefore conditional on fiscal rigidity, not a generic feature of indebted economies.

Keywords: Heterogeneous agents, Ramsey policies, monetary policy, fiscal policy, inflation, public debt.

JEL-classification: E61, E62, E63, E52, D52, H21.

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1 Introduction

Does high public debt force governments to tolerate inflation? The fiscal theory of the price level answers yes: when taxes cannot fully service the debt, the real value of nominal liabilities must fall, and inflation is the mechanism that delivers this adjustment (Cochrane 2023, Sims 2013). Leeper (2013) formalizes this as a "fiscal limit" beyond which taxation alone cannot stabilize debt; Salamaliki & Venetis (2024) and Reinhart et al. (2003) document the empirical regularity that high-debt episodes are accompanied by elevated inflation. In representative-agent models the logic is mechanical: the government budget constraint links debt, taxes, and inflation tightly, so when taxes are constrained, inflation is the residual adjustment. This paper shows that the answer changes when the tax structure can adjust: under Ramsey-optimal fiscal–monetary policy with progressive taxation, the planner never uses inflation at any debt level studied — the fiscal-theory prediction holds only when the tax structure is held fixed.

This logic, however, overlooks a channel central to actual fiscal systems but absent from representative-agent models: *progressive taxation*. When agents are heterogeneous in wealth and income, inflation and progressive taxation reach different parts of the distribution. Inflation taxes nominal savings, so its burden runs with wealth — falling on net nominal creditors regardless of their income or social marginal value. Progressive taxation, by contrast, runs with current earnings, targeting the agents the planner most wants to tax rather than those who happen to have saved. In an Aiyagari economy with precautionary saving, these are overlapping but distinct populations, and a planner who cares about who bears the adjustment faces a sharp choice between the two instruments. Whether progressive taxation is quantitatively strong enough to eliminate any role for inflation is the question this paper answers.

I study this question in a heterogeneous-agent economy using a Ramsey framework with capital accumulation, progressive labor taxation following Heathcote et al. (2017) (HSV), capital taxation, public debt, and Ramsey-optimal monetary policy. The model is calibrated to three UK fiscal regimes — 1997, 2007, and 2021 — spanning a three-fold variation in the debt-to-GDP ratio (Table 1).[1] The United Kingdom provides a natural laboratory for this question. Between 1997 and 2021, UK public debt rose from 41% to 101% of GDP — driven first by the post-2008 fiscal response and then by the Covid-19 pandemic — while the pre and post-tax income Gini remained remarkably stable (Figure 1). This combination of dramatic fiscal variation and stable inequality means that the three calibrations isolate the effect of the debt level on optimal policy, holding the underlying income distribution approximately fixed. A representative-agent model, which collapses the wealth and income distributions to a single point, cannot capture the distributional trade-off between inflation and progressive

[1] Gini values are from the ONS Households Below Average Income (HBAI) series, equivalised disposable income on a financial-year-ending basis. They differ from the World Bank series — by approximately 2.5 percentage points in 2021 — because the latter uses Eurostat EU-SILC harmonized microdata with a different equivalisation scale and reporting calendar. The qualitative cross-year comparison is robust to either source.

taxation that is at the heart of this paper.

I assume that observed fiscal outcomes and income distributions for each year reflect the optimal choices of a benevolent planner who maximizes intertemporal welfare in a heterogeneous-agent economy. The planner possesses a Social Welfare Function (SWF) that internalizes the distortions and general-equilibrium effects of all fiscal and monetary instruments.

	Debt-to-GDP (B/Y)	Gini (Pre-Tax)	Gini (Post-Tax)
United Kingdom (1997)	40.7	51.0	34.0
United Kingdom (2007)	43.1	53.5	38.6
United Kingdom (2021)	101.0	50.2	29.9

Table 1: Public debt and income inequality in the United Kingdom. Source: ONS.

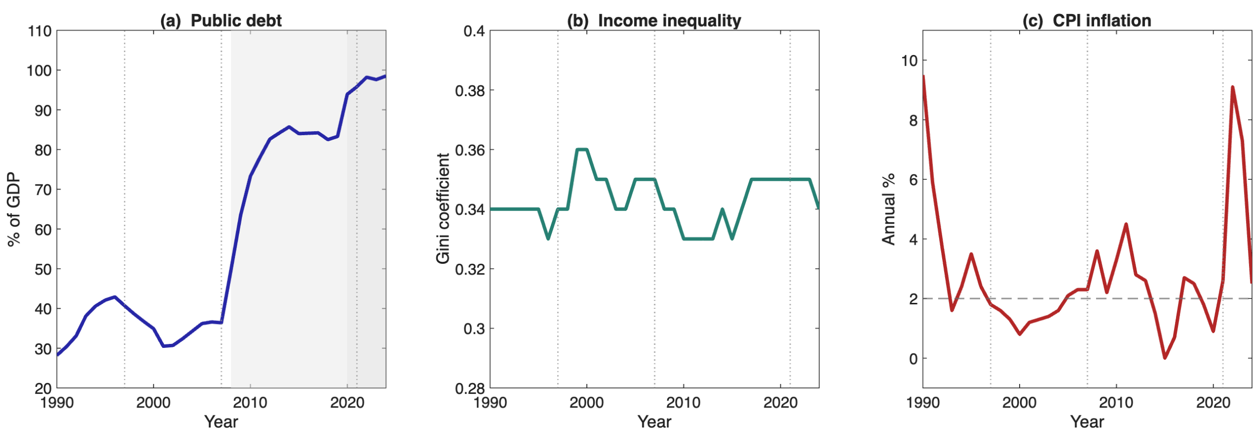

Figure 1: UK public debt, income inequality, and CPI inflation, 1990–2024. Vertical lines mark the calibration years (1997, 2007, 2021). Sources: ONS, IFS.

Given the focus on the joint dynamics of fiscal and monetary policy, I begin by estimating an SWF consistent with the observed debt and inequality measures for the UK. Following the inverse-optimum taxation approach of Heathcote & Tsujiyama (2021) and Le Grand et al. (2025), I infer the implicit social weights that rationalize observed fiscal and redistributive outcomes as optimal steady-state choices. The recovered weights rise steeply with productivity in both calibrations but are flatter under UK 2021 than under UK 2007, reflecting the more redistributive post-Covid fiscal configuration. This provides the welfare foundation for the dynamic analysis, which is solved using the truncation method of Le Grand & Ragot (2022*a*) and the Lagrangian approach of LeGrand & Ragot (2025).

The main finding is that the Ramsey planner keeps inflation at zero at every calibrated debt level, absorbing shocks entirely through the progressive tax structure — adjusting both the level κ and the progressivity τ of labor taxation. The Ramsey first-order condition for inflation makes the mechanism precise: the marginal benefit of inflation (eroding the real value of debt) is dominated by the marginal benefit of adjusting the tax structure, because progressive taxation delivers the same redistribution at a fraction of the welfare cost. The cost difference is structural rather than calibration-specific.

The zero-inflation result holds across all three debt levels, under both high- and low-persistence shocks, and is robust to alternative welfare specifications and open-economy considerations. It also holds when progressivity is removed from the Ramsey instrument set and only the tax level and capital tax are available — confirming that the full progressive tax structure, not progressivity alone, drives the result.

This finding overturns the prediction of the fiscal theory of the price level — but only when the tax structure is flexible. When I impose fiscal feedback rules that freeze the tax parameters, the same model produces inflation that scales with debt: a 6.5× increase from the lowest to the highest calibration. A Taylor-type monetary rule with frozen fiscal instruments produces even larger inflation responses. Allowing even a partial tax-structure response reduces inflation by 29%, confirming that the two instruments are close substitutes in the planner's objective. The conventional wisdom linking high debt to inflation is therefore not wrong, but conditional: it holds when governments lack the ability to adjust the tax structure at business-cycle frequency, and vanishes when they do.

Three additional findings emerge from the analysis. First, the optimal debt path itself depends on the initial fiscal position. Under high debt the planner consolidates procyclically — front-loading the adjustment through the tax structure. Under low debt the planner borrows countercyclically, smoothing the adjustment through debt issuance. The force driving this regime change is the curvature of the debt-service cost: a marginal unit of additional borrowing is more expensive to service when inherited debt is large, tilting the planner toward front-loading.

Second, the heterogeneous-agent planner's optimal response is several orders of magnitude larger than its representative-agent counterpart, with the distributional motive and the progressive-tax instrument each contributing to the amplification. Third, feeding thirty years of actual UK fiscal shocks through the model, the Ramsey planner would not have used inflation at any point, including the 2008 crisis and the 2020 pandemic — the zero-inflation result is not an artefact of a particular shock calibration.

From a normative standpoint, the central policy lesson is that the institutional architecture of fiscal policy matters as much as the level of debt for the conduct of monetary policy. Governments that retain the capacity to adjust the progressive tax schedule at business-cycle frequency preserve the planner's ability to absorb shocks without inflation; governments that do not push the burden of stabilization onto the central bank, at welfare cost to the very households the planner most wants to protect. The implication is not that progressivity should be raised in normal times, but that the *flexibility* to adjust it counter-cyclically deserves the same institutional protection that monetary independence already enjoys — a protection that high-debt economies, where the fiscal-rigidity premium on inflation is largest, stand to gain from most.

Related literature. The paper builds on the *inverse optimal taxation* literature (Bargain & Keane 2010, Bourguignon & Amadeo 2015, Saez 2001, Saez & Stantcheva 2016, Heathcote & Tsujiyama 2021, Le Grand et al. 2025), which recovers social preferences from observed fiscal systems but stops at static configurations; I embed the recovered weights in a dynamic Ramsey problem to ask whether the social preferences revealed by UK fiscal policy imply a role for inflation in shock absorption. The closest paper is LeGrand & Ragot (2025), who use the same truncation methodology to study optimal fiscal policy — capital taxes, progressive labor taxes, public debt — in an Aiyagari economy. Their model is *purely real*, with no nominal rigidity and no monetary policy; I extend their framework with a full New Keynesian block (Rotemberg costs, Phillips curve, nominal interest rate and gross inflation), jointly optimized by the Ramsey planner, and confirm and extend their debt-dynamics result to the monetary-policy case. This connects the paper to classical HA-Ramsey analyses (Aiyagari 1995, Aiyagari & McGrattan 1998, Werning 2007, Bassetto 2014, Farhi & Werning 2013, Bhandari et al. 2017) and to the optimal-debt-level literature (Floden 2001, Rohrs & Winter 2017, Chatterjee et al. 2017, 2018, Cozzi 2023), which characterize the welfare-maximizing *steady-state* debt level but not the dynamic fiscal–monetary mix conditional on an inherited debt level after a shock — the question I address.

The paper also speaks to the literature on *optimal monetary policy under heterogeneity and fiscal limits*. Kaplan et al. (2018) and Auclert (2019) show that monetary policy operates through redistributive channels absent in representative-agent models; the fiscal theory of the price level (Leeper 1991, Cochrane 2023, Sims 2013, Leeper 2013) predicts that high debt forces inflation when taxes cannot stabilize debt alone, and empirical work documents the same correlation (Salamaliki & Venetis 2024, Reinhart et al. 2003). Most recently, Kaplan (2025) shows that even fully-funded lump-sum stimulus is inflationary in HANK economies. The present paper overturns this result and qualifies the fiscal-limit prediction: the key missing ingredient is not heterogeneity per se but the *type of fiscal instrument*, since lump-sum transfers cannot target the distributional cost of a shock whereas progressive taxation can. When the progressive tax structure is constrained, inflation re-emerges as in the Kaplan framework, and the fiscal-limit logic of the FTPL applies. This also distinguishes the present paper from studies on the redistributive role of inflation (Doepke & Schneider 2006, Erosa & Ventura 2002, Albanesi 2007) and optimal inflation in HA models (Nuño & Thomas 2022, McLeay & Tenreyro 2020, Blanco 2021, McKay & Wolf 2022, Bilbiie & Ragot 2021), which generally find a role for non-zero inflation: inflation is optimal here only under fiscal rigidity, not as a generic feature of high-debt economies.

The paper is organized as follows. Section 2 presents a tractable two-type model that proves the main result analytically. Section 3 develops the full quantitative model. Section 4 derives the Ramsey problem and recovers the social welfare function. Section 5 presents the quantitative investigation. Section 6 establishes the zero-inflation result. Section 7 identifies the mechanism and tests the converse. Section 8 examines alternative welfare functions and the open-economy extension. Section 9 connects the model to UK data. Section 10 concludes.

2 A tractable model

Before turning to the full quantitative framework, I develop a tractable two-type economy that isolates the economic mechanism behind the paper's main result.

2.1 Environment

Consider an economy with two types of agents, $h \in \{P, R\}$ (Poor, Rich), with population shares S_P and $S_R = 1 - S_P$. Agent h has savings a_h (with $a_R > a_P \geq 0$) and productivity y_h (with $y_P < 1 < y_R$, normalized so that $\sum_h S_h y_h = 1$). Consumption is given by:

$$c_h = (1 - t_h)\, y_h + (1 - \pi)\, a_h,$$

where t_h is an agent-specific tax rate and π is the inflation rate. The government must finance spending $\Delta G > 0$.

The government has two instruments. The first is *inflation* $\pi \geq 0$, which taxes savings uniformly: agent h loses $\pi\, a_h$. Revenue is $\pi \sum_h S_h a_h$. The second is a *redistributive tax-and-transfer*: the government taxes the Rich at rate $t_R > 0$ and transfers to the Poor at rate $s > 0$ (i.e., $t_P = -s < 0$). Net revenue is $S_R t_R y_R - S_P s\, y_P$. This captures the key feature of the HSV tax function, which simultaneously taxes high-income agents and subsidizes low-income agents when progressivity increases.

The budget constraint requires the two instruments to jointly finance ΔG:

$$\pi \sum_h S_h a_h \; + \; S_R\, t_R\, y_R - S_P\, s\, y_P \; = \; \Delta G. \tag{1}$$

2.2 Why the planner prefers redistribution to inflation

The welfare cost of inflation per dollar of revenue is given by:

$$\mathcal{C}^{\pi} = \frac{S_R\, \omega_R\, u'(c_R)\, a_R + S_P\, \omega_P\, u'(c_P)\, a_P}{\sum_h S_h\, a_h}. \tag{2}$$

Every agent loses from inflation, so $\mathcal{C}^{\pi}$ is strictly positive.

The welfare cost of the redistributive tax, per dollar of net revenue, is:

$$\mathcal{C}^{\text{redist}} = \frac{S_R\, \omega_R\, u'(c_R)\, t_R\, y_R \; - \; S_P\, \omega_P\, u'(c_P)\, s\, y_P}{S_R\, t_R\, y_R - S_P\, s\, y_P}. \tag{3}$$

The numerator has two terms: the welfare *cost* of taxing the Rich (> 0) and the welfare *gain* from transferring to the Poor (< 0). The transfer component reduces the net welfare cost, and this is the fundamental advantage of progressive taxation over inflation.

Proposition 1 (Redistribution dominates inflation)**.** *Starting from any pure-inflation allocation ($\pi > 0$, $s = 0$), a marginal introduction of redistribution — a small transfer $ds > 0$*

to the Poor, funded by a further tax on the Rich — strictly improves welfare if and only if

$$\omega_P \, u'(c_P) \; > \; \omega_R \, u'(c_R), \tag{4}$$

that is, the social marginal value of a dollar to the Poor exceeds that of a dollar to the Rich.

Proof. A marginal transfer $ds > 0$ to the Poor requires an additional tax $dt_R = S_P \, ds \, y_P / (S_R \, y_R)$ on the Rich to maintain revenue. The welfare change is:

$$dW = S_P \, \omega_P \, u'(c_P) \, y_P \, ds - S_R \, \omega_R \, u'(c_R) \, y_R \, dt_R = S_P \, y_P \, ds \left[\omega_P \, u'(c_P) - \omega_R \, u'(c_R) \right],$$

which is positive if and only if condition (4) holds. ■

When does condition (4) hold? The condition requires that the concavity of utility is strong enough to overcome the welfare-weight difference. Since the Poor agent consumes less ($c_P < c_R$), her marginal utility is higher ($u'(c_P) > u'(c_R)$). With CRRA utility $u'(c) = c^{-\gamma}$, condition (4) becomes:

$$\left(\frac{c_R}{c_P} \right)^{\gamma} > \frac{\omega_R}{\omega_P}. \tag{5}$$

The consumption ratio raised to the power of risk aversion must exceed the welfare-weight ratio. This is a joint restriction on preferences, the consumption distribution, and the welfare function. At the Ramsey steady state, the planner has already chosen the tax structure to approximately equalize social marginal values across types, so condition (4) is approximately satisfied by construction. The quantitative model confirms this: under the UK 2021 calibration with $\gamma = 2$, the condition holds across all productivity types (Table 6 in Section 7).

Discussion. The proposition captures the core mechanism of the paper in its simplest form. Inflation is a blunt instrument: it taxes all savers proportionally to their holdings and transfers nothing back. Progressive taxation bundles revenue collection with a targeted transfer to low-income agents whose social marginal value is higher than that of the Rich — which, under concavity, is the natural case. In a representative-agent model the channel vanishes entirely: there is no one to transfer to, the redistributive instrument collapses to a proportional tax with the same welfare cost as inflation, and the planner is indifferent between the two. The instrument hierarchy is therefore *entirely a consequence of heterogeneity*.

What the full model adds. The tractable model abstracts from capital accumulation, the Phillips curve, general-equilibrium price effects, and the dynamic interaction between debt, taxes, and inflation. In the full model of Section 3, the transfer is not a free instrument — it comes at a cost through labor-supply distortions, general-equilibrium price effects, and the Rotemberg pricing friction. The quantitative analysis shows that even after accounting for all of these costs, the mechanism identified in Proposition 1 is strong enough to dominate: the Ramsey planner keeps inflation at zero at every debt level.

3 The full model

The economy has three types of actors. A continuum of households face uninsurable idiosyncratic productivity risk, choose consumption, labor supply, and savings, and are subject to a borrowing constraint. A continuum of monopolistically competitive firms produce with capital and labor and set prices subject to a Rotemberg quadratic adjustment cost, generating a New Keynesian Phillips curve. A government finances an exogenous spending requirement through progressive labor taxation, a linear capital tax, nominal public debt, and inflation — all of which the Ramsey planner chooses optimally. I follow the model and notation of Le Grand et al. (2025) and LeGrand & Ragot (2025), extending their purely real framework with a New Keynesian monetary block.

Time is discrete, $t \geq 0$. A continuum of agents with measure one is distributed over a set I according to a non-atomic measure ℓ (Green 1994).

3.1 Risk structure

At the beginning of each period, agents face an uninsurable idiosyncratic labor productivity shock y, which can take Y discrete values in the set $\mathcal{Y} \subset \mathbb{R}_+$. The productivity process follows a first-order Markov chain with transition matrix $(\pi_{yy'})_{y,y'}$.

In the stationary distribution, the fraction of agents with productivity level y is denoted by S_y and satisfies $S_y = \sum_{\tilde{y}\in\mathcal{Y}} \pi_{\tilde{y}y} S_{\tilde{y}}, \quad \forall \tilde{y} \in \mathcal{Y}$. Normalizing average productivity to one implies $\sum_{y\in\mathcal{Y}} S_y y = 1$.

In addition to idiosyncratic risk, the economy is subject to an aggregate productivity shock $Z_t = Z_0 e^{z_t}$, where z_t follows an AR(1) process. The full sequential representation of individual histories and the associated measure-theoretic notation are provided in Appendix A; in the main text I use $y_{i,t}$ for the current productivity draw and suppress dependence on histories, following Remark 1 below.

Each agent i chooses labor supply $l_{i,t}$ and earns a before-tax wage rate $\tilde{w}_t$. Hence, total before-tax labor income in period t is given by $\tilde{w}_t y_{i,t} l_{i,t}$.

3.2 Preferences

Agents derive utility from consumption and disutility from labor. Each agent i values sequences of consumption $(c_{i,t})_{t\geq 0}$ and labor $(l_{i,t})_{t\geq 0}$ according to a time-separable utility function $\sum_{t=0}^{\infty} \beta^t U(c_{i,t}, l_{i,t})$, where $\beta \in (0,1)$ is the discount factor. The period utility function $U(c_{i,t}, l_{i,t})$ is assumed to be separable in consumption and labor:

$$U(c_{i,t}, l_{i,t}) = u(c_{i,t}) - v(l_{i,t}). \tag{6}$$

The function $u : \mathbb{R}_+ \to \mathbb{R}$ represents the utility from consumption and is twice continuously differentiable, strictly increasing, and strictly concave, with $u'(0) = \infty$. The function $v : \mathbb{R}_+ \to \mathbb{R}$ represents the disutility from labor and is twice continuously differentiable, strictly increasing, and strictly convex, with $v'(0) = 0$.

3.3 Production

There is a continuum of monopolistically competitive firms indexed by $j \in [0, 1]$, each producing a differentiated variety $Y_{j,t}$. These intermediate goods are aggregated into a final composite good according to a constant-elasticity-of-substitution (CES) aggregator with elasticity of substitution ε:

$$Y_t = \left[\int_0^1 Y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} dj\right]^{\frac{\varepsilon}{\varepsilon-1}}.$$

Profit maximization by the final-good producer yields the standard demand function for each variety and the associated aggregate price index:

$$Y_{j,t} = \left(\frac{P_{j,t}}{P_t}\right)^{-\varepsilon} Y_t, \quad \text{and} \quad P_t = \left[\int_0^1 P_{j,t}^{1-\varepsilon} dj\right]^{\frac{1}{1-\varepsilon}}.$$

Intermediate firms produce using a Cobb–Douglas technology,

$$Y_{j,t} = Z_t k_{j,t}^{\alpha} n_{j,t}^{1-\alpha},$$

where Z_t is the aggregate productivity level, $k_{j,t}$ denotes the firm's capital input, and $n_{j,t}$ denotes the firm's labor input.[2] In equilibrium, the output of each firm equals the demand for its variety from the final-good producer, with intermediate goods sold at the relative price $P_{j,t}/P_t$.

Let $\tilde{w}_t$ denote the real before-tax wage, and let $\tilde{r}_t^k$ denote the real before-tax rental rate of capital (net of depreciation $\delta > 0$). The firm's cost-minimization problem yields:

$$\tilde{r}_t^k + \delta = \xi_{j,t}\alpha\frac{Y_{j,t}}{k_{j,t}} \quad \text{and} \quad \tilde{w}_t = \xi_{j,t}(1-\alpha)\frac{Y_{j,t}}{n_{j,t}}, \tag{7}$$

where $\xi_{j,t}$ is the Lagrange multiplier associated with the production constraint. Optimality implies a common value ξ_t across firms:

$$\xi_t = \frac{1}{Z_t}\left(\frac{\tilde{r}_t^k + \delta}{\alpha}\right)^{\alpha}\left(\frac{\tilde{w}_t}{1-\alpha}\right)^{1-\alpha}. \tag{8}$$

Aggregate output is $Y_t = Z_t K_{t-1}^{\alpha} L_t^{1-\alpha}$, with the factor-income identity $\xi_t Y_t = (\tilde{r}_t^k + \delta)K_{t-1} + \tilde{w}_t L_t$. In the competitive limit ($\xi_t = 1$), factor prices take the standard Cobb–Douglas forms.

Firms also face a quadratic price-adjustment cost à la Rotemberg (1982): $(\psi/2)(\Pi_{j,t} - 1)^2 Y_t$,

[2] I use $n_{j,t}$ for firm-level labor demand and $l_{i,t}$ for individual labor supply to avoid notational ambiguity.

where $\psi \geq 0$ measures price rigidity and $\Pi_{j,t} \equiv P_{j,t}/P_{j,t-1}$. The real profit of firm j is:

$$D_{j,t} = \frac{P_{j,t}}{P_t} Y_{j,t} - \frac{(1-\tau_t^D)}{Z_t} \left(\frac{\tilde{r}_t^k + \delta}{\alpha} \right)^{\alpha} \left(\frac{\tilde{w}_t}{1-\alpha} \right)^{1-\alpha} Y_{j,t} - \frac{\psi}{2} \left(\frac{P_{j,t}}{P_{j,t-1}} - 1 \right)^2 Y_t - t_t^D,$$

where t_t^D is a lump-sum tax used to finance the production subsidy τ_t^D. Substituting the equilibrium relationships yields:

$$D_{j,t} = \left(\frac{P_{j,t}}{P_t} - \xi_t (1-\tau_t^D) \right) \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} Y_t - \frac{\psi}{2} \left(\frac{P_{j,t}}{P_{j,t-1}} - 1 \right)^2 Y_t - t_t^D. \tag{9}$$

Firm j chooses its price path $(P_{j,t})_{t\geq 0}$ to maximize the expected discounted value of profits:

$$\max_{\{P_{j,t}\}_{t\geq 0}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{M_t}{M_0} D_{j,t} \right],$$

where $M_t \equiv \int_i u'(c_{i,t}) \, \ell(di)$ is the pricing kernel derived from household Euler equations, consistent with households owning the firms.

The corresponding first-order condition is:

$$\begin{aligned} &\left[(1-\varepsilon) + \varepsilon \xi_t (1-\tau_t^D) \frac{P_t}{P_{j,t}} \right] \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} \frac{Y_t}{P_t} - \psi (\Pi_t - 1) \Pi_t \frac{Y_t}{P_{j,t}} \\ &\quad + \beta \psi \mathbb{E}_t \left[\frac{M_{t+1}}{M_t} (\Pi_{t+1} - 1) \Pi_{t+1} \frac{Y_{t+1}}{Y_t} \right] \frac{Y_t}{P_{j,t}} = 0, \end{aligned}$$

where $\Pi_t \equiv P_{j,t}/P_{j,t-1}$ denotes gross inflation.

Setting $\tau_t^D = 1/\varepsilon$ ensures an efficient steady state. In a symmetric equilibrium, $P_{j,t} = P_t$ for all j, yielding:

$$(1-\varepsilon) + \varepsilon \xi_t \left(\frac{\varepsilon - 1}{\varepsilon} \right) - \psi (\Pi_t - 1) \Pi_t + \beta \psi \mathbb{E}_t \left[(\Pi_{t+1} - 1) \Pi_{t+1} \frac{Y_{t+1}}{Y_t} \frac{M_{t+1}}{M_t} \right] = 0.$$

This leads to the New Keynesian Phillips curve:

$$\Pi_t (\Pi_t - 1) = \frac{\varepsilon - 1}{\psi} (\xi_t - 1) + \beta \mathbb{E}_t \left[\Pi_{t+1} (\Pi_{t+1} - 1) \frac{Y_{t+1}}{Y_t} \frac{M_{t+1}}{M_t} \right]. \tag{10}$$

Finally, in equilibrium the firm's real profits are identical across firms and given by:

$$D_t = \left(1 - \xi_t - \frac{\psi}{2} (\Pi_t - 1)^2 \right) Y_t. \tag{11}$$

The production subsidy $\tau_t^D = 1/\varepsilon$ is set to correct the monopolistic distortion. At the calibrated steady state this implies $\xi_{ss} = 1$, and combined with $\Pi_{ss} = 1$ this yields $D_{ss} = 0$: monopoly profits are zero in equilibrium. Section 3.5 below shows that, once the government budget constraint is written out, D_t also drops algebraically from all equilibrium conditions outside the steady state, so households receive factor incomes only — the standard HANK convention (Kaplan et al. 2018, Auclert 2019).

3.4 Assets

Agents have access to two types of assets. The first is nominal public debt, with aggregate supply denoted by B_t, determined by the government. This asset pays a predetermined nominal gross interest rate before taxes. The nominal rate between periods $t-1$ and t is known and denoted by $\tilde{R}^n_{t-1}$. The corresponding real interest rate is $\tilde{R}^n_{t-1}/\Pi_t$, where Π_t denotes the gross inflation rate.

The second asset is equity (capital shares), which yields a before-tax return $\tilde{r}^k_t$, as described above. The aggregate capital stock in the economy is denoted by K_t.

Public debt and capital are assumed to be perfect substitutes and to yield the same pre-tax return $\tilde{r}_t$. This assumption is analogous to the existence of a risk-neutral mutual fund (see, e.g., Gornemann et al. (2016)) that holds all interest-bearing assets—capital and public debt—and sells its shares to households.

Let total interest-bearing assets be denoted by $A_t = K_t + B_t$. The zero-profit condition of the mutual fund implies:

$$\tilde{r}_t A_{t-1} = \tilde{r}^k_t K_{t-1} + \left(\frac{\tilde{R}^n_{t-1}}{\Pi_t} - 1 \right) B_{t-1}. \tag{12}$$

Since capital and public debt are perfect substitutes, a no-arbitrage condition must hold: the expected real return on government bonds must equal the expected real return on capital. Formally,

$$\mathbb{E}_t \left[\frac{\tilde{R}^n_t}{\Pi_{t+1}} \right] = \mathbb{E}_t \left[1 + \tilde{r}^k_{t+1} \right]. \tag{13}$$

3.5 Government and its tools

The model assumes the existence of a benevolent government that chooses a sequence of public expenditures $(G_t)_{t\geq 0}$. The government finances this spending using several instruments. First, it can issue one-period nominal public debt B_t, as described earlier. An enforcement technology is assumed to guarantee that the debt is default-free. Second, the government raises revenues through distortionary taxes on labor income and capital returns.

The labor income tax, denoted by $\mathcal{T}_{i,t}(\tilde{w}_t y_{i,t} l_{i,t})$ for pre-tax labor income $\tilde{w}_t y_{i,t} l_{i,t}$, is assumed to be possibly nonlinear and time-varying. Following Heathcote et al. (2017) (henceforth, HSV), it is defined as:

$$\mathcal{T}_{i,t}(\tilde{w}_t y_{i,t} l_{i,t}) = \tilde{w}_t y_{i,t} l_{i,t} - \kappa_t (\tilde{w}_t y_{i,t} l_{i,t})^{1-\tau_t}, \tag{14}$$

where κ_t governs the overall level of taxation and τ_t captures the progressivity of the tax schedule. Both parameters are policy instruments and may vary over time. When $\tau_t = 0$, the labor income tax is linear with rate $1-\kappa_t$; in contrast, $\tau_t = 1$ corresponds to full income redistribution, where all agents receive the same post-tax income κ_t.[3]

Capital income is taxed linearly at rate τ_t^k. Total government revenues therefore equal:

$$\int_i \mathcal{T}_{i,t}(\tilde{w}_t y_{i,t} l_{i,t})\, \ell(di) + \tau_t^k \tilde{r}_t^k K_{t-1} + \tau_t^k \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1\right) B_{t-1}.$$

Given these instruments, the government budget constraint is:

$$G_t + \frac{\tilde{R}_{t-1}^n}{\Pi_t} B_{t-1} = D_t + \int_i \mathcal{T}_{i,t}(\tilde{w}_t y_{i,t} l_{i,t})\, \ell(di) + \tau_t^k \tilde{r}_t^k K_{t-1} + \tau_t^k \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1\right) B_{t-1} + B_t. \quad (15)$$

Here D_t is the aggregate firm profit defined in (11). Substituting (14) into (15) and using the profit identity (11) together with $Y_t\xi_t = (\tilde{r}_t^k + \delta)K_{t-1} + \tilde{w}_t L_t$ yields the government budget constraint expressed in terms of pre-tax variables. Defining the post-tax quantities:

$$r_t = (1-\tau_t^k)\tilde{r}_t, \qquad r_t^k = (1-\tau_t^k)\tilde{r}_t^k, \qquad \frac{R_t^n}{\Pi_t} = (1-\tau_t^k)\left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1\right), \qquad w_t = \kappa_t(\tilde{w}_t)^{1-\tau_t}, \quad (16)$$

and using the asset-income identity $r_t A_{t-1} = r_t^k K_{t-1} + \frac{R_t^n}{\Pi_t} B_{t-1}$, the budget constraint can be written compactly as:

$$G_t + B_{t-1} + r_t A_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) = \left(1 - \frac{\psi}{2}(\Pi_t - 1)^2\right) Y_t - \delta K_{t-1} + B_t. \quad (17)$$

The intermediate steps are provided in Appendix B.

Two features of the reduced-form constraint (17) are worth noting. First, dividends D_t no longer appear: combining (11) with the factor-income identity $\xi_t Y_t = (\tilde{r}_t^k + \delta)K_{t-1} + \tilde{w}_t L_t$ shows that D_t cancels exactly — not only at the steady state but throughout the dynamics. What replaces it on the right-hand side is the net-of-adjustment-cost output $\left(1 - \frac{\psi}{2}(\Pi_t - 1)^2\right)Y_t$: the real resource cost of price rigidity is borne collectively through the government budget rather than through individual dividend flows. Second, the household budget constraint introduced in Section 3 therefore correctly omits D_t — households receive factor incomes (wages and capital returns), not residual firm profits, because the production subsidy routes all corporate income through factor markets. The pricing kernel $M_t \equiv \int_i u'(c_{i,t})\, \ell(di)$ is derived directly from household Euler equations, consistent with the standard interpretation that households own firms through the pricing kernel without requiring dividend flows in incomplete-markets equilibrium (see, e.g., Kaplan et al. 2018, Auclert 2019).

[3]The literature typically adopts either a linear tax combined with a lump-sum transfer (e.g., Dyrda & Pedroni (2022); Açikgöz et al. (2022)) or the non-linear tax structure proposed by Heathcote et al. (2017) (HSV). Recent quantitative work, such as Heathcote & Tsujiyama (2021), shows that the HSV specification provides a more realistic representation of labor income taxation.

3.6 Agents' program, resource constraints, and equilibrium

Agents' program. Consider an agent $i \in \mathcal{I}$ whose income consists of labor earnings and returns on savings. Post-tax labor income for an agent with productivity $y_{i,t}$ and labor effort $l_{i,t}$ is given by $\tilde{w}_t y_{i,t} l_{i,t} - \mathcal{T}_{i,t}(\tilde{w}_t y_{i,t} l_{i,t}) = w_t (y_{i,t} l_{i,t})^{1-\tau_t}$. Because a mutual fund aggregates all capital and public debt, agents face no portfolio choice. Let $a_{i,t}$ denote agent i's holdings of fund shares at time t. Borrowing is limited by $a_{i,t} \geq -\overline{a}$, where $\overline{a} > 0$ represents the borrowing limit. Fund claims yield the post-tax interest rate r_t, so savings payoffs are $(1 + r_t)a_{i,t-1}$, with $a_{i,t-1}$ denoting asset holdings at the end of period $t-1$.

Each agent allocates resources between consumption, savings, and labor supply. Given the exogenous path of public expenditure $(G_t)_{t\geq 0}$, the optimization problem is:

$$\max_{\{c_{i,t}, l_{i,t}, a_{i,t}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[u(c_{i,t}) - v(l_{i,t}) \Big], \tag{18}$$

$$\text{s.t.} \quad c_{i,t} + a_{i,t} \leq w_t (y_{i,t} l_{i,t})^{1-\tau_t} + (1 + r_t) a_{i,t-1}, \tag{19}$$

$$a_{i,t} \geq -\overline{a}, \quad c_{i,t} > 0, \quad l_{i,t} > 0. \tag{20}$$

Expectations $\mathbb{E}_0$ are taken over idiosyncratic and aggregate shocks, given the initial state $(a_{i,-1}, y_{i,0})$. At date 0, the agent chooses consumption $(c_{i,t})_{t\geq 0}$, labor $(l_{i,t})_{t\geq 0}$, and savings $(a_{i,t})_{t\geq 0}$ to maximize utility (18) subject to constraints (19)–(20). Decisions depend on $(a_{i,-1}, y_{i,0})$, the idiosyncratic history y_i^t, and the aggregate history z^t.

In period 0, each agent draws initial assets and productivity $(a_{i,-1}, y_{i,0})$ from the distribution $\mathbf{\Lambda}_0(a, y)$ defined on the Borel sets of $[-\overline{a}, \infty) \times \mathcal{Y}$, with $\mathbf{\Lambda}_0(a, y) : [-\overline{a}, \infty) \times \mathcal{Y} \to \mathbb{R}^+$. This setup allows the economy to start from any arbitrary, possibly stationary, distribution.

Remark 1 (Simplifying notation)**.** *For an agent with initial state $(a_{i,-1}, y_{i,0})$ and idiosyncratic history y_i^t in aggregate state z^t, the realization of any random variable $X_t((a_{i,-1}, y_{i,0}), y_i^t, z^t)$ is denoted simply by $X_{i,t}$.*

Aggregate variables are obtained by integrating over agents:

$$\int_i X_{i,t} \ell(di),$$

which is equivalent to the full sequential representation integrating over initial states (measure Λ) and idiosyncratic histories (measure θ):

$$\sum_{y^t \in \mathcal{Y}^{t+1}} \sum_{y_0 \in \mathcal{Y}} \int_{a_{-1} \in [-\overline{a}, \infty)} \theta_t(y^t) X_t((a_{-1}, y_0), y^t, z^t) \mathbf{\Lambda}(da_{-1}, y_0).$$

There thus exist functions $(c_t, l_t, a_t)_{t\geq 0}$ defined on $([-\overline{a}, \infty) \times \mathcal{Y}) \times \mathcal{Y}^{t+1} \times \mathbb{R}^{t+1}$ such that:

$$c_{i,t} = c_t((a_{i,-1}, y_{i,0}), y_i^t, z^t), \qquad l_{i,t} = l_t((a_{i,-1}, y_{i,0}), y_i^t, z^t), \qquad a_{i,t} = a_t((a_{i,-1}, y_{i,0}), y_i^t, z^t).$$

In what follows, functional arguments are suppressed and the index i is kept for brevity, following Remark 1.[4]

The optimality conditions for problem (18)–(20) are:

$$u'(c_{i,t}) = \beta \mathbb{E}_t\Big[(1 + r_{t+1})u'(c_{i,t+1})\Big] + \nu_{i,t}, \tag{21}$$

$$v'(l_{i,t}) = (1 - \tau_t) w_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}), \tag{22}$$

where $\nu_{i,t}$ is the (discounted) Lagrange multiplier on the borrowing constraint. When the constraint does not bind, $\nu_{i,t} = 0$.

Market clearing and resource constraints. Market clearing requires:

$$\int_i a_{i,t} \ell(di) = A_t = B_t + K_t, \qquad \int_i y_{i,t} l_{i,t} \ell(di) = L_t. \tag{23}$$

For simplicity, these conditions are written as integrals over agents i, though they are equivalent to integrating over initial states and idiosyncratic histories.[5]

The aggregate resource constraint is obtained by integrating the household budget constraint (19) over all agents,

$$C_t + A_t = w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + (1 + r_t) A_{t-1},$$

and combining this expression with the government budget constraint (17) and the identity $A_t = K_t + B_t$. This yields the economy-wide resource constraint:

$$C_t + G_t + K_t = \left(1 - \frac{\psi}{2}(\Pi_t - 1)^2\right) Y_t + K_{t-1} - \delta K_{t-1}, \tag{24}$$

where $C_t = \int_i c_{i,t} \ell(di)$ denotes aggregate consumption.

Equilibrium definition. The market equilibrium is defined as follows.

Definition 1 (Sequential equilibrium)**.** *A* sequential competitive equilibrium *is a collection of individual allocations* $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t \geq 0, i \in \mathcal{I}}$*; aggregate quantities* $(K_t, L_t, Y_t, D_t, \xi_t)_{t \geq 0}$*; price processes* $(r_t, \tilde{r}_t, r_t^k, \tilde{r}_t^k, R_t^n, \tilde{R}_t^n, w_t, \tilde{w}_t)_{t \geq 0}$*; fiscal policies* $(\tau_t^k, \tau_t, \kappa_t, B_t, G_t)_{t \geq 0}$*; and a monetary policy* $(\Pi_t)_{t \geq 0}$*, such that, for an initial distribution of wealth and productivity* $(a_{i,-1}, y_{i,0})_{i \in \mathcal{I}}$*, initial stocks of capital and public debt satisfying* $K_{-1} + B_{-1} = \int_i a_{i,-1} \ell(di)$*, and an initial aggregate shock* z_0*, the following conditions hold:*

1. ***Household optimization.*** *Given prices,* $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t \geq 0, i \in \mathcal{I}}$ *solve the agent's optimization problem* (18)*–*(20)*.*

[4] Existence of these policy functions follows from standard results in Açikgöz (2018), Cheridito & Sagredo (2016), and Miao (2006).

[5] Using the sequential representation, $\int_i a_{i,t} \ell(di) = \sum_{y^t \in \mathcal{Y}^{t+1}} \sum_{y_0 \in \mathcal{Y}} \int_{a_{-1} \in [-\overline{a}, \infty)} \theta_t(y^t) a_t((a_{-1}, y_0), y^t, z^t) \mathbf{\Lambda}(da_{-1}, y_0) = A_t(z^t) = A_t$, where dependence on z^t is omitted for brevity.

2. ***Market clearing.*** *Financial, labor, and goods markets clear at all dates: for every $t \geq 0$, conditions (23) and (24) hold.*

3. ***Government budget constraint.*** *For every $t \geq 0$, the government budget constraint (17) is satisfied.*

4. ***Factor prices and asset returns.*** *Factor prices $(r_t, \tilde{r}_t, r_t^k, \tilde{r}_t^k, R_t^n, \tilde{R}_t^n, w_t, \tilde{w}_t)_{t\geq 0}$ are consistent with the production relationships (7)–(8), the asset-pricing conditions (12) and (13), and the post-tax definitions (16).*

5. ***Price adjustment and inflation.*** *The inflation path $(\Pi_t)_{t\geq 0}$ satisfies the New Keynesian Phillips curve (10) for all $t \geq 0$.*

4 The Ramsey Problem

4.1 The Ramsey program

First, define the social welfare function as the expected discounted sum of individual utilities across all agents:

$$W_0 = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \int_i \omega_{i,t} \Big(u(c_{i,t}) - v(l_{i,t}) \Big) \ell(di) \right], \tag{25}$$

where $\omega_{i,t}$ represents the Pareto weight of agent i in period t. For a detailed discussion of the choice and interpretation of Pareto weights, see Le Grand et al. (2025).

In the Ramsey program, the planner determines a fiscal and monetary policy consistent with a competitive equilibrium that maximizes aggregate welfare according to the criterion in equation (25), while satisfying the government budget constraint. A ***Ramsey equilibrium*** is a set of fiscal policies, prices, individual allocations, and aggregate quantities that solve the Ramsey program. A ***Ramsey steady-state equilibrium*** is a time-invariant Ramsey equilibrium. Formally, the Ramsey program can be stated as follows.

$$\max_{\left(r_t, \tilde{r}_t, r_t^k, \tilde{r}_t^k, R_t^n, \tilde{R}_t^n, w_t, \tilde{w}_t, \tau_t^k, \tau_t, \kappa_t, B_t, G_t, \Pi_t, K_t, L_t, Y_t, D_t, \xi_t, (c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{i\in\mathcal{I}}\right)_{t\geq 0}} W_0, \tag{26}$$

$$G_t + B_{t-1} + r_t A_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) = \left(1 - \frac{\psi}{2}(\Pi_t - 1)^2\right) Y_t - \delta K_{t-1} + B_t, \tag{27}$$

$$\text{for all } i \in \mathcal{I}: \quad c_{i,t} + a_{i,t} = (1 + r_t) a_{i,t-1} + w_t (y_{i,t} l_{i,t})^{1-\tau_t}, \tag{28}$$

$$a_{i,t} \geq -\bar{a}, \quad \nu_{i,t}(a_{i,t} + \bar{a}) = 0, \quad \nu_{i,t} \geq 0, \tag{29}$$

$$u'(c_{i,t}) = \beta \mathbb{E}_t \Big[(1 + r_{t+1}) u'(c_{i,t+1}) \Big] + \nu_{i,t}, \tag{30}$$

$$v'(l_{i,t}) = (1 - \tau_t) w_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}), \tag{31}$$

$$\Pi_t(\Pi_t - 1) = \frac{\varepsilon - 1}{\psi}(\xi_t - 1) + \beta \mathbb{E}_t \left[\Pi_{t+1}(\Pi_{t+1} - 1) \frac{Y_{t+1}}{Y_t} \frac{M_{t+1}}{M_t} \right], \tag{32}$$

$$\int_i a_{i,t}\ell(di) = A_t = K_t + B_t, \quad L_t = \int_i y_{i,t} l_{i,t}\ell(di), \tag{33}$$

$$\tilde{r}_t A_{t-1} = \tilde{r}_t^k K_{t-1} + \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1\right) B_{t-1}, \tag{34}$$

$$\mathbb{E}_t\left[\frac{\tilde{R}_t^n}{\Pi_{t+1}}\right] = \mathbb{E}_t[1 + \tilde{r}_{t+1}^k], \tag{35}$$

$$w_t = \kappa_t(\tilde{w}_t)^{1-\tau_t}, \tag{36}$$

$$r_t^k = (1-\tau_t^k)\tilde{r}_t^k, \tag{37}$$

$$\xi_t = \frac{\tilde{r}_t^k + \delta}{\alpha}\frac{K_{t-1}}{Y_t}, \tag{38}$$

$$M_t = \int_i u'(c_{i,t})\ell(di), \tag{39}$$

$$Y_t = \left(\int_i a_{i,t-1}\ell(di) - B_{t-1}\right)^{\alpha}\left(\int_i y_{i,t} l_{i,t}\ell(di)\right)^{1-\alpha}. \tag{40}$$

Since the planner chooses policies and allocations to maximize welfare subject to the constraints of the economy, the individual budget constraints (28)–(31) are explicitly included in the Ramsey program. The government budget constraint (27), the New Keynesian Phillips curve (32), the aggregation conditions (33), the non-profit condition (34), the non-arbitrage condition (35), the post-tax definitions (36)–(37), the definition of ξ_t (38), the definition of M_t (39), and the production function (40) are also explicitly imposed.[6] The planner chooses sequences of prices, fiscal policies, individual allocations, and aggregate quantities to maximize welfare (26) subject to these constraints. The instruments available to the planner are: the labor income tax parameters (τ_t, κ_t), the capital income tax rate τ_t^k, the nominal public debt B_t, the government expenditure G_t, and the gross inflation rate Π_t.

Let μ_t denote the Lagrange multiplier on the government budget constraint (27), γ_t the multiplier on the Phillips curve (32), Γ_t the multiplier on the non-profit condition (34), and Ψ_t the multiplier on the non-arbitrage condition (35).

4.2 Interpretation of the Ramsey optimality conditions

The Ramsey first-order conditions are derived in full in Appendix C; three of them carry most of the paper's economic content and are displayed below. Each balances a cost against the benefits the planner derives from moving the corresponding instrument.

[6] $M_t = \int_i u'(c_{i,t})\ell(di)$ is the aggregate marginal utility of consumption, standard in heterogeneous-agent Ramsey frameworks (Açikgöz et al. 2022, Heathcote & Tsujiyama 2021). It enters the Phillips curve and the planner's FOCs as the object that translates welfare costs into nominal distortions.

The inflation condition. The planner trades the Rotemberg cost of price dispersion against two fiscal benefits, and the FOC equates them:

$$\underbrace{\mu_t \psi(\Pi_t - 1)}_{\substack{\text{Rotemberg cost}\\ \text{(price dispersion)}}} \;=\; \underbrace{(\gamma_{t-1} - \gamma_t)(2\Pi_t - 1)\, M_t}_{\substack{\text{real-wage manipulation}\\ \text{via the Phillips curve}}} \;+\; \underbrace{\Big(\Gamma_t(1-\tau_t^k)B_{t-1} - \beta^{-1}\Psi_{t-1}\Big)\frac{\tilde{R}^n_{t-1}}{\Pi_t^2 Y_t}}_{\substack{\text{erosion of real}\\ \text{debt-service cost}}}, \tag{41}$$

where $M_t \equiv \int_i u'(c_{i,t})\,\ell(di)$ is the aggregate marginal utility of consumption. The left side is the marginal cost (price-adjustment expenses reduce the tax base, internalized through μ_t). The first right-hand term is the gain from real-wage manipulation along the Phillips curve: higher inflation lowers the markup and raises the labor share. The second is the gain from eroding nominal debt: higher inflation reduces the real value of outstanding B_{t-1}. At a zero-inflation steady state both sides vanish; away from it the right-hand side is generally non-zero whenever debt is positive. Section 7.1 shows that at the Ramsey optimum both benefits are absorbed more cheaply by the tax-structure FOCs, leaving $\Pi_t = 1$.

The individual Euler equation. The Ramsey problem adds a planner-side analogue of the household Euler equation for the shadow value $\hat{\psi}_{i,t} \equiv \psi_{i,t} - \mu_t$, which measures the net welfare gain from relaxing agent i's budget constraint by one unit and is large for high-weight or borrowing-constrained agents:

$$\hat{\psi}_{i,t} \;=\; \beta\,\mathbb{E}_t\Big[(1 + r_{t+1})\,\hat{\psi}_{i,t+1}\Big] \;+\; \beta\,\mathbb{E}_t\left[\Gamma_{t+1}\left(r_{t+1} - (1-\tau^k_{t+1})\left(\frac{\tilde{R}^n_t}{\Pi_{t+1}} - 1\right)\right)\right]. \tag{42}$$

The first term propagates the shadow value across periods at the post-tax real return. The second is an asset-pricing wedge: whenever inflation or capital taxes drive the real return on capital away from the post-tax return on government bonds, the planner gains an additional margin for cross-period redistribution. Equation (42) is the channel through which the wealth distribution enters the optimal-policy problem; in a representative-agent model it collapses to a single point.

The progressivity condition. Raising the HSV progressivity parameter τ_t shifts the tax burden toward high-earning agents (a distributional gain when they carry low social weight) and reduces top-of-distribution labor supply (an efficiency cost that grows with the Frisch elasticity). The Ramsey planner balances the two:

$$\begin{aligned} 0 \;=\; &\int_j (y_{j,t}l_{j,t})^{1-\tau_t}\Big(\hat{\psi}_{j,t} + \lambda_{j,l,t}(1-\tau_t)\,u'(c_{j,t})/l_{j,t}\Big)\ln(y_{j,t}l_{j,t})\,\ell(dj) \\ &+\; \int_j \lambda_{j,l,t}\,u'(c_{j,t})/l_{j,t}\,(y_{j,t}l_{j,t})^{1-\tau_t}\,\ell(dj) \;-\; \mathcal{L}_t, \end{aligned} \tag{43}$$

where $\mathcal{L}_t$ collects the labor-supply distortion terms (full expression in equation (62) of Appendix C). The first integral is the distributional gain — log-income weighted by the shadow value $\hat{\psi}_{j,t}$; its sign depends on whether high earners carry low or high social weight. The second integral and $\mathcal{L}_t$ together are the labor-supply cost, increasing in the Frisch elasticity and in the progressivity already in place. This is the margin the Ramsey planner relies on to absorb fiscal shocks rather than inflation.

A line-by-line reading of the remaining FOCs appears in Appendix D.

4.3 Identification of the Social Welfare Function

If the observed UK tax system represents the best the government can do given its constraints, what must the government's social preferences look like? I answer this question using the inverse-optimal approach: I recover the Pareto weights $\omega_{i,t}$ that make the observed fiscal configuration an optimum of the Ramsey problem.

Specifically, I assume that the observed fiscal and monetary policies correspond to the optimal choices of a benevolent social planner who maximizes intertemporal welfare according to:

$$W_0 = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \int_i \omega_{i,t} \Big(u(c_{i,t}) - v(l_{i,t}) \Big) \ell(di) \right]. \tag{44}$$

The planner is endowed with a Social Welfare Function (SWF) and internalizes all distortions and general-equilibrium effects associated with fiscal and monetary instruments. The empirical identification therefore consists in recovering the sequence of Pareto weights $\omega_{i,t}$ that make the system of first-order conditions hold for the observed allocations and policies.

Since multiple welfare functions can, in principle, rationalize the same set of fiscal outcomes, I introduce an additional identification criterion: among all admissible SWFs consistent with the Ramsey optimality conditions, I select the one that is *closest* to the Utilitarian benchmark. Formally, this corresponds to minimizing a quadratic distance between the recovered weights and the uniform distribution, that is,

$$\min_{\{\omega_{i,t}\}} \sum_{i,t} (\omega_{i,t} - 1)^2, \tag{45}$$

subject to the planner's optimality conditions. This ensures that the inferred welfare function departs from the Utilitarian benchmark only insofar as required to rationalize the observed policies and inequality patterns. For a detailed discussion of this identification approach and its normative interpretation, see Le Grand et al. (2025).

Restriction to initial productivity. The Pareto weights $\omega_{i,t}$ that appear in the social welfare function (25) are, in their most general form, indexed jointly by agent i and date t, so they can depend on the full idiosyncratic history $y_i^t = \{y_{i,0}, y_{i,1}, \dots, y_{i,t}\}$ and the aggregate history z^t. This generality is what makes the framework powerful: it lets the planner fine-tune the social objective across agents who share the same current productivity but reached it through different paths. But it is also what makes the framework potentially vacuous: with one weight per history, the planner has enough degrees of freedom to rationalize almost any allocation (Le Grand et al. 2025). I address this by restricting the welfare weights to depend only on the agent's *initial* productivity state y_0, so the weight function collapses from a history-indexed object to a finite vector $\boldsymbol{\omega} = (\omega_y)_{y \in \mathcal{Y}} \in \mathbb{R}^{|\mathcal{Y}|}$, with one entry per point on the productivity grid.

Why initial productivity, not initial wealth. One might ask whether welfare weights should also condition on initial asset holdings a_0, as in a Mirrlees programme. I condition on y_0 only for the following reason: in the Ramsey steady state, a_0 is the beginning-of-period wealth draw from the model's own ergodic distribution, so it is endogenously determined by the fiscal and monetary policies that the planner chooses. Conditioning the welfare weight on a_0 would therefore create a circularity: the weight function would change whenever the planner adjusts policy, making the SWF a moving target rather than a stable normative criterion. In contrast, y_0 is drawn from an exogenous Markov process that the planner takes as given. This is consistent with the standard inverse-optimal taxation approach of Bourguignon & Amadeo (2015) and Heathcote & Tsujiyama (2021), where welfare weights are conditioned on exogenous type characteristics (productivity) rather than on endogenous outcomes (wealth).

Recovery procedure. Under the y_0-only restriction, the weight function reduces to a finite vector $\boldsymbol{\omega} = (\omega_y)_{y \in \mathcal{Y}}$. I parameterize this vector by the two-parameter exponential $\omega(y_0) = \exp(\theta_1 \log y_0 + \theta_2 (\log y_0)^2)$, strictly positive by construction. The two parameters are then pinned down by the labor-tax-level FOC (which determines κ) and the progressivity FOC (which determines τ), with the remaining FOCs used to eliminate the auxiliary multipliers; the implementation, with the calibrated steady states in hand, is detailed in Section 5.4. The recovery is a moment-matching procedure rather than a statistical estimator, so I use the term "recovered" rather than "estimated".

5 Quantitative Investigation

I calibrate the model to three UK fiscal regimes: 2007 (low debt, pre-crisis), 2021 (high debt, post-pandemic), and 1997 (very low debt, pre-financial-crisis). The three calibration years bracket a three-fold variation in the debt-to-GDP ratio ($B/Y = 0.35$, 0.42, 1.13), providing a broad empirical foundation for the dynamic analysis in Sections 6–9.

5.1 Calibration strategy and the steady-state assumption

A natural concern is that none of the three benchmark years represents a literal macroeconomic steady state. The year 2007 preceded the global financial crisis; 2021 followed the

COVID-19 fiscal expansion; and even 1997, while preceding both events, marked the beginning of a transition from Conservative to New Labour fiscal policy. I do not claim that the UK economy was on a balanced growth path in any of these years.

Instead, I interpret each calibration as representing a *fiscal regime* — a constellation of debt level, tax progressivity and inequality that prevailed for a sustained period. The comparison across regimes isolates the effect of the debt-to-GDP ratio on optimal policy, holding the model structure fixed. The three-year gradient ($B/Y = 0.35$ in 1997, 0.42 in 2007, 1.13 in 2021) spans more than three-fold variation in the fiscal stance and brackets the relevant range of UK fiscal history over the past thirty years (Figure 1, Panel a).

Two features of the design strengthen this interpretation. First, the pre-tax income Gini is remarkably stable across all three calibration years (0.50–0.54; see Table 1), so the cross-year variation in optimal policy is driven by debt, not by changes in inequality. Second, the addition of UK 1997 provides a reference point with unambiguously low debt ($B/Y = 0.35$) that was essentially flat for twelve years (1985–1997), making it the closest approximation to a literal steady state among the three calibration years. This addresses the concern that the 2007 calibration may be contaminated by the approaching financial crisis.

I acknowledge that the elevated debt in 2021 is partly the result of transitory shocks (the pandemic, the energy-price episode) rather than a permanent shift in fundamentals. The results should therefore be read as characterizing optimal policy *conditional on the observed debt level*, not as predicting that the UK will remain at $B/Y \approx 1$ in perpetuity. The relevant policy question — how should fiscal and monetary authorities coordinate when debt is high? — is well-posed regardless of whether the high-debt state is permanent or transitory.

Why 2007 as the primary low-debt benchmark. The 2007/2021 pair brackets the fiscal regime change of the past two decades — UK public debt rose from roughly 43% to 101% of GDP between these two years — and each calibration is computed independently against contemporary UK moments, so the parameters $(\rho_y, \sigma_y, \tau, \kappa, \tau^k)$ differ across calibrations in addition to the debt level (Section 5.2). The qualitative predictions hold across all three calibrations despite these parameter differences, so the mechanism is driven by the debt level rather than by any particular parameter configuration. UK 1997 generalizes the results across the full debt range and reveals patterns that a two-year comparison cannot: the twelve-fold progressivity gradient ($\hat{\tau} = 0.016 \to 0.037 \to 0.194$) and the qualitative regime change from countercyclical borrowing (1997, 2007) to procyclical consolidation (2021) are visible only because all three periods are present. UK 1997 enters the baseline IRFs (Figure 5), the 3×3 synthesis (Table 7), and the historical validation (Section 9).

5.2 The UK economies in 2007 and 2021

Both calibrations share the same quarterly frequency, functional forms, and structural parameters ($\alpha = 0.36$, $\delta = 0.025$, $\gamma = 2$, $\varphi = 0.5$, $\varepsilon = 6$, $\psi = 100$). The period utility function is $u(c) = (c^{1-\gamma} - 1)/(1-\gamma)$ and $v(l) = \chi^{-1} l^{1+1/\varphi}/(1 + 1/\varphi)$, with $\beta = 0.99$ targeting an annual capital-to-output ratio of $K/Y = 2.56$, the UK average over 1990–2019 in the Penn World Tables (version 10.01). Production is Cobb–Douglas, $Y_t = Z_t K_{t-1}^{\alpha} L_t^{1-\alpha}$, with TFP following a standard AR(1).

Idiosyncratic productivity follows $\log y_t = \rho_y \log y_{t-1} + \eta_{y,t}$, discretized with seven states (Rouwenhorst 1995). For each year, the parameters $(\rho_y, \sigma_y, \tau, \kappa, \tau^k)$ are calibrated via the Simulated Method of Moments to match three targets from ONS data: the debt-to-GDP ratio, the post-tax Gini coefficient, and the variance of pre-tax income. Table 2 reports all parameter values; Table 3 compares model-implied moments with the data.

		UK 2007		UK 2021	
Parameter	**Description**	**Value**	**Target / Source**	**Value**	**Target / Source**
Preference parameters					
β	Discount factor	0.99	$K/Y = 2.56$ (PWT 10.01)	0.99	$K/Y = 2.56$ (PWT 10.01)
γ	Risk aversion	2.0	Standard	2.0	Standard
φ	Frisch elasticity	0.5	Chetty et al. (2011)	0.5	Chetty et al. (2011)
χ	Labor scaling	0.058	Normalized $L = 0.39$	0.053	Normalized $L = 0.35$
Technology and fiscal parameters					
α	Capital share	0.36	ONS (Profit share)	0.36	ONS (Profit share)
δ	Depreciation rate	0.025	Standard	0.025	Standard
ε	CES elasticity	6.0	Bilbiie & Ragot (2021)	6.0	Bilbiie & Ragot (2021)
ψ	Rotemberg cost	100	Bilbiie & Ragot (2021)	100	Bilbiie & Ragot (2021)
τ^k	Capital income tax	0.38	Bank of England	0.35	Bank of England
τ	Labor tax progressivity	0.05	Moment-matching	0.22	Moment-matching
κ	Labor tax scaling	0.62	Moment-matching	0.775	Moment-matching
Idiosyncratic productivity parameters					
ρ_y	Autocorrelation	0.995	Moment-matching	0.992	Moment-matching
σ_y	Std. deviation	0.07	Moment-matching	0.09	Moment-matching

Table 2: Calibration parameters for the UK economies in 2007 and 2021.

UK 2007. The calibration yields $\rho_y = 0.995$, $\sigma_y = 0.07$, $\tau = 0.05$, $\kappa = 0.62$, and $\tau^k = 0.38$. The model reproduces the targeted statistics closely: pre-tax Gini 0.53 (target 0.535), post-tax Gini 0.384 (target 0.386), and $B/Y = 42.5\%$ (target 43.1%).

UK 2021. The calibration yields $\rho_y = 0.992$, $\sigma_y = 0.09$, $\tau = 0.22$, $\kappa = 0.775$, and $\tau^k = 0.35$. The model matches the pre-tax Gini closely (0.50 vs. 0.502) and the debt ratio reasonably (112% vs. 101%). The post-tax Gini is 0.357 against a target of 0.299 — a gap of approximately 6 points. This reflects a limitation of the two-parameter HSV tax function, which cannot capture means-tested benefits.[7] I prioritize matching the pre-tax distribution because the dynamic analysis operates through pre-tax prices and the tax schedule. The model nonetheless captures the key qualitative feature — that the 2021 fiscal system is substantially more redistributive than the 2007 system. If anything, a richer tax

[7]The 2007 calibration matches the post-tax Gini much more closely (0.384 vs. 0.386), because the pre-crisis UK system relied less on means-tested transfers.

function that matched the post-tax Gini more closely would imply even greater steady-state redistribution, likely strengthening the zero-inflation result by making the progressive tax structure an even more effective substitute for inflation.

	UK 2007		**UK 2021**	
	Model	**Data**	**Model**	**Data**
B/Y	42.5%	43%	112%	101%
Gini (pre-tax income)	0.53	0.535	0.50	0.502
Gini (post-tax income)	0.384	0.386	0.357	0.299

Table 3: Model-implied moments for key fiscal and inequality indicators.

5.3 A Third Calibration: UK 1997

To span the full range of UK fiscal experience, I add UK 1997 ($B/Y = 0.35$, income Gini = 0.34) as a third calibration. The two targets pull in opposite directions using the income-risk parameter σ_y alone: lowering σ_y reduces inequality toward the 1997 Gini but also lowers precautionary saving, pushing the debt ratio negative. I resolve this by tuning β upward (from 0.99 to 0.9935) to increase aggregate saving while preserving income dispersion. The final parameter set inherits all structural parameters from UK 2007 (α, δ, γ, ϕ, κ, τ, ρ_y) and adjusts three: $\sigma_y = 0.057$, $\tau_k = 0.40$, and $\beta = 0.9935$. The resulting steady state matches both targets: $B/(4Y) = 0.3515$ (vs. target 0.35, off by 1.5%) and Gini = 0.341 (vs. target 0.34, off by 0.2%). Table 4 summarizes the tuned parameters and resulting moments.

	UK 1997	UK 2007	UK 2021
Tuned parameters			
σ_y (income risk)	0.057	0.070	0.091
β (discount factor)	0.9935	0.990	0.990
τ_k (capital tax)	0.40	0.38	0.35
Resulting moments			
$B/(4Y)$ (annual debt/GDP)	0.35	0.42	1.13
Gini (post-tax income)	0.34	0.36	0.30

Table 4: UK 1997 calibration: tuned parameters and resulting moments.

5.4 Numerical Tools and Recovery of Pareto Weights

With the three calibrations in hand, the remaining task is to translate them into dynamic responses. This subsection summarizes the numerical machinery that underlies the impulse responses reported in Sections 6–9. The identification procedure recovers Pareto weights that satisfy the first-order conditions of the social planner (i.e., equations (51)–(62) in Appendix C) and are as close as possible to the utilitarian benchmark. The methodology follows Le Grand et al. (2025).

The Ramsey problem in Section 4.1 features a joint distribution over wealth and Lagrange multipliers, which is high-dimensional and interacts with the planner's instruments in ways that make conventional perturbation methods around a steady state infeasible (Reiter 2009, Boppart et al. 2018, Bayer et al. 2019, Auclert et al. 2021). I therefore rely on the Lagrangian aggregation method of Le Grand & Ragot (2022*a*), further developed in Le Grand & Ragot (2022*b*), LeGrand & Ragot (2023, 2025) and Le Grand et al. (2025). The method groups agents who share the same idiosyncratic productivity history over the last N periods into "aggregate" agents whose averaged allocations exactly reproduce the steady state of the underlying Bewley economy and approximate its dynamics under small aggregate shocks. I adopt $N = 5$ throughout (larger values leave the results virtually unchanged) and use the refined-truncation approach of Le Grand & Ragot (2022*b*) to keep the number of histories tractable. Appendix E states the sequential representation, the truncated model, and the planner's first-order conditions in closed form.

Recovering the weights. With the truncated steady-state allocation in hand, the planner's first-order conditions involve many objects: the agent-level multipliers ψ_i, $\lambda_{c,i}$, $\lambda_{l,i}$; the aggregate multipliers μ (government budget), γ (Phillips curve), Γ, Ψ (asset pricing); and the weights $\boldsymbol{\omega}$. The recovery proceeds in two steps. First, the FOCs with respect to $a_{i,t}$, $l_{i,t}$, B_t, r_t, $\tilde{R}_t^n$, $\tilde{r}_t^k$, and Π_t are used to express the auxiliary multipliers as linear functions of $\boldsymbol{\omega}$, leaving only the labor-tax-level FOC and the progressivity FOC to be imposed. Second, substituting the parametric form $\omega(y_0) = \exp(\theta_1 \log y_0 + \theta_2 (\log y_0)^2)$ into these two FOCs yields a system of two scalar equations in (θ_1, θ_2):

$$\widetilde{\boldsymbol{L}}_1\, \boldsymbol{\omega}(\theta_1, \theta_2) \;=\; \widetilde{\boldsymbol{b}}_1, \qquad \widetilde{\boldsymbol{L}}_2\, \boldsymbol{\omega}(\theta_1, \theta_2) \;=\; \widetilde{\boldsymbol{b}}_2, \tag{46}$$

where $\widetilde{\boldsymbol{L}}_1, \widetilde{\boldsymbol{L}}_2$ are row vectors and $\widetilde{\boldsymbol{b}}_1, \widetilde{\boldsymbol{b}}_2$ are scalar constants, all computed from the steady-state allocation (full expressions in Appendix E, equations (108)–(109)). The system is nonlinear in (θ_1, θ_2), so uniqueness is not automatic; Appendix F verifies that the recovered (θ_1^*, θ_2^*) is the unique global solution.

Recovered weights across the three UK calibrations. Figure 2 reports the recovered weight vectors $\boldsymbol{\omega}^*$ for UK 2007 and UK 2021 alongside the utilitarian benchmark.

Three features of Figure 2 are worth emphasizing. First, the weights rise monotonically with productivity (left panel) and differ across calibrations in an economically meaningful way: $\omega(y_{0,\max}) = 11.9$ in the low-debt 2007 economy versus 7.6 in the high-debt 2021 economy. The pre-crisis UK fiscal system, with lower debt and less redistribution (post-tax Gini 38.6 vs. 29.9), implies a welfare function that places relatively greater weight on the productive tails of the income distribution, consistent with a more growth-oriented fiscal stance when fiscal space is ample. This systematic variation across fiscal regimes would not arise if the identification were vacuous.

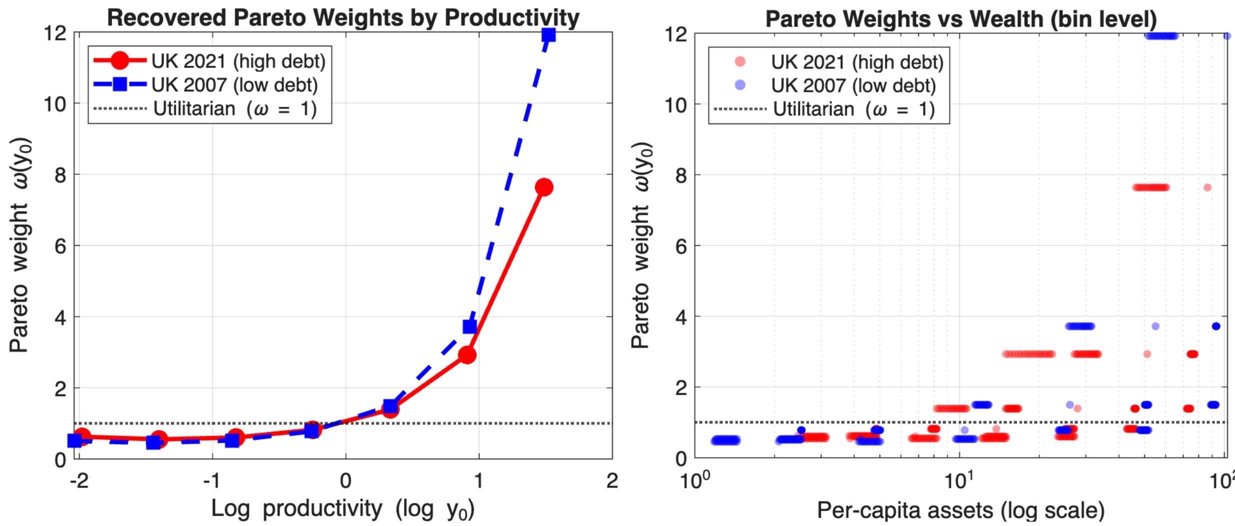

Figure 2: Recovered Pareto weights. **Left**: weights by productivity type y_0. **Right**: weights by mean wealth holding $\bar{a}(y_0)$. UK 2021 (red), UK 2007 (blue), utilitarian (dashed).

Second, the relationship between weights and wealth (right panel) is *non-monotonic*: mean asset holdings peak at intermediate productivity types and decline for the very highest types, since the most productive agents are rare and cycle through the income process before accumulating much wealth. As a result, the highest-weight agents are not the highest-wealth agents. This distinction matters for the mechanism behind the zero-inflation result analysed in Section 7: it is the welfare-weighted asset position $\omega \cdot \bar{a}$, not assets alone, that determines the social cost of an inflation tax on nominal wealth.

Third, the parametric exponential form is not the only way to recover the weights. Section 8 re-estimates them via a non-parametric minimum-norm correction of the utilitarian benchmark that imposes the same two Ramsey first-order conditions as linear constraints, and Appendix F explores additional unrestricted bin-level specifications. Both alternative procedures deliver weight vectors of the same qualitative shape and the same 2007-versus-2021 contrast, confirming that the parametric assumption is innocuous.

Connection to the welfare weight literature. The inverse-optimal approach builds on the theoretical foundation of Le Grand et al. (2025), who develop a "Bewley theory of the Social Welfare Function" in heterogeneous-agent economies. The resulting SWF takes the linear-Pareto form $W_0 = \sum_i \omega_i V_i$, where the weights ω_i emerge endogenously from the interaction of heterogeneous views rather than being imposed *a priori*.

This micro-foundation has two important implications for the recovered profile. The steeply increasing weights need not be interpreted as a normative preference for inequality. They can instead reflect the fiscal system required to rationalize observed redistribution.

In this sense, the steeper profile in UK 2007 ($\omega_{\max} = 11.9$) relative to UK 2021 ($\omega_{\max} = 7.6$) captures the less redistributive fiscal configuration of the pre-crisis period.

Second, the recovered weights are *sufficient statistics* for optimal stabilization policy. Le Grand et al. (2025) show that, under first-order perturbation, the impulse responses depend on the marginal welfare weights at the steady state, not on the global curvature of the welfare aggregator. This is why a concave transformation $\Phi(U^{1-\eta})/(1-\eta)$ produces analytically identical results for any η (Appendix G): different degrees of aggregate inequality aversion leave the linearized dynamics unchanged, provided the marginal weights coincide. The recovered weights therefore characterize the planner's response to the shocks studied here, regardless of the specific functional form of the social aggregator.

The existing inverse-optimal literature — including Heathcote & Tsujiyama (2021), Bourguignon & Amadeo (2015), and Le Grand et al. (2025) — recovers welfare weights from static fiscal configurations but does not investigate their consequences for dynamic stabilization policy. I depart by asking whether the recovered social preferences shape the *optimal response to aggregate shocks*. Sections 7 and Appendix G show that the choice of welfare function has first-order consequences: alternative specifications alter fiscal response magnitudes, while leaving the qualitative mechanisms — zero inflation, progressivity as the primary adjustment margin — unchanged.

6 Optimal Policy Responses

I now turn to the dynamic implications of the calibrated model. The numerical strategy of Section 5.4 combines the truncation method with a first-order perturbation of the Ramsey system (26)–(40). The planner's instruments are the fiscal variables $(\tau_t^k, \tau_t, \kappa_t, B_t)$ and, when monetary policy is active, the nominal instruments $(\Pi_t, \tilde{R}_t^n)$. The exogenous driver in the main experiments is government spending: following Chari & Kehoe (1999) and LeGrand & Ragot (2025), G_t is an exogenous expenditure requirement and $g_t \equiv \ln(G_t/G_{ss})$ follows the AR(1) process:

$$g_t \;=\; \rho_g \, g_{t-1} \;+\; \varepsilon_{g,t}, \tag{47}$$

where $\rho_g \in [0, 1)$ is the persistence parameter and $\varepsilon_{g,t}$ is a mean-zero innovation with standard deviation σ_g. I consider two values of persistence in order to bracket the empirically relevant range: a high-persistence case ($\rho_g = 0.97$) that captures slow-moving expenditure shifts of the kind observed in the post-GFC period, and a low-persistence case ($\rho_g = 0.01$) that captures essentially one-off shocks such as the immediate Covid fiscal response. Appendix H confirms the results under aggregate productivity shocks.

What follows isolates the fiscal-only benchmark, the joint fiscal–monetary optimum that delivers the zero-inflation result, and the regime change across the three UK debt levels.

6.1 Fiscal policy only

As a benchmark, I shut down monetary policy and study the optimal fiscal response. Figure 3 reports the impulse responses to a government-spending shock under both persistence regimes.

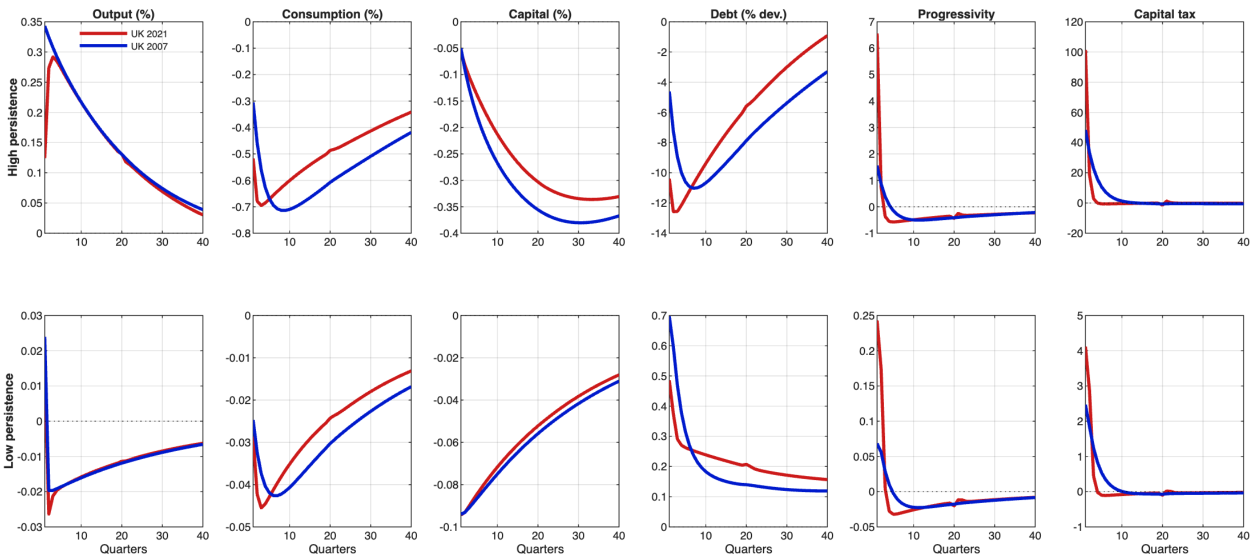

Figure 3: Government spending shock, fiscal policy only. Top: high persistence ($\rho_g = 0.97$); bottom: low persistence ($\rho_g = 0.01$). UK 2021 (red), UK 2007 (blue).

Two features are common to both calibrations. First, the rise in government spending crowds out private investment, generating a temporary decline in the capital stock K_t: higher public expenditure substitutes for private saving in the household's portfolio, depressing investment until the shock dissipates. Private consumption C_t also falls on impact as the planner reallocates resources toward public expenditure while smoothing the welfare cost intertemporally. Second, persistence governs how the planner finances the shock. Under low persistence the shock is transitory and standard tax-smoothing dominates: in *both* calibrations the planner allows public debt to rise modestly as a buffer and tax instruments barely move. Under high persistence, by contrast, the two calibrations diverge sharply, and inherited debt becomes the dimension that matters for fiscal space.

The mechanism behind the high-persistence response is as follows. Under high persistence, the decline in capital is long-lasting, so additional debt issuance would require higher taxes in periods when the tax base is already weak (i.e., periods when the capital is still low). Anticipating this constraint, the planner restrains borrowing and opts for more immediate fiscal adjustment through higher taxes. The response of debt is therefore smaller and shorter-lived.

UK 2021 (high debt). Persistence governs the direction of the debt response; the inherited debt level governs only the magnitude of the tax-side adjustment. Under high persistence the planner front-loads the fiscal adjustment, raising HSV progressivity to a peak of 6.5 pp on impact alongside a sharp increase in the capital tax, while public debt falls on impact and recovers only slowly. Under low persistence the shock is transitory: public debt rises modestly as a buffer and the tax-side response is small in absolute magnitude, although still meaningfully larger than the corresponding UK 2007 response.

UK 2007 (low debt). The qualitative path of public debt mirrors UK 2021 within each persistence regime: under high persistence debt also falls on impact, and under low persistence it rises. What differs is the scale of the tax-side adjustment. Under high persistence progressivity peaks at only 1.6 pp, about a quarter of the UK 2021 adjustment, and the capital-tax response is correspondingly smaller. Under low persistence the tax-side movement is again much smaller than in UK 2021. In every case the planner relies more heavily on debt and less on distortionary taxes than the high-debt economy does, but the qualitative direction of the debt response is identical to UK 2021's.

Tables 12 and 13 in Appendix I report the full set of moments and peak responses for the fiscal-only case. This persistence-versus-debt-level decomposition changes once monetary policy is added in Section 6.2: with the additional nominal instrument available, the calibrations also diverge along the debt margin, with UK 2021 consolidating and UK 2007 borrowing countercyclically under high persistence.

6.2 The zero-inflation result

When the planner can adjust both fiscal and monetary instruments — including inflation Π_t and the nominal interest rate $\tilde{R}_t^n$ alongside taxes and debt — the composition and magnitude of the optimal response change substantially. Figure 4 reports the impulse responses.

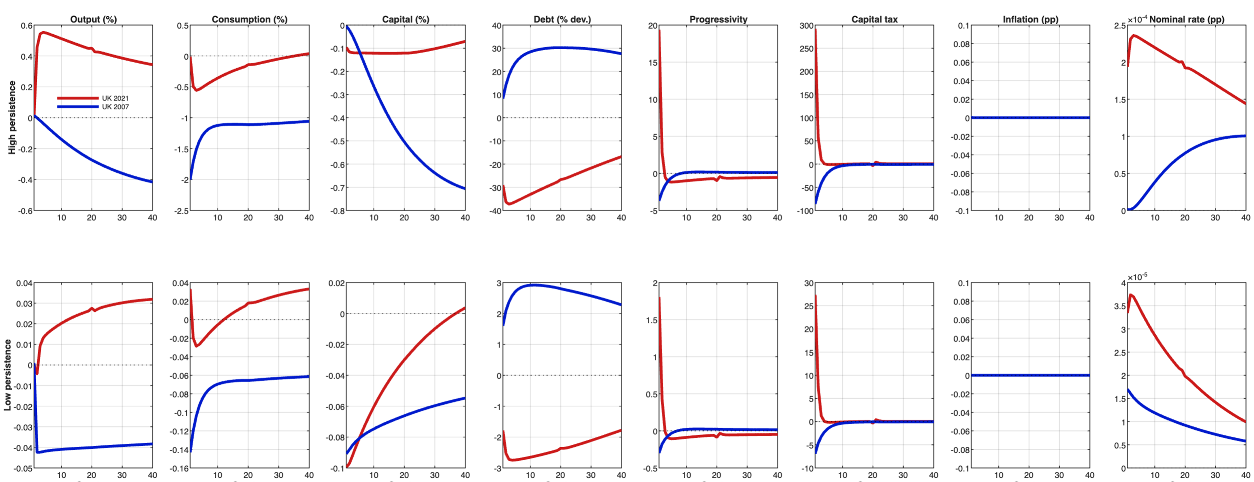

Figure 4: Government spending shock, optimal fiscal and monetary policy. Top: high persistence ($\rho_g = 0.97$); bottom: low persistence ($\rho_g = 0.01$). UK 2021 (red), UK 2007 (blue).

The headline result is that *optimal inflation is zero in both calibrations*. The planner uses the nominal interest rate to anchor prices and relies entirely on fiscal instruments to absorb the real distortions caused by the shock. Beyond this common feature, the composition of the fiscal adjustment differs sharply across calibrations, and the difference traces directly to the inherited debt level.

UK 2021 (high debt). The spending shock crowds out private investment and depresses the capital stock. Consumption smoothing would prescribe debt issuance to provide households with a safe store of value, but the elevated debt-to-GDP ratio rules this out: additional borrowing would require higher future taxes precisely when the capital base, and the tax base it supports, is weakened. Anticipating this trade-off, the planner front-loads the fiscal adjustment. Progressivity rises by 19.4 pp on impact (against 6.5 pp under fiscal-only policy), and the planner relies on the tax structure rather than on borrowing to absorb the shock. The cost is a sharper short-run contraction in private demand; the benefit is the preservation of debt sustainability without recourse to inflation.

UK 2007 (low debt). With ample fiscal space, the planner takes the opposite route. Public debt rises by 17.8% to finance the temporary spending expansion, and progressivity *falls* rather than rises, broadening the tax base in order to stimulate labor supply, investment, and aggregate demand during the contraction. Lower distortionary taxes boost disposable income and consumption; the budget deficit widens; the debt ratio rises temporarily. This countercyclical stance is feasible precisely because the inherited debt level leaves room for additional borrowing without threatening sustainability.

Two regimes, one inflation target. The contrast reveals two distinct fiscal regimes: countercyclical debt issuance in the low-debt economy, front-loaded tax-based consolidation in the high-debt economy. In both, monetary policy plays an anchoring role: the planner adjusts the nominal interest rate to neutralize the inflationary pressure created by the shock, holding $\Pi_t \approx 1$ throughout. The introduction of monetary instruments also weakens the dependence of debt dynamics on shock persistence relative to the fiscal-only benchmark of Section 6.1, but the high-debt economy remains constrained: limited fiscal capacity transforms the response from countercyclical to procyclical and amplifies the short-run output cost of stabilization.

The second-moment evidence corroborates this asymmetry along three dimensions. Capital taxation is strongly procyclical in both calibrations, but its volatility is markedly higher under high debt, reflecting the heavier reliance on fiscal adjustment. Progressivity is countercyclical in both economies and the correlation is stronger under high debt: the planner raises progressivity in downturns to redistribute the burden of adjustment, and does so more aggressively when fiscal space is tight. Public debt is countercyclical and highly persistent in both calibrations, yet less volatile under high debt, confirming that the high-debt planner cannot afford large debt fluctuations. Tables 14 and 15 in Appendix I report the full set of moments, correlations, and peak responses for readers interested in the underlying magnitudes.

6.3 Three calibrations, two fiscal regimes

The two-calibration comparison establishes the zero-inflation result, but it cannot reveal whether the optimal fiscal response varies smoothly with debt or jumps at a threshold. Adding UK 1997 to the comparison answers this question — and uncovers a qualitative regime change in fiscal adjustment that the two-calibration design hides. Figure 5 reports the impulse responses to a government-spending shock under high persistence for all three calibrations, and Table 5 summarizes the corresponding peak responses.

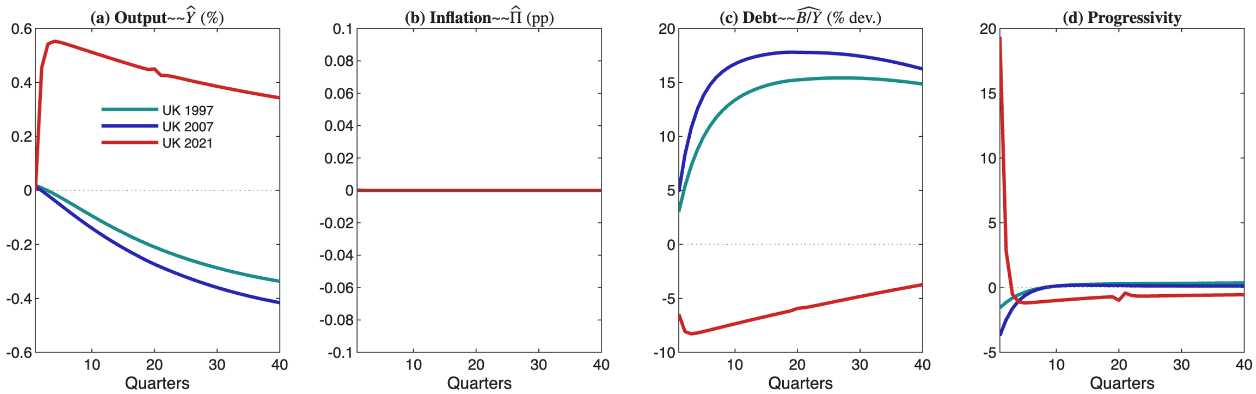

Figure 5: Baseline IRFs, G shock ($\rho_g = 0.97$). UK 1997 (teal), UK 2007 (blue), UK 2021 (red). Inflation is virtually zero in all three calibrations.

	Output $\hat{Y}$ (%)	Inflation $\lvert\hat{\Pi}\rvert$ (pp)	Debt $\widehat{B/Y}$ (%)	Progressivity $\hat{\tau}$
UK 1997 ($B/Y = 0.35$)	0.337	≈ 0	15.42	0.016
UK 2007 ($B/Y = 0.42$)	0.416	≈ 0	17.80	0.037
UK 2021 ($B/Y = 1.13$)	0.553	≈ 0	8.28	0.194

Table 5: Peak IRFs across three calibrations ($\rho_g = 0.97$).

The regime change. The qualitative nature of fiscal adjustment changes discretely along the debt gradient. Under UK 1997 and UK 2007, the planner borrows countercyclically: debt rises on impact ($\widehat{B/Y} = 15.4\%$ and 17.8% respectively) and progressivity barely moves ($\hat{\tau} \leq 0.04$). Under UK 2021, the planner consolidates procyclically: debt *falls* on impact ($\widehat{B/Y} = 8.3\%$, with the sign of the impact response reversed) while progressivity jumps twelve-fold ($\hat{\tau} = 0.19$). The threshold lies between $B/Y = 0.42$ and $B/Y = 1.13$ — precisely the range over which the UK transitioned from low to high debt.

The economic logic is visible in the Ramsey FOC for public debt, which requires the shadow cost of public funds μ_t to satisfy $\mu_t = \beta(1+\tilde{r}_{t+1})\mu_{t+1}$ (equation (56) in Appendix C).

At high debt, interest payments on the outstanding stock are large, so issuing additional debt raises μ_{t+1} substantially — the marginal cost of future fiscal adjustment is high. The planner therefore avoids borrowing and instead front-loads the adjustment through higher progressivity and capital taxes, consolidating on impact. At low debt, the interest-payment channel is weak, μ_{t+1} responds little to additional borrowing, and the planner exploits debt as a shock absorber — the standard tax-smoothing logic of Barro (1979). The regime change is therefore driven by the *curvature of the debt-service cost*: at high B/Y, a marginal unit of debt is more expensive to service, tilting the planner's intertemporal trade-off toward front-loading.

Inflation stays at zero throughout. Despite this discrete change in the fiscal-adjustment composition, inflation is virtually zero in all three calibrations — as panel (b) of Figure 5 makes visually clear, the inflation response is indistinguishable from the zero line at any economically meaningful scale. This confirms that the zero-inflation result is not an artefact of any single calibration: it survives both the smoothly-varying responses at low debt and the qualitatively different procyclical-consolidation regime at high debt.

7 The Mechanism

Why does the Ramsey planner refuse to use inflation? The Ramsey first-order condition for inflation, equation (41) in Section 4.2, equates its marginal cost to two benefits: real-wage manipulation and the erosion of real debt. At the optimum, both benefits are absorbed by the progressivity and tax-level conditions, leaving inflation with no residual work to do. Section 7.1 establishes this and quantifies it (the tax structure is 70× cheaper per unit of redistribution); Section 7.2 shows the mechanism is heterogeneity-driven; Section 7.3 tests the converse — freezing the tax structure brings inflation back, and it scales with debt — and consolidates the results within a single 3×3 frame.

7.1 The instrument hierarchy

The Ramsey planner keeps inflation at zero because the progressive tax structure — both the progressivity parameter τ and the level parameter κ — is a strictly cheaper instrument for redistribution than the inflation tax. The decomposition in Table 19 (Appendix J) confirms that τ and κ respond with nearly identical magnitudes on impact ($100\Delta\hat{\tau} = 19.4$, $100\Delta\hat{\kappa} = 19.1$). The inflation FOC from Section 4.2, equation (41), makes this precise: the marginal Rotemberg cost equals the sum of real-wage manipulation and real-debt erosion. At the Ramsey optimum the left-hand side is zero ($\Pi_t = 1$), not because the right-hand side is zero — the debt-erosion benefit of inflation is positive in the high-debt economy — but because the *progressivity and level conditions* (equations (60)–(62)) absorb the redistributive motive more cheaply. The key link is the shadow cost μ_t: when the tax-structure FOCs are satisfied, the marginal value of relaxing the government budget constraint through progressivity exceeds the marginal value of relaxing it through inflation. The inflation condition then has no residual work to do.

Distinction from representative-agent NK Ramsey. In a standard representative-agent New Keynesian model with Rotemberg costs and flexible fiscal instruments, the Ramsey planner also sets $\Pi_t \approx 1$ — the well-known zero-inflation result of the NK-Ramsey literature (Schmitt-Grohé & Uribe 2004). But the reason is fundamentally different. In the RA model, there is no distributional motive: the right-hand side of (41) is small because the single agent can absorb the shock through intertemporal substitution, so the planner has little reason to manipulate either prices or the real value of debt. In the HA model, the distributional motive is large — the spending shock creates heterogeneous welfare costs across agents with different wealth and income — but the planner satisfies this motive through the tax structure rather than through inflation. The HA zero-inflation result is therefore a *substitution* result (progressivity crowds out inflation), not an *irrelevance* result (inflation has no benefit). The constrained experiments in Section 7.3 confirm this distinction: when the tax structure is frozen, the distributional motive that was being absorbed by progressivity spills over into inflation.

The cost difference is quantitative. The inflation FOC establishes that the planner avoids inflation because the tax structure absorbs the redistributive motive more cheaply. The remainder of this subsection quantifies the cost difference. The key comparison is between two channels of fiscal adjustment: inflation, which operates through the *wealth distribution* (eroding accumulated savings), and progressive taxation, which operates through the *income distribution* (shifting the tax burden toward high earners and transferring to low earners). In an economy with incomplete markets and precautionary saving, these are overlapping but distinct populations, and the welfare-weighted cost of each instrument depends on how the burden is distributed across agents with different social marginal values.

Figures 6 and 7 document this quantitatively, panel by panel.

The HSV tax schedule is strongly progressive (Figure 6). Panel (a) plots the average tax rate by productivity type for both calibrations. In UK 2021 the lowest productivity type receives a large net subsidy — the effective tax rate is -333%, meaning post-tax income exceeds pre-tax labor income by more than a factor of four — while the highest productivity type pays 56%. The same monotone profile holds for UK 2007, although flatter, consistent with the less redistributive pre-crisis fiscal system. Panel (b) reports the immediate effect on net income of a marginal progressivity increase of $\Delta\tau = +0.01$: low-productivity agents gain $+2.2\%$ in net income, while high-productivity agents lose -0.7%. The progressivity parameter is therefore an income-redistribution dial that moves resources from high to low earners, with modest absolute losses at the top. Panel (c) reports the recovered welfare weights

$\omega(y_0)$, which rise steeply with productivity in both calibrations and supply the metric for the welfare-cost comparisons below.

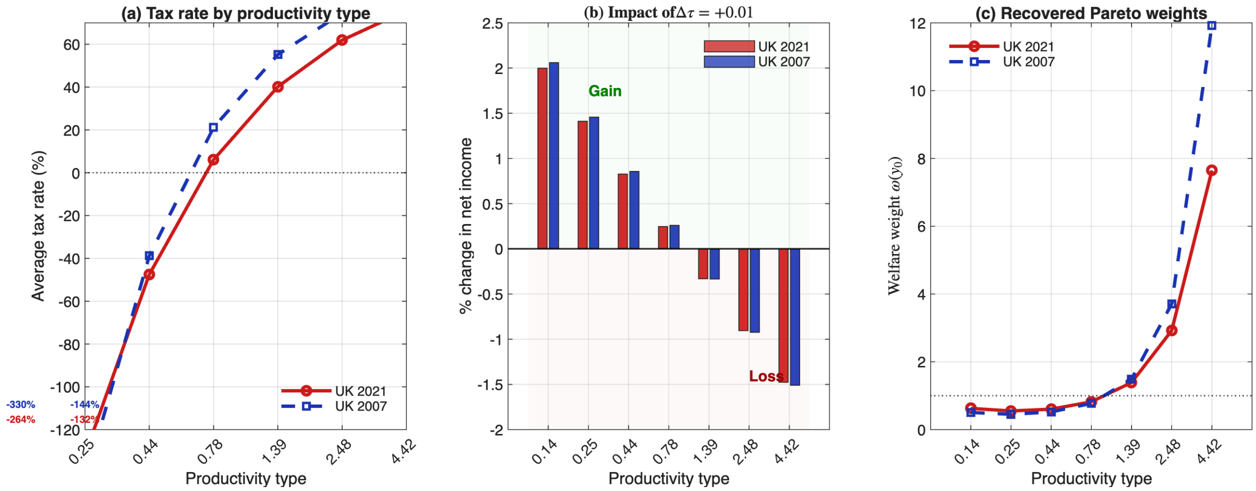

Figure 6: (a) Average tax rate by productivity type. (b) Change in net income from $\Delta\tau = +0.01$. (c) Recovered welfare weights $\omega(y_0)$. UK 2021 (red), UK 2007 (blue).

Inflation, by contrast, taxes wealth (Figure 7). Panel (a) shows the percentage loss of real wealth at each productivity type from a 1 pp inflation shock in UK 2021. The loss is nearly proportional to mean asset holdings, so the top type loses 23.8$\times$ more than the bottom type in absolute terms. Panel (b) shows the same incidence for a marginal progressivity increase: in contrast with inflation, the burden falls on high-earning agents (top of the productivity ladder) rather than on high-wealth agents (whose wealth need not coincide with their productivity). Panel (c) aggregates each instrument's incidence across the population: it reports the share of the total burden borne by each productivity type, weighted by the population share $S(y_0)$.

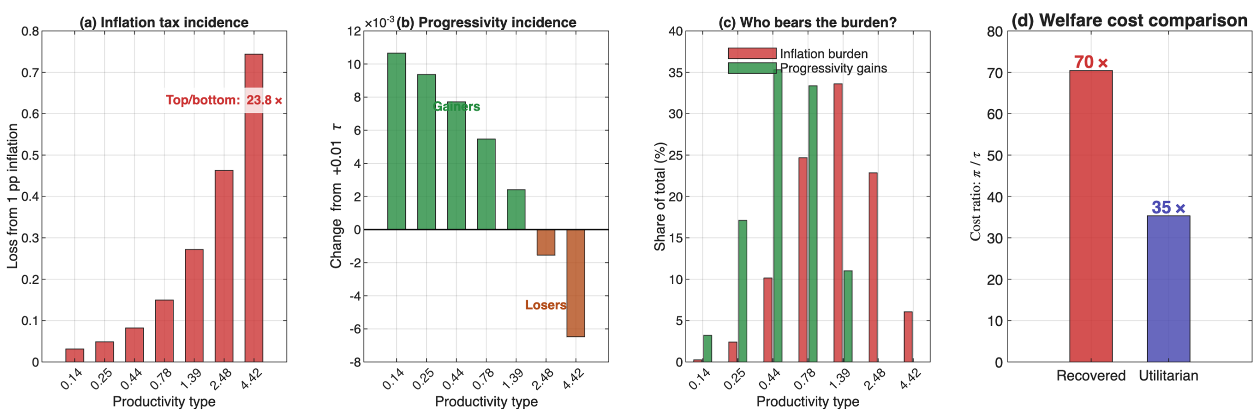

Figure 7: (a) Inflation loss by type. (b) Progressivity gain/loss by type. (c) Population-weighted burden shares. (d) Welfare-weighted cost ratio. UK 2021.

Progressivity loads the burden cleanly on the top of the income distribution; inflation spreads it more broadly, with intermediate productivity types bearing a disproportionate share because they hold the most nominal wealth.

The headline cost ratio. Panel (d) combines the population-weighted incidence with the recovered welfare weights to compute the welfare cost per unit of redistribution. Under the recovered weights, inflation is at least 70× costlier than a progressivity change per unit of redistribution; under utilitarian weights the ratio drops to 35×, but inflation remains substantially more expensive.[8] This is the structural advantage of progressive taxation over inflation that drives the zero-inflation result.

7.2 The role of heterogeneity

Three complementary exercises confirm that the instrument hierarchy is driven by agent heterogeneity and vanishes in a representative-agent economy.

RA benchmark. Solving the same Ramsey problem in a representative-agent version of the model, the HA output response exceeds the RA response by a factor of approximately 8,300 (Table 18 in Appendix J). This ratio is large because the RA responses are near zero — the representative agent can absorb the spending shock through intertemporal substitution alone, so the RA planner barely moves any instrument. The economically meaningful comparison is the HA response magnitudes themselves: output deviates by 0.55%, progressivity by 19.4 percentage points, and debt by 8.3% of GDP. In the HA model, the spending shock tightens borrowing constraints for some agents, creating distributional costs that the planner actively offsets through the tax structure. In a representative-agent model the wealth and income distributions collapse to a single point, inflation and taxation are equivalent, and the instrument hierarchy disappears. The amplification decomposes into two pieces: removing progressivity from the planner's toolkit still produces responses 3,400× larger than RA — the distributional motive operating through the remaining aggregate instruments — and restoring progressivity multiplies the response by a further 2.4×, giving the headline 8,300× figure (3,400 × 2.4 ≈ 8,300). Figure 22 and the decomposition in Table 19 (Appendix J) provide the full comparison.

Ramsey without τ. A clean test of progressivity's role is to keep full Ramsey optimization but fix $\tau = \tau_{ss}$, letting the planner optimize over κ, τ^k, B, and Π. Table 19 reports the key result: *inflation remains at zero* even without τ. The planner compensates through κ and τ^k, and restoring τ front-loads the adjustment but does not change the inflation outcome. The zero-inflation result is driven by the full tax structure, not by progressivity alone.

[8] The 70× ratio is computed for the τ margin alone. Since the Ramsey planner adjusts both τ and κ simultaneously (Table 19 in Appendix J), the effective advantage of the full tax structure is at least as large — a conservative lower bound.

Joint distribution of welfare weights and wealth. The mechanism rests on the claim that inflation disproportionately hurts the agents the planner values most. Table 6 reports the joint distribution of recovered welfare weights $\omega(y_0)$, mean steady-state asset holdings $\bar{a}(y_0)$, and the welfare-weighted inflation cost $\omega \cdot \bar{a}$ across productivity types for UK 2021.

	Productivity type y_0						
	1 (low)	2	3	4	5	6	7 (high)
y_0	0.14	0.25	0.44	0.78	1.39	2.48	4.42
$\omega(y_0)$	0.63	0.55	0.60	0.82	1.39	2.92	7.64
$\bar{a}(y_0)$	0.02	0.14	0.63	1.71	2.59	1.90	0.53
$\omega \cdot \bar{a}$	0.01	0.08	0.38	1.40	3.60	5.55	4.05
$S(y_0)$ (%)	1.5	9.4	23.5	31.3	23.5	9.4	1.5

Table 6: Welfare weights, mean asset holdings, and inflation cost by productivity type, UK 2021 steady state.

The raw correlation between $\omega(y_0)$ and $\bar{a}(y_0)$ is near zero (-0.04): mean assets peak at type 5 ($\bar{a} = 2.59$) and decline for the highest types because the most productive agents are rare and cycle through the income process. Nevertheless, the welfare-weighted inflation cost $\omega \cdot \bar{a}$ is heavily concentrated at the top of the distribution: types 5–7, which comprise 34% of the population, bear 73% of the total welfare-weighted inflation cost. This concentration arises because the welfare weight ω rises steeply enough (from 0.63 to 7.64) to more than offset the decline in assets for the highest types. The planner therefore avoids inflation not because high-weight agents hold the most assets in levels, but because the product $\omega \cdot \bar{a}$ — which determines the welfare cost of a given inflation rate — is disproportionately large for the agents the planner values most.

These findings relate to Kaplan (2025), who shows that even fully-funded lump-sum stimulus is inflationary in HANK economies. Lump-sum transfers cannot target the distributional cost of a shock — they distribute revenue uniformly, missing the opportunity to exploit the heterogeneity in social marginal values that progressive taxation leverages. The present model overturns the Kaplan result because the progressive tax structure provides exactly this targeting. The constrained experiments below confirm this interpretation: when the tax structure is frozen, inflation re-emerges as in the Kaplan framework.

7.3 When fiscal tools are constrained: inflation re-emerges

The preceding subsections showed that the tax structure is cheaper than inflation and that Ramsey without τ still produces zero inflation. The definitive test is the converse: what happens when the *entire* tax structure is removed from the planner's control? The experiments below also speak to a practical concern, since governments rarely adjust progressivity at business-cycle frequency.

7.3.1 Fiscal feedback rules

I replace Ramsey-optimal fiscal conditions with simple feedback rules. Each tax instrument $\Xi_t \in \{\kappa_t, \tau_t^k, \tau_t\}$ evolves according to:

$$\Xi_t = \Xi + \phi_\Xi (B_{t-1} - B), \tag{48}$$

where ϕ_Ξ governs responsiveness to debt deviations from steady state. Figure 8 reports the impulse responses; Appendix H reports the corresponding results for productivity shocks.

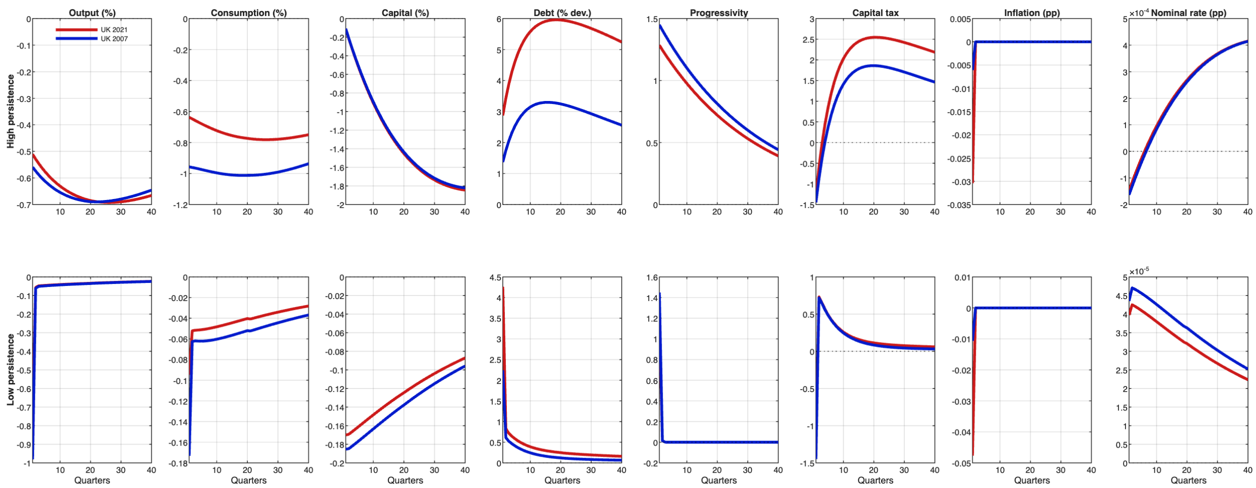

Figure 8: Government spending shock under fiscal rules (48). Top: high persistence; bottom: low persistence. UK 2021 (red), UK 2007 (blue).

Three features are common to both calibrations. First, the decline in capital is more pronounced than under Ramsey because the progressivity-and-capital-tax channel that would have cushioned the decline is shut by construction; consumption falls more sharply as well, as agents perceive a permanent income loss with no offsetting redistribution. Second, public debt becomes the primary shock absorber. The social planner lowers the nominal interest rate to alleviate debt-servicing costs and temporarily support aggregate demand, but the monetary easing also encourages additional borrowing, further increasing public debt in the short run. Debt peaks a few quarters after the initial shock, reflecting the cumulative impact of higher expenditures and lower revenues. Third, to contain this debt buildup and ensure long-run fiscal sustainability, the planner subsequently reduces inflation, thereby raising the real burden of outstanding debt and anchoring expectations. The resulting optimal inflation path departs from the zero-inflation benchmark of Section 6.2 and illustrates the dynamic interplay between monetary accommodation and debt stabilization.

UK 2021 (high debt) versus UK 2007 (low debt). The two calibrations differ in how fiscal capacity shapes this monetary–fiscal interaction. In UK 2021, the elevated debt-to-GDP ratio constrains fiscal space, forcing a more front-loaded reliance on monetary policy and inflation to restore equilibrium: the inflation deviation is several times larger than under UK 2007 (exact magnitudes in Table 16, Appendix I). In UK 2007 fiscal instruments could adjust more freely and the government possessed greater borrowing capacity to finance the temporary spending increase, so the burden of stabilization falls more on debt issuance and less on inflation. Consequently, the increase in public debt required to maintain budget equilibrium in UK 2021 is both larger and more persistent than in UK 2007, despite a comparable magnitude of fiscal shock.

Frozen fiscal tools shift the burden onto monetary policy. With fiscal parameters held fixed, the persistence of the shock plays a limited role in the qualitative trajectory of macroeconomic variables: the lack of fiscal flexibility forces monetary policy to absorb most of the adjustment regardless of whether the shock is short- or long-lived (compare Figures 3 and 8). The muted fiscal response magnifies the wealth effect, dampening labor supply and deepening the downturn in real activity, with the cost concentrated where debt is high. In a representative-agent model the same configuration would produce inflation as pure seigniorage; in the HA model, inflation reflects the *absence of distributional tools*.

7.3.2 Monetary rule with fixed fiscal instruments

The results obtained thus far underscore the pivotal role of prices as an adjustment mechanism from an optimal-policy perspective, particularly in settings where fiscal tools cannot be freely optimized but may adjust endogenously over the business cycle. This insight is especially relevant in the current macroeconomic environment, characterized by historically high public debt levels, where optimal fiscal policy would typically call for substantial tax adjustments to accommodate shocks; as discussed in Section 7.3, such adjustments may be politically or institutionally constrained.

The configuration can also be interpreted from the standpoint of a monetary authority — such as a central bank — that lacks direct control over fiscal instruments but retains discretion over inflation and the nominal interest rate. To analyse this case, I solve the Ramsey problem (26)–(40) after closing the first-order conditions for the fiscal instruments (equations (60), (61), and (62)). Fiscal parameters are held constant at their steady-state values, $(\tau_t^k, \tau_t, \kappa_t) = (\tau^k, \tau, \kappa)$, as reported in Table 2. The resulting monetary policy takes the form of a Taylor-type rule:

$$\tilde{R}_t^n = R^n + \phi_{\Pi,t}(\Pi_t - 1). \tag{49}$$

Figure 9 reports the resulting impulse responses. Two features are common to both calibrations. First, inflation departs sharply from the zero-inflation benchmark of Section 6.2: with fiscal instruments frozen, the central bank's Taylor rule becomes the sole stabilization device, and inflation absorbs the shock rather than the tax structure. Second, the monetary easing generates a sharp initial decline in public debt while capital and consumption decline as the inflation episode distorts intertemporal choices.

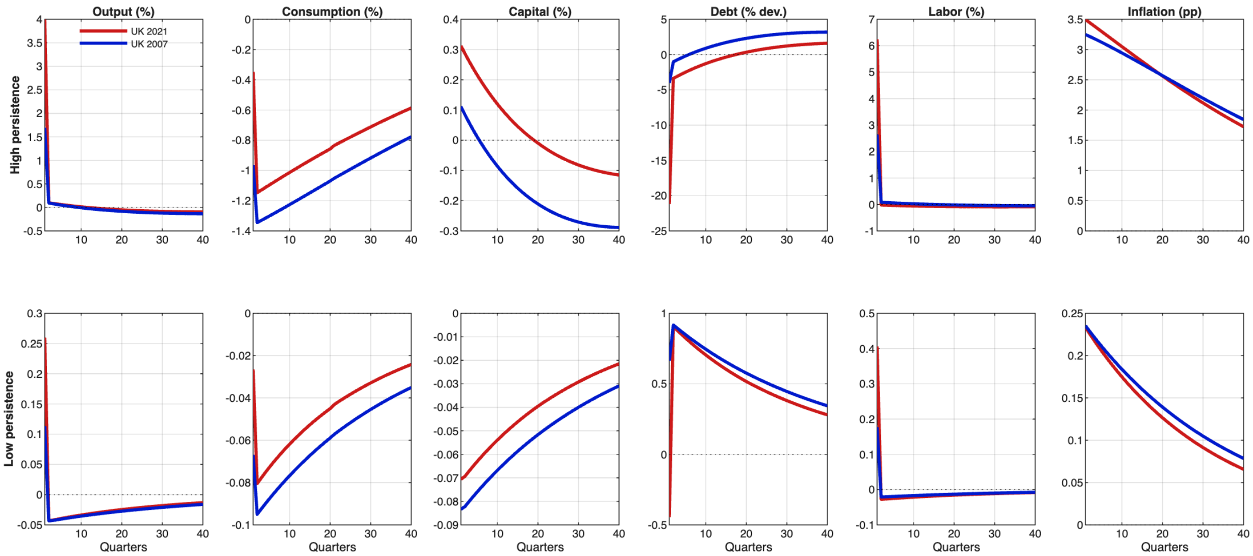

Figure 9: Government spending shock under monetary rule (49). Top: high persistence; bottom: low persistence. UK 2021 (red), UK 2007 (blue).

UK 2021 (high debt) versus UK 2007 (low debt). The two calibrations differ in real-side magnitudes rather than in the inflation response itself. Under high persistence, inflation rises by 3.5 pp in UK 2021 and 3.3 pp in UK 2007 (Table 17 in Appendix I) — two responses of similar magnitude, since the Taylor rule's reaction to the shock is structural and does not depend on the inherited debt level. Output, however, expands much more in UK 2021 (4.0%) than in UK 2007 (1.7%), and the deleveraging of public debt is several times larger (21% versus 4%). The high-debt economy benefits disproportionately because it carries more nominal liabilities to erode through inflation, and the resulting fiscal space translates into a stronger real expansion.

The role of persistence. The magnitude of the inflation response is shaped sharply by the persistence of the shock. Under low persistence the rise in inflation is modest — about 0.2 pp in both calibrations — and short-lived: a small temporary inflation deviation facilitates consumption smoothing, public debt rises only slightly before reverting to steady state, and the mild inflationary response reduces real debt-servicing costs without jeopardizing long-run price stability. Under high persistence the inflation deviation is an order of magnitude larger (about 3–3.5 pp), and the dynamics differ qualitatively. The initial price increase erodes the real value of outstanding nominal debt, allowing the government to absorb the shock without immediate tax adjustments; public debt declines on impact and only gradually returns to its steady-state path as inflation is reined in. This "inflate-then-refinance" pattern mirrors the optimal response of a high-debt economy facing limited fiscal flexibility: inflation provides short-term relief from debt overhang but is swiftly corrected to prevent credibility losses.

Implications. Monetary policy alone cannot replicate the distributional adjustment that the Ramsey planner achieves through progressive taxation: the Taylor rule restores nominal stability eventually, but along the adjustment path high-debt economies experience larger real disturbances and a substantial inflation episode that the full Ramsey allocation avoids.

Even partial flexibility displaces inflation. Reintroducing progressivity as a debt-stabilizing margin while keeping the level parameter frozen (and holding the capital-tax response unchanged) cuts the peak inflation deviation in UK 2021 from 0.030 pp to 0.022 pp — roughly a 29% reduction (Figure 10). Output, consumption, and capital paths are essentially unchanged: the substitution operates entirely between two debt-stabilizing tools. Figure 11 consolidates the peak inflation deviations across all three regimes and both calibrations, showing that any flexibility in the progressive tax structure displaces inflation as the residual debt-stabilization instrument, smoothly with the strength of the response.

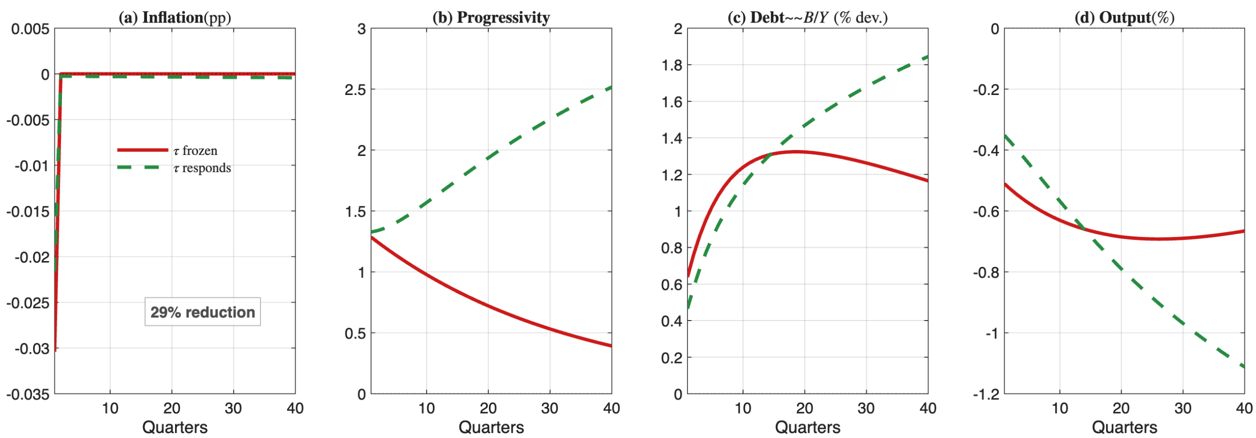

Figure 10: UK 2021 fiscal rules: baseline with progressivity frozen (solid red) vs. partially responsive progressivity (dashed green); the capital-tax response is held the same in both cases.

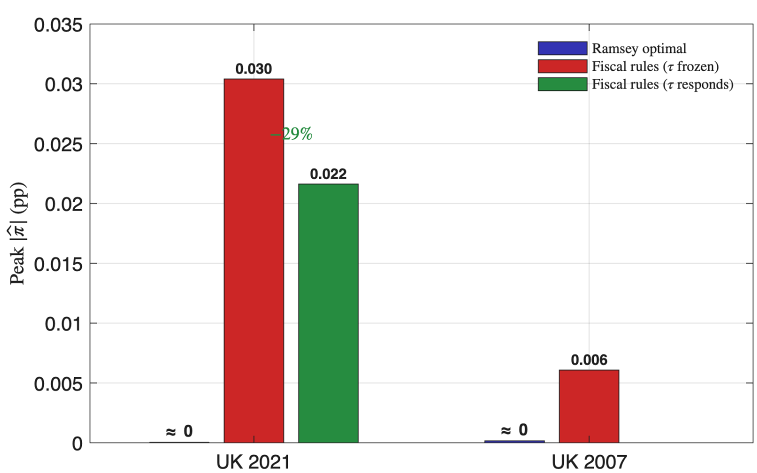

Figure 11: Peak $|\hat{\pi}|$ across policy regimes (data-driven from IRFs).

Synthesis across three debt levels and three regimes. The preceding analysis develops the paper's main finding from two complementary angles: under full Ramsey the planner keeps inflation at zero regardless of the inherited debt level, and when the tax structure is frozen inflation re-emerges in proportion to debt. Table 7 unifies these findings within a single 3×3 frame. The columns vary the inherited debt-to-GDP ratio across the three UK calibrations; the rows vary the fiscal toolkit available to the planner. Row 1 reports peak inflation under the baseline Ramsey allocation, in which the planner optimizes freely over the HSV progressivity, the level parameter, the capital tax, and public debt. Row 2 augments this set with a non-distortionary lump-sum transfer T_t. Row 3 instead removes the tax-progressivity margin entirely: progressivity and the level parameter are held fixed at their steady-state values, and only the capital tax responds to debt. All other features of the environment — the Rotemberg pricing friction, the recovered welfare weights, the persistent government-spending shock — are held constant across the nine cells, so any variation in the table can be attributed cleanly to the joint variation in inherited debt and instrument set.

	UK 1997 ($B/Y = 0.35$)	UK 2007 ($B/Y = 0.42$)	UK 2021 ($B/Y = 1.13$)
Full Ramsey	0	0	0
Ramsey + lump-sum transfer	0	0	0
Fiscal rules (τ frozen)	**0.005**	**0.006**	**0.030**

Table 7: Peak inflation ($|\hat{\pi}|$, percentage points): three regimes, three calibrations ($\rho_g = 0.97$).

The three rows answer increasingly sharp versions of the paper's central question. Row 1 shows that the optimizing planner does *not* tolerate more inflation at higher debt: peak inflation is essentially zero in all three calibrations, with no monotone relationship to the debt level. Row 2 shows that enriching the toolkit with a non-distortionary lump-sum transfer leaves inflation untouched — the transfer crowds out the distortionary instruments rather than inflation, because both are dominated by progressive taxation in the welfare ranking that determines instrument choice. Row 3 reverses the conclusion by removing the tax-progressivity margin: once progressivity and the level parameter are held fixed, inflation absorbs the residual financing burden of the shock and its magnitude scales sharply with debt, from 0.005 pp at UK 1997 to 0.030 pp at UK 2021. The fiscal-theory prediction is recovered exactly, but only when the conditions that generate it are imposed by hand.

The paper's headline is therefore conditional on fiscal rigidity. Section 8 confirms the conclusion is robust to alternative welfare specifications and open-economy channels, and Section 9 validates the mechanism against UK macroeconomic data over 1990–2024.

8 Alternative Welfare Functions and Open Economy

The zero-inflation result of Section 6 and the distributional mechanism of Section 7 are derived under the recovered welfare weights and a closed-economy assumption. Two natural questions arise: would a different welfare function change the planner's instrument choice? And does the closed-economy framework bias the comparison between inflation and progressive taxation? This section addresses each in turn.

8.1 Inequality aversion and the welfare function

The recovered welfare weights rationalize observed UK fiscal policy as optimal, but their profile — steeply increasing in productivity — may raise concerns about the generality of the zero-inflation result. I therefore ask whether a planner with stronger inequality aversion would still choose not to use inflation as a stabilization instrument. I consider three robustness exercises: Atkinson weights projected onto the set of weights satisfying the Ramsey first-order conditions, endogenous welfare weights that respond to consumption dispersion around the recovered steady state, and non-parametric minimum-norm weights defined as the closest-to-utilitarian weights satisfying the Ramsey FOCs.

The most direct alternative is the Atkinson SWF (Atkinson 1970), with $\omega_h \propto c_h^{-\eta}$, where $\eta > 0$ measures inequality aversion. Because the exogenous Atkinson weights do not generally satisfy the Ramsey first-order conditions at the steady state, I project them onto the constraint set by choosing the vector closest, in quadratic distance, to the original Atkinson profile among all vectors satisfying the planner's optimality conditions. Figure 12 compares the recovered weights with the corrected Atkinson profiles for $\eta = 0.5$, 1.0, and 2.0. The corrected weights remain increasing in productivity, but the profile becomes flatter as inequality aversion rises. Thus, stronger inequality aversion reduces the relative weight assigned to highly productive agents, while the Ramsey restrictions still require sufficient weight on those agents because they generate the tax base that finances redistribution.

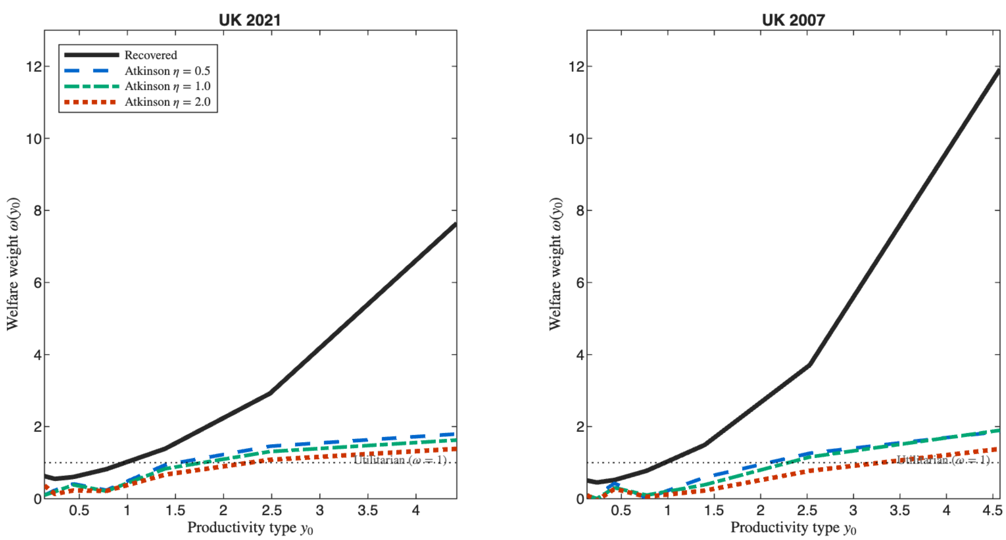

Figure 12: Recovered weights (solid) and FOC-corrected Atkinson weights at $\eta = 0.5$, 1.0, 2.0 (dashed).

The previous exercise changes the steady-state welfare weights. I therefore also consider an endogenous specification that preserves the recovered steady state while allowing the planner's marginal weights to respond to consumption dispersion:

$$\omega_h(c_h) = \omega_h^{ss} \left(\frac{c_h}{c_h^{ss}} \right)^{-\eta}.$$

Under this formulation, a fall in a household's consumption raises its welfare weight at the margin. The recovered steady-state weights serve as the anchor, while η governs the strength of the inequality-aversion channel away from steady state. Figure 13 reports the impulse responses for fixed weights ($\eta = 0$), low aversion ($\eta = 0.2$), and high aversion ($\eta = 2.0$). Inflation remains at zero in all cases. Stronger inequality aversion mainly dampens the real and fiscal responses: under $\eta = 2.0$, output, consumption, and the HSV tax instruments move by up to 70% less than under the baseline.

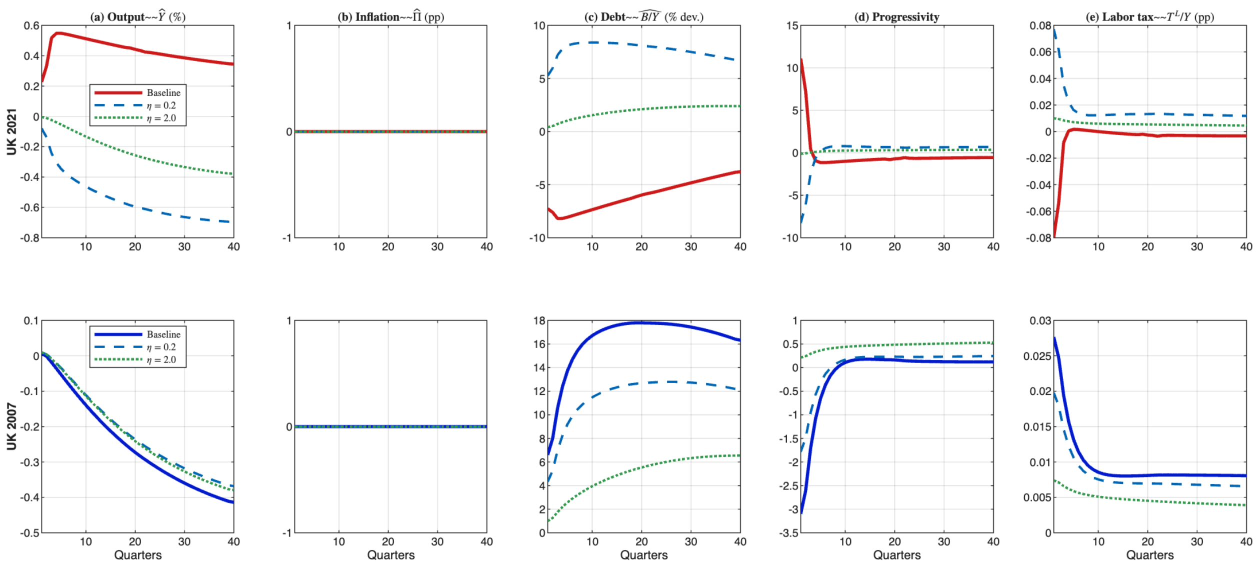

Figure 13: Endogenous inequality aversion, $\rho_g = 0.97$. Solid: baseline. Dashed: $\eta = 0.2$. Dash-dot: $\eta = 2.0$.

The main qualitative change concerns public debt. Under recovered weights, UK 2021 consolidates on impact because additional borrowing is costly at high debt levels. Under strong inequality aversion, debt instead rises on impact: the planner values the insurance role of public debt more highly and is willing to issue bonds to smooth the consumption of constrained households. Hence inequality aversion changes the use of debt, but not the use of inflation.

As a final benchmark, I compute non-parametric minimum-norm weights: the closest-to-utilitarian weights satisfying the Ramsey first-order conditions. These weights are increasing in productivity but much flatter than the recovered profile, with $\omega_{\max} \approx 2.0$ instead of 7.6. Figure 14 shows the resulting profile. Despite the much flatter weights, the impulse responses remain close to the baseline. The scale of the real and fiscal responses changes, but inflation remains inactive.

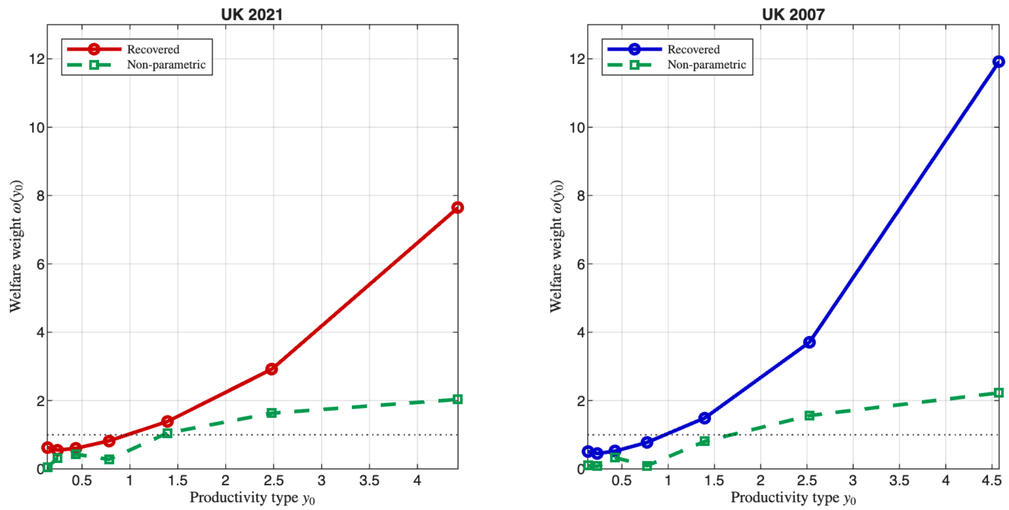

Figure 14: Recovered weights (solid) vs. non-parametric minimum-norm weights (dashed).

These exercises lead to the same conclusion. Making the planner more inequality-averse flattens the welfare-weight profile, dampens some real and fiscal responses, and can even reverse the sign of the debt response in the high-debt economy; but inflation remains at zero throughout, and the impulse responses themselves trace visually almost indistinguishable profiles across welfare specifications (Appendix G.1), with only the magnitudes scaling as the weights become flatter. The advantage of progressive taxation over inflation derives from the structure of the wealth and income distributions, not from the particular recovered welfare weights. The zero-inflation result is therefore robust to the choice of welfare function, provided the chosen weights satisfy the Ramsey first-order conditions and support a well-defined equilibrium: it is a structural feature of the economy with progressive taxation, not an artefact of the inverse-optimal weights.

8.2 The closed-economy assumption

The mechanism identified in Section 7 operates through the domestic wealth and income distributions: inflation taxes wealth, progressive taxation targets income, and the planner prefers the latter because it has lower welfare cost. A natural question is whether opening the economy to international capital flows would change this ranking.

The closed-economy assumption is in fact *conservative* for the paper's conclusion. With foreign-held debt ($\sim 30\%$ of UK gilts in 2021), a fraction of the inflation tax falls on non-resident bondholders — reducing the domestic welfare cost of inflation and making it *more* attractive, not less. Despite this, the closed-economy planner already chooses zero inflation; with foreign debt, the planner would choose zero inflation a fortiori. Accounting for foreign holdings, the cost ratio falls from $70\times$ to $49\times$ — still an order of magnitude.

To verify this quantitatively, I introduce a Laubach (2009)-style debt-elastic fiscal cost

($\phi_B = 0.001$, implying 25 bp per percentage point of debt). The cost enters the government's resource constraint only; the no-arbitrage condition, the planner's FOCs, and the recovered Pareto weights are unchanged. I refer to the baseline as the *closed economy* (CE) and to the version with the debt-elastic borrowing cost as the *small open economy* (SOE). Table 8 reports the result: all CE–SOE differences are below 1.3% under high persistence and below 1.0% under low persistence. Sensitivity across $\phi_B \in \{0.001, \dots, 0.05\}$ — up to 50× the Laubach benchmark — confirms differences below 0.3%. The debt-elastic channel is quantitatively negligible because it is dwarfed by the direct fiscal responses the planner uses.

Appendix K reports the full impulse responses behind these comparisons. Figures 23a and 23b show that the CE and SOE paths are visually indistinguishable for both UK 2021 and UK 2007, under both persistence regimes, and Figure 24 confirms that the invariance survives even at the 50× Laubach calibration and that, with a 50% foreign debt share, inflation would still be 49× costlier than progressivity.

	UK 2021			UK 2007		
	CE	SOE	Δ%	CE	SOE	Δ%
Output (%)	0.553	0.551	0.27	0.416	0.418	0.51
Debt (% dev.)	8.28	8.26	0.26	17.80	18.02	1.24

Table 8: CE vs. SOE peak IRF magnitudes ($\phi_B = 0.001$).

9 UK 1990–2024: Model versus Data

The preceding sections established the theoretical result and stress-tested it. I now bring the model to UK data over the past three decades and ask: would the Ramsey planner have used inflation when confronted with the actual sequence of UK fiscal shocks?

I construct an empirical G-shock series from ONS quarterly data, $u_t^{\text{emp}} = (G_t/Y_t) - \overline{G/Y}^{1995\text{–}2007}$, where the pre-crisis average $\overline{G/Y} = 0.398$ anchors the zero level. I back out innovations $\varepsilon_t^{\text{emp}} = u_t^{\text{emp}} - \rho_u u_{t-1}^{\text{emp}}$ with $\rho_u = 0.97$ and use linear superposition to compute model-implied paths by convolving the innovation series with the stored unit-shock IRFs. Figure 15 reports the results.

Figure 15 has four panels. Panel (a) plots the empirical UK G/Y shock series anchored at the pre-crisis average; the two largest expenditure spikes are the 2008–2009 fiscal response to the global financial crisis and the 2020–2021 Covid-19 fiscal expansion. The remaining three panels report the model-implied responses to this shock series under each calibration. Panel (b) shows the progressivity response: under the UK 2021 calibration the Ramsey planner would have raised progressivity to 1.6 in 2009Q4 and 1.8 in 2020Q2, reflecting the larger fiscal financing need at high debt; UK 1997 and UK 2007 show much smaller responses ($\sim$ 0.1–0.4). Panel (c) reports the model-implied inflation path: across all three calibrations and thirty years of UK fiscal shocks, inflation is virtually zero, confirming that the Ramsey planner does not need inflation as a fiscal financing tool even when confronted with the actual UK shock history. Panel (d) reports the model-implied B/Y deviation: positive for UK 1997 and UK 2007 (the planner accumulates debt countercyclically) and negative for UK 2021 (the planner runs down debt procyclically), broadly tracking the empirical pattern.

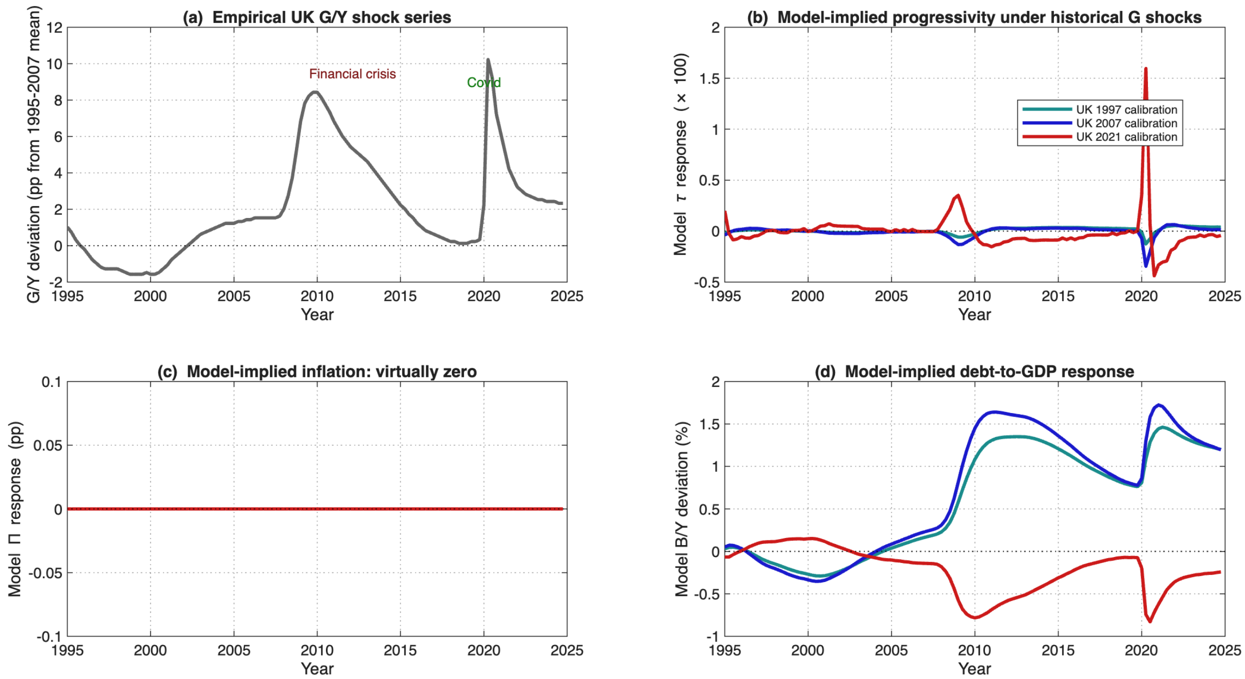

Figure 15: Model-implied paths under historical UK G shocks, 1995–2024.

UK 2021's elevated debt reflects both a gradual post-crisis buildup (from approximately 43% in 2007 to 83% in 2019) and the Covid jump layered on top (83% to 96%); the calibration captures the cumulative outcome, not the specific pathway, and the three-year design therefore brackets the relevant range of UK fiscal history. The historical exercise reinforces the paper's main conclusion: even under thirty years of actual UK fiscal shocks, the Ramsey planner does not use inflation.

A counterfactual for the 2022 UK cost-push inflation episode, constructed by combining the Ramsey-vs-rules damping factor with the observed CPI shock, is reported in Appendix L.

10 Concluding Remarks

Using a heterogeneous-agent Ramsey model calibrated to three UK fiscal regimes (1997, 2007, 2021), this paper has examined optimal fiscal–monetary coordination under different levels of public indebtedness.

Three findings emerge. First, under full Ramsey optimization with progressive taxation, the planner keeps inflation at zero regardless of the debt level. This holds across all welfare function specifications that produce a well-defined equilibrium and under open-economy considerations with a debt-elastic sovereign premium. Second, progressive taxation is the planner's cheapest redistribution instrument by a wide margin, and the heterogeneous-agent model therefore amplifies optimal fiscal responses by several orders of magnitude relative to a representative-agent benchmark. Third, when fiscal tools are constrained — either through mechanical feedback rules or institutional rigidity — inflation emerges as the residual financing tool, and its magnitude scales with the initial debt level. A counterfactual exercise for the 2022 UK inflation episode illustrates the quantitative implications: a Ramsey planner would have damped the observed 10.7% CPI peak to 2.0%.

The policy implication is direct: high public debt need not require higher inflation, provided the fiscal authority retains flexibility over the tax structure. The conventional wisdom from representative-agent models — that high debt mechanically pushes toward inflation — rests on the absence of distributional instruments. In heterogeneous-agent economies with progressive taxation, the planner has a superior tool that renders inflation unnecessary at any debt level the UK has experienced.

Several natural extensions remain. Solving the Ramsey problem with global or higher-order methods would test whether the zero-inflation result survives larger shocks and would quantify the welfare cost of off-equilibrium policy paths. Relaxing full commitment, or introducing political-economy frictions in tax-schedule adjustment, would constrain the planner's ability to front-load the fiscal response and likely push the optimum toward more inflation. A final limitation concerns the asset-market structure. The model follows the standard Heterogeneous-Agent literature of treating capital and public debt as perfect substitutes held through a mutual fund. This eliminates the household portfolio problem and implies a no-arbitrage condition equating the expected real return on government debt with the expected return on capital. As a result, the model abstracts from an equity premium and likely overstates the cost of public borrowing relative to an environment in which safe government debt carries a lower return than risky capital. This limitation works against the main result. A lower government borrowing rate would expand the planner's fiscal space and make inflation even less attractive as an adjustment margin. The zero-inflation result is therefore obtained under a conservative asset-pricing assumption. Moreover, the mechanism emphasized in the paper does not depend on the level of the risk-free rate, but on the welfare-weighted incidence of the two instruments: inflation taxes wealth, while progressive taxation taxes income. Introducing an explicit portfolio choice and an equity premium is therefore an important extension, but it is unlikely to overturn the central ranking between inflation and progressive taxation.

During the preparation of this work the author used ChatGPT (OpenAI) in order to assist with grammar and language editing. After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix

A Sequential representation and measure-theoretic notation

This subsection collects the measure-theoretic objects underlying the model of Section 3 that the truncation below approximates. The economy is populated by a unit mass of agents indexed by i on a probability space $([0,1], \mathcal{I}, \ell)$, where ℓ is the Lebesgue measure on $[0,1]$. At each date t, an agent's idiosyncratic productivity lies in a finite set $\mathcal{Y} \subset \mathbb{R}_+$, and beginning-of-period wealth lies in $[-\overline{a}, \infty)$.

A full idiosyncratic history at date t is the sequence $y_i^t = (y_{i,0}, y_{i,1}, \ldots, y_{i,t}) \in \mathcal{Y}^{t+1}$ of all past and current productivity realizations for agent i, and the corresponding aggregate-state history is $z^t = (z_0, z_1, \ldots, z_t)$. The cross-section of date-0 states $(a_{i,-1}, y_{i,0})$ is described by a probability measure $\mathbf{\Lambda}$ on $[-\overline{a}, \infty) \times \mathcal{Y}$, and the date-$t$ probability of a particular idiosyncratic history y^t conditional on the initial productivity draw is denoted $\theta_t(y^t)$ and follows from the transition kernel of the productivity process.

Equilibrium policy functions (c_t, l_t, a_t) are measurable maps from the product space $([-\overline{a}, \infty) \times \mathcal{Y}) \times \mathcal{Y}^{t+1} \times \mathbb{R}^{t+1}$ into the relevant choice sets, so any agent-level variable can be written as

$$X_{i,t} = X_t\Big((a_{i,-1}, y_{i,0}), y_i^t, z^t\Big).$$

Aggregate quantities are then defined as integrals over the cross-sectional distribution, with the equivalence

$$\int_i X_{i,t}\, \ell(di) \;=\; \sum_{y^t \in \mathcal{Y}^{t+1}} \sum_{y_0 \in \mathcal{Y}} \int_{a_{-1} \in [-\overline{a}, \infty)} \theta_t(y^t)\, X_t\Big((a_{-1}, y_0), y^t, z^t\Big)\, \mathbf{\Lambda}(\mathrm{d}a_{-1}, y_0).$$

The left-hand integral over agents and the right-hand integral over $(\mathbf{\Lambda}, \theta_t)$ coincide because the cross-section is generated by drawing date-0 states from $\mathbf{\Lambda}$ and idiosyncratic histories from θ_t.

The history space $\mathcal{Y}^{t+1}$ grows exponentially with t, so tracking the joint distribution at every date is computationally infeasible. The truncation procedure introduced below replaces full histories by their last N realizations and assigns every agent sharing the same truncated history the same allocation, producing a finite-dimensional state space that exactly aggregates the full economy in steady state and delivers accurate local dynamics under small aggregate shocks.

B Derivation of the Government Budget Constraint

Substituting (14) into (15) gives:

$$G_t + \frac{\tilde{R}^n_{t-1}}{\Pi_t} B_{t-1} = D_t + \tilde{w}_t L_t - \kappa_t \int_i (\tilde{w}_t y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \tau^k_t \tilde{r}^k_t K_{t-1} + \tau^k_t \left(\frac{\tilde{R}^n_{t-1}}{\Pi_t} - 1 \right) B_{t-1} + B_t,$$

where $L_t = \int_i y_{i,t} l_{i,t} \, \ell(di)$ and D_t denotes firms' aggregate real profits, given by (11). Using $Y_t \xi_t = (\tilde{r}^k_t + \delta) K_{t-1} + \tilde{w}_t L_t$, this yields:

$$D_t = \left(1 - \frac{\psi}{2}(\Pi_t - 1)^2\right) Y_t - \xi_t Y_t = \left(1 - \frac{\psi}{2}(\Pi_t - 1)^2\right) Y_t - \left[(\tilde{r}^k_t + \delta) K_{t-1} + \tilde{w}_t L_t\right].$$

Substituting this expression into the budget constraint yields:

$$\begin{aligned} G_t + \frac{\tilde{R}^n_{t-1}}{\Pi_t} B_{t-1} = {} & \left(1 - \frac{\psi}{2}(\Pi_t - 1)^2\right) Y_t - \left[(\tilde{r}^k_t + \delta) K_{t-1} + \tilde{w}_t L_t\right] + \tilde{w}_t L_t \\ & - \kappa_t \int_i (\tilde{w}_t y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \tau^k_t \tilde{r}^k_t K_{t-1} + \tau^k_t \left(\frac{\tilde{R}^n_{t-1}}{\Pi_t} - 1 \right) B_{t-1} + B_t. \end{aligned}$$

Next, define post-tax rates for capital, bonds, and labor income:

$$r_t = (1 - \tau^k_t) \tilde{r}_t, \quad r^k_t = (1 - \tau^k_t) \tilde{r}^k_t, \quad \frac{R^n_t}{\Pi_t} = (1 - \tau^k_t) \left(\frac{\tilde{R}^n_{t-1}}{\Pi_t} - 1 \right), \quad \text{and} \quad w_t = \kappa_t (\tilde{w}_t)^{1-\tau_t}.$$

Using these definitions, the government budget constraint becomes:

$$G_t + \frac{R^n_t}{\Pi_t} B_{t-1} + B_{t-1} + r^k_t K_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) = \left(1 - \frac{\psi}{2}(\Pi_t - 1)^2\right) Y_t - \delta K_{t-1} + B_t.$$

Finally, using (12):

$$(1 - \tau^k_t) \tilde{r}_t A_{t-1} = (1 - \tau^k_t) \tilde{r}^k_t K_{t-1} + (1 - \tau^k_t) \left(\frac{\tilde{R}^n_{t-1}}{\Pi_t} - 1 \right) B_{t-1},$$

which, together with the definitions in (16), implies $r_t A_{t-1} = r^k_t K_{t-1} + \frac{R^n_t}{\Pi_t} B_{t-1}$. The final government budget constraint can therefore be written as:

$$G_t + B_{t-1} + r_t A_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) = \left(1 - \frac{\psi}{2}(\Pi_t - 1)^2\right) Y_t - \delta K_{t-1} + B_t.$$

C The Lagrangian and First Order Conditions of the Ramsey Problem

The Ramsey problem (26)–(40) can be rewritten as:

$$
\begin{aligned}
\mathcal{L} = {} & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i \omega_{i,t} \left(u\left(c_{i,t}\right) - v\left(l_{i,t}\right)\right) \ell(di) \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i \left(\lambda_{i,c,t} - \left(1+r_t\right) \lambda_{i,c,t-1}\right) u'\left(c_{i,t}\right) \ell(di) \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i \lambda_{i,l,t} \left(v'(l_{i,t}) - (1-\tau_t) w_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t})\right) \ell(di) \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\gamma_t - \gamma_{t-1}\right) \Pi_t \left(\Pi_t - 1\right) Y_t M_t \\
& + \frac{\varepsilon - 1}{\psi} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \gamma_t \left(\xi_t - 1\right) Y_t M_t \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu_t \left(G_t + (1-\delta) B_{t-1} + (r_t + \delta) A_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) - \left(1 - \frac{\psi}{2}\left(\Pi_t - 1\right)^2\right) Y_t - B_t\right) \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Gamma_t \left((1-\tau_t^k) \tilde{r}_t^k K_{t-1} + (1-\tau_t^k) \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1\right) B_{t-1} - r_t A_{t-1}\right) \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Psi_t \left(\mathbb{E}_t [1 + \tilde{r}_{t+1}^k] - \mathbb{E}_t \left[\frac{\tilde{R}_t^n}{\Pi_{t+1}}\right]\right).
\end{aligned}
\tag{50}
$$

The instruments are $\Pi_t, \tilde{R}_t^n, \tilde{r}_t^k, a_{i,t}, B_t, l_{i,t}, w_t, r_t$, and τ_t.

FOC with respect to Π_t. Differentiating (50) with respect to Π_t yields:

$$
0 = -(\gamma_t - \gamma_{t-1})(2\Pi_t - 1) Y_t M_t - \mu_t \psi (\Pi_t - 1) Y_t - \Gamma_t (1 - \tau_t^k) B_{t-1} \frac{-\tilde{R}_{t-1}^n}{\Pi_t^2} + \Psi_{t-1} \beta^{-1} \frac{-\tilde{R}_{t-1}^n}{\Pi_t^2}.
$$

Rearranging gives:

$$
\underbrace{\mu_t \psi (\Pi_t - 1)}_{\text{Total cost due to inflation}} = \underbrace{(\gamma_{t-1} - \gamma_t)(2\Pi_t - 1) M_t}_{\text{Manipulation of real wages via the Phillips curve}} + \underbrace{\left(\Gamma_t (1 - \tau_t^k) B_{t-1} - \beta^{-1} \Psi_{t-1}\right) \frac{\tilde{R}_{t-1}^n}{\Pi_t^2 Y_t}}_{\text{Reduction of real interest payments on debt}}.
$$

Hence,

$$
0 = \mu_t \psi (\Pi_t - 1) + (\gamma_t - \gamma_{t-1})(2\Pi_t - 1) M_t - \left(\Gamma_t (1 - \tau_t^k) B_{t-1} - \beta^{-1} \Psi_{t-1}\right) \frac{\tilde{R}_{t-1}^n}{\Pi_t^2 Y_t}. \tag{51}
$$

FOC with respect to $\tilde{R}^n_t$. Differentiating (50) with respect to $\tilde{R}^n_t$ yields:

$$0 = -\beta\Gamma_{t+1}(1-\tau^k_{t+1})\frac{B_t}{\Pi_{t+1}} + \Psi_t\frac{1}{\Pi_{t+1}}. \tag{52}$$

FOC with respect to $\tilde{r}^k_t$. Differentiating (50) with respect to $\tilde{r}^k_t$ yields:

$$-\Gamma_t(1-\tau^k_t)K_{t-1} + \left(\frac{\varepsilon-1}{\psi}\right)\gamma_t\frac{1}{\alpha}K_{t-1}M_t - \beta^{-1}\Psi_{t-1} = 0.$$

Thus,

$$\beta^{-1}\Psi_{t-1} = \left(\left(\frac{\varepsilon-1}{\psi}\right)\gamma_t\frac{1}{\alpha}M_t - \Gamma_t(1-\tau^k_t)\right)K_{t-1}. \tag{53}$$

FOC with respect to $a_{i,t}$. Differentiating (50) with respect to $a_{i,t}$ yields:

$$\begin{aligned}
0 &= \beta^t\int_j \omega_{j,t}u'(c_{j,t})\frac{\partial c_{j,t}}{\partial a_{i,t}}\ell(dj) \\
&- \beta^t\int_j (\lambda_{j,c,t} - (1+r_t)\lambda_{j,c,t-1})\,u''(c_{j,t})\frac{\partial c_{j,t}}{\partial a_{i,t}}\ell(dj) \\
&+ \beta^t(1-\tau_t)w_t\int_j \lambda_{j,l,t}(y_{j,t})^{1-\tau_t}(l_{j,t})^{-\tau_t}u''(c_{j,t})\frac{\partial c_{j,t}}{\partial a_{i,t}}\ell(dj) \\
&- \beta^t\left((\gamma_t-\gamma_{t-1})\Pi_t(\Pi_t-1)Y_t - \left(\frac{\varepsilon-1}{\psi}\right)\gamma_t(\xi_t-1)Y_t\right)\int_j u''(c_{j,t})\frac{\partial c_{j,t}}{\partial a_{i,t}}\ell(dj) \\
&+ \beta^t\mu_t\left(\left(1-\frac{\psi}{2}(\Pi_t-1)^2\right)\frac{\partial Y_t}{a_{i,t}} - (r_t+\delta)\frac{\partial A_{t-1}}{\partial a_{i,t}}\right) \\
&+ \beta^{t+1}\mathbb{E}_t\left[\int_j \omega_{j,t+1}u'(c_{j,t+1})\frac{\partial c_{j,t+1}}{\partial a_{i,t}}\ell(dj)\right] \\
&- \beta^{t+1}\mathbb{E}_t\left[\int_j (\lambda_{j,c,t+1} - (1+r_{t+1})\lambda_{j,c,t})\,u''(c_{j,t+1})\frac{\partial c_{j,t+1}}{\partial a_{i,t}}\ell(dj)\right] \\
&+ \beta^{t+1}(1-\tau_{t+1})w_{t+1}\mathbb{E}_t\left[\int_j \lambda_{j,l,t+1}(y_{j,t+1})^{1-\tau_{t+1}}(l_{j,t+1})^{-\tau_{t+1}}u''(c_{j,t+1})\frac{\partial c_{j,t+1}}{\partial a_{i,t}}\ell(dj)\right] \\
&- \beta^{t+1}\mathbb{E}_t\left[\left((\gamma_{t+1}-\gamma_t)\Pi_{t+1}(\Pi_{t+1}-1)Y_{t+1} - \left(\frac{\varepsilon-1}{\psi}\right)\gamma_{t+1}(\xi_{t+1}-1)Y_{t+1}\right)\int_j u''(c_{j,t+1})\frac{\partial c_{j,t+1}}{\partial a_{i,t+1}}\ell(dj)\right] \\
&+ \beta^{t+1}\mathbb{E}_t\left[\mu_{t+1}\left(\left(1-\frac{\psi}{2}(\Pi_{t+1}-1)^2\right)\frac{\partial Y_{t+1}}{a_{i,t}} - (r_{t+1}+\delta)\frac{\partial A_t}{\partial a_{i,t}}\right)\right] \\
&- \beta^{t+1}\mathbb{E}_t\left(\left((\gamma_{t+1}-\gamma_t)\Pi_{t+1}(\Pi_{t+1}-1) - \left(\frac{\varepsilon-1}{\psi}\right)\gamma_{t+1}(\xi_{t+1}-1)\right)M_{t+1}\frac{\partial Y_{t+1}}{\partial a_{i,t}}\right) \\
&+ \beta^{t+1}\mathbb{E}_t[\Gamma_{t+1}r_{t+1}] - \beta^{t+1}\mathbb{E}_t\left[\Gamma_{t+1}\left((1-\tau^k_{t+1})\tilde{r}^k_{t+1}\right)\right]
\end{aligned}$$

$$+\underbrace{\left(\frac{\varepsilon-1}{\psi}\right)\beta^{t+1}\mathbb{E}_t\left[\gamma_{t+1}Y_{t+1}M_{t+1}\frac{\tilde{r}^k_{t+1}+\delta}{\alpha}\frac{(1-\alpha)}{Y_{t+1}}\right]}_{\left(\frac{\varepsilon-1}{\psi}\right)\beta^{t+1}\mathbb{E}_t\left[\gamma_{t+1}Y_{t+1}M_{t+1}\xi_{t+1}(1-\alpha)\frac{1}{K_t}\right]}.$$

Observe that:

$$\left(\frac{\varepsilon-1}{\psi}\right)\beta^{t+1}\mathbb{E}_t\left[\gamma_{t+1}(\xi_{t+1}-1)M_{t+1}\alpha K_t^{\alpha-1}L_{t+1}^{1-\alpha}\frac{K_t}{K_t}\right]+\left(\frac{\varepsilon-1}{\psi}\right)\beta^{t+1}\mathbb{E}_t\left[\gamma_{t+1}Y_{t+1}M_{t+1}\xi_{t+1}(1-\alpha)\frac{1}{K_t}\right]$$
$$=\left(\frac{\varepsilon-1}{\psi}\right)\beta^{t+1}\mathbb{E}_t\left[\gamma_{t+1}(\xi_{t+1}-\alpha)M_{t+1}\frac{Y_{t+1}}{K_t}\right].$$

Also note that $K_{t-1}=A_{t-1}-B_{t-1}$ is used to simplify the FOCs. The last term above arises from $\xi_t=\left(\frac{\tilde{r}^k_t+\delta}{\alpha}\right)\frac{K_{t-1}}{Y_t}$, so $\frac{\partial\xi_{t+1}}{\partial a_{i,t}}=(1-\alpha)\frac{\tilde{r}^k_{t+1}+\delta}{\alpha}\frac{1}{Y_{t+1}}$.

Define:

$$\psi_{i,t}=\omega_{i,t}u'(c_{i,t})-\Bigg[\lambda_{i,c,t}-(1+r_t)\lambda_{i,c,t-1}-\lambda_{i,l,t}(1-\tau_t)w_t(y_{i,t})^{1-\tau_t}(l_{i,t})^{-\tau_t}$$
$$+\left((\gamma_t-\gamma_{t-1})\Pi_t(\Pi_t-1)-\left(\frac{\varepsilon-1}{\psi}\right)\gamma_t(\xi_t-1)\right)Y_t\Bigg]u''(c_{i,t}). \tag{54}$$

Using (28) and (40), one obtains:

$$\frac{\partial c_{j,t}}{\partial a_{i,t}}=-1_{i=j},\qquad \frac{\partial A_t}{\partial a_{i,t}}=1,$$
$$\frac{\partial c_{j,t+1}}{\partial a_{i,t}}=(1+r_{t+1})1_{i=j},\qquad \frac{\partial Y_{t+1}}{\partial a_{i,t}}=\alpha K_t^{\alpha-1}L_{t+1}^{1-\alpha}.$$

Hence,

$$\psi_{i,t}=\beta\mathbb{E}_t\left[(1+r_{t+1})\psi_{i,t+1}\right]+\beta\mathbb{E}_t\left[\mu_{t+1}\left(\left(1-\frac{\psi}{2}\left(\Pi_{t+1}-1\right)^2\right)\underbrace{\alpha K_t^{\alpha-1}L_{t+1}^{1-\alpha}}_{\left(\frac{\tilde{r}^k_{t+1}+\delta}{\xi_{t+1}}\right)}-r_{t+1}-\delta\right)\right]$$
$$+\left(\frac{\varepsilon-1}{\psi}\right)\beta\mathbb{E}_t\left[\gamma_{t+1}(\xi_{t+1}-\alpha)M_{t+1}\frac{Y_{t+1}}{K_t}\right]-\beta\mathbb{E}_t\left[\left(\gamma_{t+1}-\gamma_t\right)\Pi_{t+1}\left(\Pi_{t+1}-1\right)\alpha\frac{M_{t+1}Y_{t+1}}{K_t}\right]$$
$$-\beta\mathbb{E}_t\left[\Gamma_{t+1}(1-\tau^k_{t+1})\tilde{r}^k_{t+1}\right]+\beta\mathbb{E}_t\left[\Gamma_{t+1}r_{t+1}\right]. \tag{55}$$

FOC with respect to B_t. Before deriving the first-order condition with respect to B_t, substitute $K_{t-1} = A_{t-1} - B_{t-1}$. Differentiating (50) with respect to B_t yields:

$$0 = \mu_t - \beta \mathbb{E}_t \left[\mu_{t+1} \left(1 - \delta + \left(1 - \frac{\psi}{2} (\Pi_{t+1} - 1)^2 \right) \underbrace{\alpha K_t^{\alpha-1} L_{t+1}^{1-\alpha}}_{\left(\frac{\tilde{r}^k_{t+1}+\delta}{\xi_{t+1}}\right)} \right) \right]$$
$$- \left(\frac{\varepsilon - 1}{\psi}\right) \beta \mathbb{E}_t \left[\gamma_{t+1}(\xi_{t+1} - \alpha) M_{t+1} \frac{Y_{t+1}}{K_t} \right] + \beta \mathbb{E}_t \left[(\gamma_{t+1} - \gamma_t) \Pi_{t+1} (\Pi_{t+1} - 1) \alpha \frac{M_{t+1} Y_{t+1}}{K_t} \right]$$
$$- \beta \mathbb{E}_t \left[\Gamma_{t+1}(1 - \tau^k_{t+1}) \left(\frac{\tilde{R}^n_t}{\Pi_{t+1}} - 1 \right) \right] + \beta \mathbb{E}_t \left[\Gamma_{t+1}(1 - \tau^k_{t+1}) \tilde{r}^k_{t+1} \right].$$

Thus,

$$\begin{aligned} \mu_t = \beta \mathbb{E}_t &\left[\mu_{t+1} \left(1 - \delta + \left(1 - \frac{\psi}{2} (\Pi_{t+1} - 1)^2 \right) \left(\frac{\tilde{r}^k_{t+1} + \delta}{\xi_{t+1}} \right) \right) \right] \\ &+ \beta \mathbb{E}_t \left[\left(\left(\frac{\varepsilon - 1}{\psi} \right) \gamma_{t+1}(\xi_{t+1} - \alpha) - \alpha (\gamma_{t+1} - \gamma_t) \Pi_{t+1} (\Pi_{t+1} - 1) \right) \frac{M_{t+1} Y_{t+1}}{K_t} \right] \\ &+ \beta \mathbb{E}_t \left[\Gamma_{t+1}(1 - \tau^k_{t+1}) \left(\frac{\tilde{R}^n_t}{\Pi_{t+1}} - 1 - \tilde{r}^k_{t+1} \right) \right]. \end{aligned} \tag{56}$$

Combining (55)–(56) yields:

$$\psi_{i,t} - \mu_t = \beta \mathbb{E}_t [\psi_{i,t}(1 + r_{t+1})] - \beta \mathbb{E}_t [\mu_{t+1}(1 - \delta)] + \beta \mathbb{E}_t [\mu_{t+1}(-r_{t+1} - \delta)] - \beta \mathbb{E}_t \left[\Gamma_{t+1}(1 - \tau^k_{t+1}) \tilde{r}^k_{t+1} \right]$$
$$+ \beta \mathbb{E}_t [\Gamma_{t+1} r_{t+1}] - \beta \mathbb{E}_t \left[\Gamma_{t+1}(1 - \tau^k_{t+1}) \left(\frac{\tilde{R}^n_t}{\Pi_{t+1}} - 1 \right) \right] + \beta \mathbb{E}_t \left[\Gamma_{t+1}(1 - \tau^k_{t+1}) \tilde{r}^k_{t+1} \right].$$

Defining:

$$\hat{\psi}_{i,t} = \psi_{i,t} - \mu_t, \tag{57}$$

one obtains:

$$\hat{\psi}_{i,t} = \beta \mathbb{E}_t \left[\hat{\psi}_{i,t}(1 + r_{t+1}) \right] + \beta \mathbb{E}_t \left[\Gamma_{t+1} \left(r_{t+1} - (1 - \tau^k_{t+1}) \left(\frac{\tilde{R}^n_t}{\Pi_{t+1}} - 1 \right) \right) \right]. \tag{58}$$

FOC with respect to $l_{i,t}$. Differentiating (50) with respect to $l_{i,t}$ yields:

$$0 = \int_j \psi_{j,t} \frac{\partial c_{j,t}}{\partial l_{i,t}} \ell(dj) - \omega_{i,t} v'(l_{i,t}) - \lambda_{i,l,t} v''(l_{i,t})$$
$$+ \lambda_{i,l,t}(1 - \tau_t)(-\tau_t) w_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t - 1} u'(c_{i,t}) - \mu_t \Big(w_t (1 - \tau_t)(y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t}$$

$$
- \left(1 - \frac{\psi}{2}\left(\Pi_t - 1\right)^2\right)\frac{\partial Y_t}{\partial l_{i,t}}\Bigg) - \left(\gamma_t - \gamma_{t-1}\right)\Pi_t\left(\Pi_t - 1\right)M_t\frac{\partial Y_t}{\partial l_{i,t}} + \left(\frac{\varepsilon - 1}{\psi}\right)\gamma_t(\xi_t - 1)M_t\frac{\partial Y_t}{\partial l_{i,t}}
$$

$$
\underbrace{+\left(\frac{\varepsilon-1}{\psi}\right)\gamma_t Y_t M_t \underbrace{\alpha\frac{\tilde{w}_t}{1-\alpha}\frac{1}{Y_t}y_{i,t}}_{\frac{\partial \xi_t}{\partial l_{i,t}}}}_{\left(\frac{\varepsilon-1}{\psi}\right)\gamma_t \alpha M_t \frac{\xi_t Y_t}{L_t}y_{i,t}}.
$$

Note that $\xi_t = \left(\frac{\tilde{w}_t}{1-\alpha}\right)\frac{L_t}{Y_t}$ and $L_t = \int_i y_{i,t}l_{i,t}\ell(di)$. Therefore, $\frac{\partial \xi_t}{\partial l_{i,t}} = \alpha\frac{\tilde{w}_t}{1-\alpha}\frac{1}{Y_t}y_{i,t}$. Using (28) and (40), respectively, $\frac{\partial c_{j,t}}{\partial l_{i,t}} = (1-\tau_t)w_t(y_{i,t})^{1-\tau_t}(l_{i,t})^{-\tau_t}1_{i=j}$ and $\frac{\partial Y_t}{\partial l_{i,t}} = (1-\alpha)K_{t-1}^{\alpha}L_t^{-\alpha}y_{i,t} = F_{L,t}y_{i,t} = \frac{\tilde{w}_t}{\xi_t}y_{i,t}$. So:

$$
\begin{aligned}
\omega_{i,t}v'(l_{i,t}) + \lambda_{i,l,t}v''(l_{i,t}) = {} & (1-\tau_t)w_t(y_{i,t})^{1-\tau_t}(l_{i,t})^{-\tau_t}\hat{\psi}_{i,t} \\
& + \lambda_{i,l,t}(1-\tau_t)(-\tau_t)w_t(y_{i,t})^{1-\tau_t}(l_{i,t})^{-\tau_t-1}u'(c_{i,t}) + \mu_t\left(1 - \frac{\psi}{2}\left(\Pi_t - 1\right)^2\right)F_{L,t}y_{i,t} \\
& - \left(\gamma_t - \gamma_{t-1}\right)\Pi_t\left(\Pi_t - 1\right)M_t\frac{(1-\alpha)Y_t y_{i,t}}{L_t} \\
& \underbrace{+\left(\frac{\varepsilon-1}{\psi}\right)\gamma_t(\xi_t - 1)M_t\frac{(1-\alpha)Y_t y_{i,t}}{L_t} + \left(\frac{\varepsilon-1}{\psi}\right)\gamma_t\xi_t M_t\frac{\alpha Y_t y_{i,t}}{L_t}}_{\left(\frac{\varepsilon-1}{\psi}\right)\gamma_t(\xi_t-(1-\alpha))M_t\frac{Y_t y_{i,t}}{L_t}}.
\end{aligned}
$$

where $\hat{\psi}_t^i = \psi_t^i - \mu_t$. After some manipulation:

$$
\begin{aligned}
\frac{\omega_{i,t}v'(l_{i,t}) + \lambda_{i,l,t}v''(l_{i,t})}{(1-\tau_t)w_t(y_{i,t})^{1-\tau_t}(l_{i,t})^{-\tau_t}} = {} & \hat{\psi}_{i,t} - \lambda_{i,l,t}\tau_t\frac{u'(c_{i,t})}{l_{i,t}} + \mu_t\frac{\left(1 - \frac{\psi}{2}\left(\Pi_t - 1\right)^2\right)F_{L,t}}{(1-\tau_t)w_t(y_{i,t})^{-\tau_t}(l_{i,t})^{-\tau_t}} \\
+ \left[\left(\frac{\varepsilon-1}{\psi}\right)\gamma_t(\xi_t - (1-\alpha)) - (1-\alpha)(\gamma_t - \gamma_{t-1})\Pi_t(\Pi_t - 1)\right] & \frac{Y_t M_t}{L_t(1-\tau_t)w_t(y_{i,t})^{-\tau_t}(l_{i,t})^{-\tau_t}}.
\end{aligned} \tag{59}
$$

FOC with respect to w_t. Differentiating (50) with respect to w_t yields:

$$
\begin{aligned}
0 = & \int_j \left(\psi_{j,t}\frac{\partial c_{j,t}}{\partial w_t} + \lambda_{j,l,t}(1-\tau_t)(y_{j,t})^{1-\tau_t}(l_{j,t})^{-\tau_t}u'(c_{j,t})\right)\ell(dj) \\
& - \mu_t\int_j (y_{j,t}l_{j,t})^{1-\tau_t}\ell(dj) + \left(\frac{\varepsilon-1}{\psi}\right)\gamma_t Y_t M_t\frac{\partial \xi_t}{\partial w_t}.
\end{aligned}
$$

Using (28), $\frac{\partial c_{j,t}}{\partial w_t} = (y_{j,t}l_{j,t})^{1-\tau_t}$. Since $w_t = \kappa(\tilde{w}_t)^{1-\tau_t}$, it follows that $\tilde{w}_t = \left(\frac{w_t}{\kappa}\right)^{\frac{1}{1-\tau_t}}$. Hence $\xi_t = \left(\frac{\left(\frac{w_t}{\kappa}\right)^{\frac{1}{1-\tau_t}}}{1-\alpha}\right)\frac{L_t}{Y_t}$, so:

$$
\frac{\partial \xi_t}{\partial w_t} = \frac{\frac{1}{1-\tau_t}\left(\frac{w_t}{\kappa}\right)^{\frac{\tau_t}{1-\tau_t}}\frac{1}{\kappa}\frac{L_t}{Y_t}}{1-\alpha}.
$$

Substituting back yields:

$$
\begin{aligned}
0 = &\int_j (y_{j,t} l_{j,t})^{1-\tau_t} \left(\hat{\psi}_{j,t} + \lambda_{j,l,t}(1-\tau_t) u'(c_{j,t})/l_{j,t} \right) \ell(dj) \\
&+ \left(\frac{\varepsilon - 1}{\psi} \right) \gamma_t M_t \frac{1}{1-\tau_t} \left(\frac{w_t}{\kappa} \right)^{\frac{\tau_t}{1-\tau_t}} \frac{1}{\kappa} \frac{L_t}{1-\alpha}.
\end{aligned}
\tag{60}
$$

FOC with respect to r_t. Differentiating (50) with respect to r_t yields:

$$
0 = \int_j \left(\psi_{j,t} \frac{\partial c_{j,t}}{\partial r_t} + \lambda_{j,c,t-1} u'(c_{j,t}) \right) \ell(dj) - \mu_t \int_j a_{j,t-1} \ell(dj) + \Gamma_t (B_{t-1} + K_{t-1}).
$$

Using (28), $\frac{\partial c_{j,t}}{\partial r_t} = a_{j,t-1}$. Hence,

$$
0 = \int_j \left(\hat{\psi}_{j,t} a_{j,t-1} + \lambda_{j,c,t-1} u'(c_{j,t}) \right) \ell(dj) + \Gamma_t (B_{t-1} + K_{t-1}).
\tag{61}
$$

FOC with respect to τ_t. Differentiating (50) with respect to τ_t yields:

$$
\begin{aligned}
0 = &\int_j \psi_{j,t} \frac{\partial c_{j,t}}{\partial \tau_t} \ell(dj) + w_t \int_j \lambda_{j,l,t} \frac{\partial}{\partial \tau_t} \left((1-\tau_t)(y_{j,t} l_{j,t})^{1-\tau_t} \right) (u'(c_{j,t})/l_{j,t}) \ell(dj) \\
&- \mu_t w_t \int_j \frac{\partial}{\partial \tau_t} \left((y_{j,t} l_{j,t})^{1-\tau_t} \right) \ell(dj) + \left(\frac{\varepsilon - 1}{\psi} \right) \gamma_t Y_t M_t \frac{\partial \xi_t}{\partial \tau_t}.
\end{aligned}
$$

Using (28), $\frac{\partial c_{j,t}}{\partial \tau_t} = w_t \frac{\partial}{\partial \tau_t}((y_{j,t} l_{j,t})^{1-\tau_t})$. Let $y = (y_{j,t} l_{j,t})^{1-\tau_t}$. Then $\ln y = (1-\tau_t) \ln(y_{j,t} l_{j,t})$ and $\frac{\partial y}{\partial \tau_t} = -\ln(y_{j,t} l_{j,t})(y_{j,t} l_{j,t})^{1-\tau_t}$, so:

$$
\frac{\partial}{\partial \tau_t} \left((y_{j,t} l_{j,t})^{1-\tau_t} \right) = -\ln(y_{j,t} l_{j,t})(y_{j,t} l_{j,t})^{1-\tau_t}.
$$

Similarly, with $y = \left(\frac{w_t}{\kappa} \right)^{\frac{1}{1-\tau_t}}$, $\frac{\partial y}{\partial \tau_t} = \frac{1}{(1-\tau_t)^2} \left(\frac{w_t}{\kappa} \right)^{\frac{1}{1-\tau_t}} \ln \left(\frac{w_t}{\kappa} \right)$, implying:

$$
\frac{\partial \xi_t}{\partial \tau_t} = \frac{1}{(1-\tau_t)^2} \left(\frac{w_t}{\kappa} \right)^{\frac{1}{1-\tau_t}} \ln \left(\frac{w_t}{\kappa} \right) \frac{L_t}{(1-\alpha) Y_t}.
$$

Therefore,

$$
\begin{aligned}
0 = &\int_j \hat{\psi}_{j,t} w_t \ln(y_{j,t} l_{j,t})(y_{j,t} l_{j,t})^{1-\tau_t} \ell(dj) + w_t \int_j \lambda_{j,l,t} (1-\tau_t) (u'(c_{j,t})/l_{j,t}) \ln(y_{j,t} l_{j,t})(y_{j,t} l_{j,t})^{1-\tau_t} \ell(dj) \\
&+ w_t \int_j \lambda_{j,l,t} (u'(c_{j,t})/l_{j,t})(y_{j,t} l_{j,t})^{1-\tau_t} \ell(dj) - \left(\frac{\varepsilon - 1}{\psi} \right) \gamma_t Y_t M_t \frac{1}{(1-\tau_t)^2} \left(\frac{w_t}{\kappa} \right)^{\frac{1}{1-\tau_t}} \ln \left(\frac{w_t}{\kappa} \right) \frac{L_t}{(1-\alpha) Y_t}.
\end{aligned}
$$

Equivalently,

$$0=\int_j (y_{j,t}l_{j,t})^{1-\tau_t}\left(\hat{\psi}_{j,t}+\lambda_{j,l,t}\left(1-\tau_t\right)\left(u'(c_{j,t})/l_{j,t}\right)\right)\ln(y_{j,t}l_{j,t})\ell(dj) \tag{62}$$
$$+\int_j \lambda_{j,l,t}(u'(c_{j,t})/l_{j,t})(y_{j,t}l_{j,t})^{1-\tau_t}\ell(dj)-\left(\frac{\varepsilon-1}{\psi}\right)\gamma_t M_t\frac{1}{(1-\tau_t)^2}\left(\frac{w_t^{\frac{\tau_t}{1-\tau_t}}}{\kappa^{\frac{1}{1-\tau_t}}}\right)\ln\left(\frac{w_t}{\kappa}\right)\frac{L_t}{(1-\alpha)}.$$

D Interpretation of the Ramsey Optimality Conditions

This appendix provides the economic interpretation of the planner's first-order conditions summarized in Section 4. The formal derivations appear in Appendix C.

Inflation condition. The central trade-off of monetary policy in the presence of price adjustment costs is:

$$\underbrace{\mu_t\psi(\Pi_t-1)}_{\text{cost of price dispersion}}=\underbrace{(\gamma_{t-1}-\gamma_t)(2\Pi_t-1)M_t}_{\text{real-wage manipulation}}+\underbrace{\left(\Gamma_t(1-\tau_t^k)B_{t-1}-\beta^{-1}\Psi_{t-1}\right)\frac{\tilde{R}_{t-1}^n}{\Pi_t^2Y_t}}_{\text{reduction of real debt payments}}, \tag{63}$$

which equates the marginal cost of inflation to two benefits: the ability to reallocate resources through nominal rigidities, and the fiscal advantage of eroding the real value of nominal debt. In the limiting case of flexible prices ($\psi\to 0$) and no nominal debt ($B_t=0$), inflation ceases to have any real allocative role.

Nominal-real return link. Internal policy consistency requires:

$$0=-\beta\Gamma_{t+1}(1-\tau_{t+1}^k)\frac{B_t}{\Pi_{t+1}}+\Psi_t\frac{1}{\Pi_{t+1}},\qquad \beta^{-1}\Psi_{t-1}=\left[\left(\frac{\varepsilon-1}{\psi}\right)\frac{\gamma_t}{\alpha}M_t-\Gamma_t(1-\tau_t^k)\right]K_{t-1}. \tag{64}$$

These conditions link the shadow values of bond and capital returns, ensuring that expected real yields are consistent across nominal assets.

Individual Euler equation. The shadow value $\psi_{i,t}$ associated with the individual budget constraint evolves as:

$$\psi_{i,t}=\beta\mathbb{E}_t\left[(1+r_{t+1})\psi_{i,t+1}\right]+\beta\mathbb{E}_t\left[\mu_{t+1}\left(\left(1-\frac{\psi}{2}(\Pi_{t+1}-1)^2\right)\frac{\tilde{r}_{t+1}^k+\delta}{\xi_{t+1}}-r_{t+1}-\delta\right)\right] \tag{65}$$
$$+\left(\frac{\varepsilon-1}{\psi}\right)\beta\mathbb{E}_t\left[\gamma_{t+1}(\xi_{t+1}-\alpha)M_{t+1}\frac{Y_{t+1}}{K_t}\right]-\beta\mathbb{E}_t\left[\left(\gamma_{t+1}-\gamma_t\right)\Pi_{t+1}(\Pi_{t+1}-1)\alpha\frac{M_{t+1}Y_{t+1}}{K_t}\right]$$
$$-\beta\mathbb{E}_t\left[\Gamma_{t+1}(1-\tau_{t+1}^k)\tilde{r}_{t+1}^k\right]+\beta\mathbb{E}_t\left[\Gamma_{t+1}r_{t+1}\right].$$

Defining $\hat{\psi}_{i,t} = \psi_{i,t} - \mu_t$ yields:

$$\hat{\psi}_{i,t} = \beta \mathbb{E}_t \left[\hat{\psi}_{i,t}(1 + r_{t+1}) \right] + \beta \mathbb{E}_t \left[\Gamma_{t+1} \left(r_{t+1} - (1 - \tau^k_{t+1}) \left(\frac{\tilde{R}^n_t}{\Pi_{t+1}} - 1 \right) \right) \right], \tag{66}$$

which highlights how agents with higher $\hat{\psi}_{i,t}$ — those facing tighter constraints or higher social weight — are prioritized in the planner's intertemporal redistribution.

Public debt. The FOC with respect to B_t,

$$\begin{aligned} \mu_t = \beta \mathbb{E}_t & \left[\mu_{t+1} \left(1 - \delta + \left(1 - \frac{\psi}{2} (\Pi_{t+1} - 1)^2 \right) \frac{\tilde{r}^k_{t+1} + \delta}{\xi_{t+1}} \right) \right] \\ & + \beta \mathbb{E}_t \left[\left(\left(\frac{\varepsilon - 1}{\psi} \right) \gamma_{t+1} (\xi_{t+1} - \alpha) - \alpha (\gamma_{t+1} - \gamma_t) \Pi_{t+1} (\Pi_{t+1} - 1) \right) \frac{M_{t+1} Y_{t+1}}{K_t} \right] \\ & + \beta \mathbb{E}_t \left[\Gamma_{t+1} (1 - \tau^k_{t+1}) \left(\frac{\tilde{R}^n_t}{\Pi_{t+1}} - 1 - \tilde{r}^k_{t+1} \right) \right], \end{aligned} \tag{67}$$

governs optimal debt smoothing: when frictions vanish, it simplifies to $\mu_t = \beta(1+r_{t+1})\mathbb{E}_t[\mu_{t+1}]$.

Labor and progressivity. The labor wedge and the two aggregate conditions for w_t and τ_t are:

$$0 = \int_j (y_{j,t} l_{j,t})^{1-\tau_t} \left(\hat{\psi}_{j,t} + \lambda_{j,l,t} (1 - \tau_t) u'(c_{j,t}) / l_{j,t} \right) \ell(dj) + \left(\frac{\varepsilon - 1}{\psi} \right) \gamma_t M_t \frac{1}{1 - \tau_t} \left(\frac{w_t}{\kappa} \right)^{\frac{\tau_t}{1-\tau_t}} \frac{1}{\kappa} \frac{L_t}{1 - \alpha}, \tag{68}$$

$$\begin{aligned} 0 = & \int_j (y_{j,t} l_{j,t})^{1-\tau_t} \left(\hat{\psi}_{j,t} + \lambda_{j,l,t} \left(1 - \tau_t \right) \left(u'(c_{j,t}) / l_{j,t} \right) \right) \ln(y_{j,t} l_{j,t}) \ell(dj) \\ & + \int_j \lambda_{j,l,t} (u'(c_{j,t}) / l_{j,t}) (y_{j,t} l_{j,t})^{1-\tau_t} \ell(dj) - \left(\frac{\varepsilon - 1}{\psi} \right) \gamma_t M_t \frac{1}{(1 - \tau_t)^2} \left(\frac{w_t^{\frac{\tau_t}{1-\tau_t}}}{\kappa^{\frac{1}{1-\tau_t}}} \right) \ln \left(\frac{w_t}{\kappa} \right) \frac{L_t}{(1 - \alpha)}. \end{aligned} \tag{69}$$

These describe how wage-setting and tax progressivity propagate heterogeneity. The equilibrium return condition, $0 = \int_j \left(\hat{\psi}_{j,t} a_{j,t-1} + \lambda_{j,c,t-1} u'(c_{j,t}) \right) \ell(dj) + \Gamma_t (B_{t-1} + K_{t-1})$, ensures asset-pricing consistency.

E Truncation Method for the Ramsey Problem with Nominal Rigidities

This appendix develops a truncation framework for the Ramsey problem studied in the main text, which features nominal rigidities (Rotemberg adjustment), inflation Π_t, the New

Keynesian Phillips block via the auxiliary objects (γ_t, ξ_t, M_t), and fiscal/asset-pricing wedges. The goal is to construct a finite state-space (truncated) economy that (i) exactly aggregates the heterogeneous-agent model in steady state and (ii) delivers accurate local dynamics under small aggregate shocks, while preserving the planner's first-order conditions (FOCs) (51)–(62).

E.1 The truncated model

Truncated histories. The key step of the aggregation procedure consists in assigning the same wealth and allocation to all agents who share the same idiosyncratic history over the recent past. This recent past is characterized by a fixed number of periods, called the *truncation length* and denoted by N. The truncation length is a parameter of the model. The N-period history is referred to as a *truncated history*.

For an individual history $y^t = \{\ldots, y^t_{t-N}, y^t_{t-N+1}, \ldots, y^t_{t-1}, y^t_t\}$, the truncated version is the N-length vector:

$$y^N := \{y^t_{t-N}, y^t_{t-N+1}, \ldots, y^t_{t-1}, y^t_t\}.$$

Thus, the truncated history of an agent i whose idiosyncratic history is y^t can be represented as:

$$y^t = \left\{ \underbrace{\ldots, y^t_{t-N-2}, y^t_{t-N-1}, y^t_{t-N}}_{\zeta_{y^N}}, \underbrace{y^t_{t-N+1}, \ldots, y^t_{t-1}, y^t_t}_{=y^N} \right\},$$

where the parameter ζ_{y^N} captures the residual heterogeneity associated with the truncated history y^N. The variable y^t_{t-k} represents the idiosyncratic state k periods before date t. The procedure to compute the set of parameters $(\zeta_{y^N})_{y^N}$ will be detailed below. In what follows, the main elements needed to implement this aggregation process are outlined.

Population shares and aggregation of individual choices. First, consider the measure of agents who share the same truncated history y^N. An agent with history $\tilde{y}^N$ at time $t-1$ will have a different truncated history at time t depending on the realization of the idiosyncratic state $y_{i,t}$. The probability of transitioning from truncated history $\tilde{y}^N$ to y^N is denoted by $\Pi_{\tilde{y}^N y^N}$, with $\sum_{y^N \in \mathcal{Y}^N} \Pi_{\tilde{y}^N y^N} = 1$. It can be derived from the transition probabilities of the underlying idiosyncratic process as:

$$\Pi_{\tilde{y}^N y^N} = 1_{y^N \succeq \tilde{y}^N} \Pi_{\tilde{y}^N_0 y^N_0} \geq 0,$$

where $1_{y^N \succeq \tilde{y}^N}$ equals one if y^N is a possible continuation of $\tilde{y}^N$ and zero otherwise. Using these elements, the share of agents with truncated history y^N at time t, denoted by S_{t,y^N}, evolves according to:

$$S_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^N} S_{t-1,\tilde{y}^N} \Pi_{\tilde{y}^N y^N}, \tag{70}$$

where the initial shares $(S_{-1,y^N})_{y^N \in \mathcal{Y}^N}$ are given and satisfy $\sum_{y^N \in \mathcal{Y}^N} S_{-1,y^N} = 1$.

The aggregation of the model then assigns to each truncated history y^N the average of all individual choices—whether for consumption, savings, or labor supply—of the group of agents sharing that same truncated history. Consider a generic variable $X_t(y^t, s^t)$ and denote by X_{t,y^N} the average value of X assigned to the group with truncated history y^N. Formally:

$$X_{t,y^N} = \frac{1}{S_{t,y^N}} \sum_{y^t \in \mathcal{Y}^{t+1} | (y^t_{t-N+1}, \ldots, y^t_{t-1}, y^t_t) = y^N} X_t(y^t, s^t)\, \mu_t(y^t), \tag{71}$$

where $\mu_t(y^t)$ denotes the measure of agents with history y^t.

Equation (71) can be applied to key individual-level variables such as consumption, savings, labor supply, and the Lagrange multiplier on the borrowing constraint. This leads to the aggregate quantities c_{t,y^N}, $\tilde{a}_{t,y^N}$, l_{t,y^N}, and ν_{t,y^N}, respectively.

When applying the aggregation procedure to beginning-of-period wealth, one must account for the fact that agents with truncated history y^N at time t may have originated from multiple truncated histories at $t-1$. Specifically, the beginning-of-period wealth for truncated history y^N, denoted $\tilde{\tilde{a}}_{t,y^N}$, is the weighted average of wealth among all groups $\tilde{y}^N$ at $t-1$ that can transition into y^N:

$$\tilde{\tilde{a}}_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^N} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N}\, \tilde{a}_{t-1,\tilde{y}^N}. \tag{72}$$

This expression ensures consistency in the aggregation of beginning-of-period asset holdings across truncated histories.

Nonlinear aggregation coefficients. Because the planner's FOCs involve nonlinear functions (notably $u(\cdot)$, $v(\cdot)$, $u'(\cdot)$, $v'(\cdot)$, powers of (yl), and cross-terms with $u''(\cdot)$), coefficients are introduced that map the sum of individual nonlinearities within each truncated group to functions of group-level averages. For each truncated history y^N, a set of steady-state coefficients is defined (ζ_{y^N}) satisfying the following relations:

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} u(c_{i,t}) = \zeta^{u,0}_{y^N}\, u(c_{t,y^N}), \qquad \sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} v(l_{i,t}) = \zeta^{v,0}_{y^N}\, v(l_{t,y^N}), \tag{73}$$

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} u'(c_{i,t}) = \zeta^{u,1}_{y^N}\, u'(c_{t,y^N}), \qquad \sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} v'(l_{i,t}) = \zeta^{v,1}_{y^N}\, v'(l_{t,y^N}), \tag{74}$$

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} (y_{i,t} l_{i,t})^{1-\tau_t} = \zeta^{y}_{y^N}\, (y_0^N l_{t,y^N})^{1-\tau_t}, \qquad \sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} \frac{u'(c_{i,t})}{l_{i,t}} = \zeta^{u,\ell}_{y^N}\, \frac{u'(c_{t,y^N})}{l_{t,y^N}}. \tag{75}$$

Here, y_0^N denotes the most recent idiosyncratic component in the truncated history y^N.

For log-product terms appearing in the progressivity condition, similarly define:

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} (y_{i,t} l_{i,t})^{1-\tau_t} \ln(y_{i,t} l_{i,t}) = \zeta^{y\ln}_{y^N}\, (y_0^N l_{t,y^N})^{1-\tau_t} \ln(y_0^N l_{t,y^N}). \tag{76}$$

To ensure the aggregation preserves the structure of the planner's intertemporal and intratemporal optimality conditions, coefficients are also defined that link the group-level Euler equations to their individual counterparts. For the consumption Euler equation, introduce $\xi^{u,E}_{y^N}$ such that:

$$\zeta^{u,E}_{y^N}\, u'(c_{t,y^N}) = \beta\, \mathbb{E}_t\left[R_{t+1} \sum_{\tilde{y}^N \in \mathcal{Y}^N} \Pi_{t+1,\, y^N \tilde{y}^N}\, \xi^{u,E}_{\tilde{y}^N}\, u'(c_{t+1,\tilde{y}^N})\right] + \nu_{t,y^N}. \tag{77}$$

This condition ensures that aggregate marginal utilities satisfy a properly defined truncated-history Euler equation.

Similarly, for the intratemporal labor-leisure tradeoff, impose:

$$\zeta^{v,1}_{y^N}\, v'(l_{t,y^N}) = (1-\tau_t)\, w_t\, \zeta^{y}_{y^N}\, (y_0^N l_{t,y^N})^{1-\tau_t}\, \zeta^{u,1}_{y^N}\, \frac{u'(c_{t,y^N})}{l_{t,y^N}}. \tag{78}$$

Finally, for the aggregation of expressions that involve second-order derivatives of utility, such as $u''(c_{i,t})$ multiplied by linear combinations of Lagrange multipliers, coefficients $\{\zeta^{u,2}_{y^N}, \zeta^{u\ell,2}_{y^N}\}$ are introduced such that group-level FOCs can be expressed in terms of $u''(c_{t,y^N})$ and $u'(c_{t,y^N})/l_{t,y^N}$, maintaining the same structural form as the individual FOCs of the planner.

Aggregate technology and government constraints. Aggregate production is $Y_t = K^{\alpha}_{t-1} L^{1-\alpha}_t$ with $L_t = \sum_{y^N} S_{t,y^N}\, y_0^N\, l_{t,y^N}$ and $K_{t-1} = \int_i a_{i,t-1}\, di - B_{t-1} = \sum_{y^N} S_{t,y^N} \tilde{\tilde{a}}_{t,y^N} - B_{t-1}$. The (post-tax) household budget identity aggregated by truncated group is:

$$c_{t,y^N} + \tilde{a}_{t,y^N} = R_t\, \tilde{\tilde{a}}_{t,y^N} + w_t\, \xi^{y}_{y^N} \left(y_0^N\, l_{t,y^N}\right)^{1-\tau_t}, \tag{79}$$

and government resource feasibility is the same as in the main text (with nominal terms and Rotemberg adjustment), using the aggregate Y_t, B_t, and G_t. Market clearing for assets and labor is:

$$K_{t-1} + B_{t-1} = \sum_{y^N} S_{t-1,y^N} \tilde{a}_{t-1,y^N}, \qquad L_t = \sum_{y^N} S_{t,y^N} y_0^N\, l_{t,y^N}. \tag{80}$$

Steady state and identification of the ξ's. In steady state the policy functions and the invariant wealth distribution are computed of the underlying Bewley economy, then aggregate within each y^N-group to obtain $(c_{y^N}, l_{y^N}, \tilde{a}_{y^N})$. The coefficients $\{\xi^{u,0}, \xi^{v,0}, \xi^{u,1}, \xi^{v,1}, \xi^{y}, \xi^{y\,\ln}, \xi^{u,\ell}, \xi^{u,2}, \xi^{u\ell,2}\}$ are pinned down by identities (73)–(76) (and their u'' analogs) evaluated at steady state, ensuring exact aggregation of the nonlinear terms that enter the planner's FOCs.

Aggregate shocks and fixed-coefficient approximation. The standard truncation assumption is adopted:

Assumption A. *(i) The aggregation coefficients $\{\xi_\cdot\}$ are held fixed at their steady-state values when solving dynamics with small aggregate shocks. (ii) The set of credit-constrained truncated histories is time-invariant.*

These assumptions ensure that the steady state of the truncated model coincides with that of the full model, and that truncated allocations converge to the full equilibrium as N increases.

E.2 Ramsey program in the truncated economy

Objective and constraints. Let ω_{y^N} denote the Pareto weight attached to truncated history y^N. The planner's objective in the truncated economy is:

$$W_0 = \mathbb{E}_0 \sum_{t\geq 0} \beta^t \sum_{y^N \in \mathcal{Y}^N} S_{t,y^N} \left(\omega_{y^N} \left[\xi^{u,0}_{y^N} u(c_{t,y^N}) - \xi^{v,0}_{y^N} v(l_{t,y^N}) \right] \right), \tag{81}$$

subject to: (i) the evolution of population shares (70); (ii) the definition of beginning-of-period wealth by truncated group (72); (iii) aggregate Euler equations (77) and (78); (iv) the group budget identities (79); (v) market clearing (80); (vi) the aggregate production function; (vii) the nominal Phillips block and related definitions from the main text (including M_t and ξ_t); and (viii) the government resource constraint (with Rotemberg term);

The planner chooses $\{\Pi_t, \tilde{R}^n_t, \tilde{r}^k_t, B_t, r_t, w_t, \tau_t\}$ and group allocations $\{c_{t,y^N}, l_{t,y^N}, \tilde{a}_{t,y^N}\}_{y^N}$.

Aggregated multipliers and social value of liquidity. Define the (truncated) social value of liquidity for group y^N as:

$$\begin{aligned}\psi_{t,y^N} := {}& \omega_{y^N}\, \xi^{u,0}_{y^N}\, u'\!\left(c_{t,y^N}\right) \\ & - \Big[\lambda_{c,t,y^N}\, \xi^{u,E}_{y^N} - (1+r_t)\, \tilde{\lambda}_{c,t,y^N}\, \xi^{u,E}_{y^N} - \lambda_{l,t,y^N}\, \xi^{y}_{y^N} (1-\tau_t)\, w_t\, (y^N_0)^{1-\tau_t}\, l^{-\tau_t}_{t,y^N}\, \xi^{u,1}_{y^N} \\ & \quad + \Big((\gamma_t - \gamma_{t-1})\, \Pi_t(\Pi_t - 1) - \tfrac{\varepsilon - 1}{\psi}\, \gamma_t(\xi_t - 1) \Big)\, Y_t \Big]\, u''\!\left(c_{t,y^N}\right),\end{aligned} \tag{82}$$

and the net value $\hat{\psi}_{t,y^N} := \psi_{t,y^N} - \mu_t$, where μ_t is the multiplier on the aggregate resource constraint.

FOCs of the truncated Ramsey problem. The FOCs in the truncated model mirror the structure of (51)–(62) with group-level aggregation coefficients. They are listed below.

Inflation Π_t.

$$0 = \mu_t\, \psi\, (\Pi_t - 1) + (\gamma_t - \gamma_{t-1})(2\Pi_t - 1)\, M_t - \Big(\Gamma_t (1 - \tau^k_t) B_{t-1} - \beta^{-1} \Psi_{t-1} \Big) \frac{\tilde{R}^n_{t-1}}{\Pi^2_t Y_t}. \tag{83}$$

Nominal bond return $\tilde{R}^n_t$.

$$0 = -\beta\,\Gamma_{t+1}(1-\tau^k_{t+1})\,\frac{B_t}{\Pi_{t+1}} + \Psi_t\,\frac{1}{\Pi_{t+1}}. \tag{84}$$

Capital return $\tilde{r}^k_t$.

$$\beta^{-1}\Psi_{t-1} = \Big(\Big(\tfrac{\varepsilon-1}{\psi}\Big)\gamma_t\,\tfrac{1}{\alpha}M_t - \Gamma_t(1-\tau^k_t)\Big)K_{t-1}. \tag{85}$$

Group assets $\tilde{a}_{t,y^N}$. Using (95), the net social value of liquidity satisfies:

$$\begin{aligned}\hat{\psi}_{t,y^N} &= \beta\,\mathbb{E}_t\Bigg[(1+r_{t+1})\sum_{\tilde{y}^N}\Pi_{y^N\tilde{y}^N}\,\hat{\psi}_{t+1,\tilde{y}^N}\Bigg] + \beta\,\mathbb{E}_t\Big[\mu_{t+1}\Big(\Big(1-\tfrac{\psi}{2}(\Pi_{t+1}-1)^2\Big)\tfrac{\tilde{r}^k_{t+1}+\delta}{\xi_{t+1}} - r_{t+1}-\delta\Big)\Big]\\ &+\Big(\tfrac{\varepsilon-1}{\psi}\Big)\beta\,\mathbb{E}_t\Big[\gamma_{t+1}(\xi_{t+1}-\alpha)M_{t+1}\tfrac{Y_{t+1}}{K_t}\Big] - \beta\,\mathbb{E}_t\Big[(\gamma_{t+1}-\gamma_t)\Pi_{t+1}(\Pi_{t+1}-1)\,\alpha\,\tfrac{M_{t+1}Y_{t+1}}{K_t}\Big]\\ &-\beta\,\mathbb{E}_t\Big[\Gamma_{t+1}(1-\tau^k_{t+1})\tilde{r}^k_{t+1}\Big] + \beta\,\mathbb{E}_t[\Gamma_{t+1}r_{t+1}]\,.\end{aligned} \tag{86}$$

Public debt B_t.

$$\begin{aligned}\mu_t &= \beta\,\mathbb{E}_t\Big[\mu_{t+1}\Big(1-\delta+\Big(1-\tfrac{\psi}{2}(\Pi_{t+1}-1)^2\Big)\tfrac{\tilde{r}^k_{t+1}+\delta}{\xi_{t+1}}\Big)\Big]\\ &+\beta\,\mathbb{E}_t\Big[\Big(\Big(\tfrac{\varepsilon-1}{\psi}\Big)\gamma_{t+1}(\xi_{t+1}-\alpha)-\alpha(\gamma_{t+1}-\gamma_t)\Pi_{t+1}(\Pi_{t+1}-1)\Big)\tfrac{M_{t+1}Y_{t+1}}{K_t}\Big]\\ &+\beta\,\mathbb{E}_t\Big[\Gamma_{t+1}(1-\tau^k_{t+1})\Big(\tfrac{\tilde{R}^n_t}{\Pi_{t+1}}-1-\tilde{r}^k_{t+1}\Big)\Big]\,.\end{aligned} \tag{87}$$

Labor l_{t,y^N}. With the coefficients in (74)–(75),

$$\begin{aligned}\frac{\omega_{y^N}\,\xi^{v,0}_{y^N}v'(l_{t,y^N})+\lambda_{l,t,y^N}\,\xi^{v,1}_{y^N}v''(l_{t,y^N})}{(1-\tau_t)w_t\,\xi^y_{y^N}(y^N_0)^{1-\tau_t}l^{-\tau_t}_{t,y^N}} &= \hat{\psi}_{t,y^N} - \lambda_{l,t,y^N}\,\tau_t\,\xi^{u,\ell}_{y^N}\,\frac{u'(c_{t,y^N})}{l_{t,y^N}}\\ &+\mu_t\,\frac{\Big(1-\tfrac{\psi}{2}(\Pi_t-1)^2\Big)F_{L,t}}{(1-\tau_t)w_t\,\xi^y_{y^N}(y^N_0)^{-\tau_t}l^{-\tau_t}_{t,y^N}}\\ &+\frac{\Big(\Big(\tfrac{\varepsilon-1}{\psi}\Big)\gamma_t(\xi_t-(1-\alpha))-(1-\alpha)(\gamma_t-\gamma_{t-1})\Pi_t(\Pi_t-1)\Big)Y_tM_t/L_t}{(1-\tau_t)w_t\,\xi^y_{y^N}(y^N_0)^{-\tau_t}l^{-\tau_t}_{t,y^N}}.\end{aligned} \tag{88}$$

Wage w_t.

$$0 = \sum_{y^N}S_{t,y^N}\,\xi^y_{y^N}(y^N_0l_{t,y^N})^{1-\tau_t}\Big(\hat{\psi}_{t,y^N}+\lambda_{l,t,y^N}(1-\tau_t)\,\xi^{u,\ell}_{y^N}\,\frac{u'(c_{t,y^N})}{l_{t,y^N}}\Big)$$

$$+\left(\tfrac{\varepsilon-1}{\psi}\right)\gamma_t M_t \frac{1}{1-\tau_t}\left(\frac{w_t}{\kappa}\right)^{\frac{\tau_t}{1-\tau_t}}\frac{1}{\kappa}\frac{L_t}{1-\alpha}. \tag{89}$$

Safe return r_t.

$$0=\sum_{y^N} S_{t,y^N}\left(\hat{\psi}_{t,y^N}\,\tilde{\tilde{a}}_{t,y^N}+\lambda_{c,t-1,y^N}\,\xi^{u,1}_{y^N}\,u'(c_{t,y^N})\right)+\Gamma_t(B_{t-1}+K_{t-1}). \tag{90}$$

Progressivity τ_t.

$$\begin{aligned}0=&\sum_{y^N} S_{t,y^N}\,\xi^{y}_{y^N}(y_0^N l_{t,y^N})^{1-\tau_t}\left(\hat{\psi}_{t,y^N}+\lambda_{l,t,y^N}(1-\tau_t)\,\xi^{u,\ell}_{y^N}\frac{u'(c_{t,y^N})}{l_{t,y^N}}\right)\ln(y_0^N l_{t,y^N})\\&+\sum_{y^N} S_{t,y^N}\,\lambda_{l,t,y^N}\,\xi^{y}_{y^N}(y_0^N l_{t,y^N})^{1-\tau_t}\,\xi^{u,\ell}_{y^N}\frac{u'(c_{t,y^N})}{l_{t,y^N}}\\&-\left(\tfrac{\varepsilon-1}{\psi}\right)\gamma_t M_t\frac{1}{(1-\tau_t)^2}\left(\frac{w_t^{\frac{\tau_t}{1-\tau_t}}}{\kappa^{\frac{1}{1-\tau_t}}}\right)\ln\!\left(\frac{w_t}{\kappa}\right)\frac{L_t}{1-\alpha}.\end{aligned} \tag{91}$$

It must furthermore hold that:

$$\tilde{\lambda}_{c,t,y^N}=\sum_{\tilde{y}^N\in\mathcal{Y}^{\infty N}}\frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}}\Pi_{t,\tilde{y}^N y^N}\lambda_{c,t-1,\tilde{y}^N}, \tag{92}$$

$$\tilde{a}_{t,y^N}\geq 0 \text{ and } \tilde{\tilde{a}}_{t,y^N}=\sum_{\tilde{y}^N\in\mathcal{Y}^{\infty N}}\frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}}\Pi_{t,\tilde{y}^N y^N}\tilde{a}_{t-1,\tilde{y}^N}. \tag{93}$$

Link with the primitive FOCs. Equations (83)–(93) reproduce the structure of the primitive Ramsey FOCs (51)–(62), with individual sums replaced by truncated-history aggregates and with the nonlinear aggregation coefficients $\{\xi_\cdot\}$ ensuring that (i) the truncated steady state coincides with the full model and (ii) local dynamics under small shocks are well-approximated.

E.3 Computation and identification of Pareto weights in the truncated model

Given a truncation length N and a discrete set $\mathcal{Y}$ of idiosyncratic states, the steady-state allocation of the underlying Bewley economy is computed and the coefficients are recovered $\{\xi_\cdot\}$. The truncated system (81)–(91) is then formed, evaluated at the observed (or calibrated) steady state, and the planner's steady-state FOCs are expressed as linear restrictions in the Pareto weights $\{\omega_{y^N}\}_{y^N\in\mathcal{Y}^N}$. Using the identification criterion described in Section 4.3, the set of Pareto weights is selected that (i) satisfy those restrictions and (ii) are closest to the utilitarian benchmark.

As discussed in the main text, refined truncation can be used to emphasize histories that occur with higher probability by increasing their effective memory length, which improves accuracy while keeping the number of aggregated states linear in the maximum truncation length. The quality of the approximation improves with N, and the steady-state replication is exact by construction under Assumption A.

E.4 Matrix expression

This section provides closed-form formulas for preference multipliers discussed in Section E.1. First, denote: $\circ$ as the Hadamard product, $\otimes$ the Kronecker product, and $\times$ the usual matrix product. For a vector V, $\mathrm{diag}(V)$ is the diagonal matrix with V on the diagonal.

Truncated histories are indexed by y^N, with N_{tot} total histories. Let $\boldsymbol{S} = (S_{y^N})_{y^N}$ be the steady-state shares and define the stacked vectors from the Bewley steady state:

$$\boldsymbol{a},\ \boldsymbol{c},\ \boldsymbol{l},\ \boldsymbol{\lambda_c},\ \boldsymbol{\lambda_l},\ u'(\boldsymbol{c}),\ u''(\boldsymbol{c}),\ v'(\boldsymbol{l}),\ v''(\boldsymbol{l}).$$

Let $\boldsymbol{y} = (y_0^N)_{y^N}$ collect current idiosyncratic productivities, $\boldsymbol{I}$ be the $N_{\text{tot}} \times N_{\text{tot}}$ identity, and $\boldsymbol{\Pi}$ the transition matrix over truncated histories. As before, $\boldsymbol{P}$ is diagonal with ones on unconstrained histories.

Also stack the NK objects at steady state:

$$\boldsymbol{Y},\ \boldsymbol{L},\ \boldsymbol{K},\ \boldsymbol{\Pi}^{(\text{infl})},\ \boldsymbol{\gamma},\ \boldsymbol{\xi},\ \boldsymbol{M},$$

and define the Rotemberg–markup composite (per history) that appears in the primitive FOCs:

$$\boldsymbol{\Theta} := \Big((\boldsymbol{\gamma} - \boldsymbol{\gamma}^-) \circ \boldsymbol{\Pi}^{(\text{infl})} \circ (\boldsymbol{\Pi}^{(\text{infl})} - \boldsymbol{1}) - \tfrac{\varepsilon - 1}{\psi} \boldsymbol{\gamma} \circ (\boldsymbol{\xi} - \boldsymbol{1}) \Big) \circ \boldsymbol{Y},$$

where $\boldsymbol{\gamma}^-$ is the (steady-state) one-step lag of $\boldsymbol{\gamma}$. (At a strict steady state one typically has $\boldsymbol{\gamma} = \boldsymbol{\gamma}^-$ and $\boldsymbol{\Pi}^{(\text{infl})} = \boldsymbol{1}$, but $\boldsymbol{\Theta}$ is kept explicit for generality.)

Stacked steady-state relations. The share recursion and aggregated individual budget read:

$$\boldsymbol{S} = \boldsymbol{\Pi} \boldsymbol{S}, \tag{94}$$

$$\boldsymbol{S} \circ \boldsymbol{c} + \boldsymbol{S} \circ \boldsymbol{a} = (1 + r)\, \boldsymbol{\Pi} \Big(\boldsymbol{S} \circ \boldsymbol{a} \Big) + w\ \boldsymbol{S} \circ \boldsymbol{\xi}^y \circ (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau}, \tag{95}$$

with credit constraints $(\boldsymbol{I} - \boldsymbol{P})\boldsymbol{a} = \boldsymbol{0}$. Here $\boldsymbol{\xi}^y$ is the (steady-state) nonlinearity coefficient from (75).

Nonlinear aggregation coefficients. Let $\boldsymbol{D_x}$ denote $\mathrm{diag}(\boldsymbol{x})$. The truncated Euler-aggregation coefficients are obtained exactly as in the Appendix E.1:

$$\boldsymbol{\xi}^{u,E} = \Big[\Big(\boldsymbol{I} - \beta(1 + r) \boldsymbol{\Pi}^\top \Big)\ \boldsymbol{D}_{u'(\boldsymbol{c})} \Big]^{-1}\ \boldsymbol{\nu}, \tag{96}$$

$$\boldsymbol{\xi}^{v,1} = (1-\tau)\, w\ (\boldsymbol{y}\circ\boldsymbol{l})^{1-\tau} \circ \boldsymbol{\xi}^{y} \circ \boldsymbol{\xi}^{u,1} \circ u'(\boldsymbol{c})\ ./\ (\boldsymbol{l}\circ v'(\boldsymbol{l})), \tag{97}$$

with $\boldsymbol{\xi}^{u,0}, \boldsymbol{\xi}^{u,1}, \boldsymbol{\xi}^{v,0}, \boldsymbol{\xi}^{y}$ computed from the definitions in (73)–(75) (and $\boldsymbol{\xi}^{y\,\ln}$ analogously for the progressivity FOC).

E.4.1 Matrix expressions for the FOCs

Define the sized (share-weighted) variables:

$$\overline{\boldsymbol{\lambda}}_l := \boldsymbol{S}\circ\boldsymbol{\lambda}_l, \quad \overline{\boldsymbol{\psi}} := \boldsymbol{S}\circ\widehat{\boldsymbol{\psi}}, \quad \overline{\boldsymbol{\Pi}} := \boldsymbol{S}\circ\boldsymbol{\Pi}^\top\circ(1/\boldsymbol{S}), \quad \overline{\boldsymbol{\varpi}} := \boldsymbol{S}\circ\boldsymbol{\omega}, \quad \overline{\boldsymbol{\lambda}}_c := \boldsymbol{S}\circ\boldsymbol{\lambda}_c,$$

and shortcuts $\tilde{\boldsymbol{\xi}}^{u,1} := \boldsymbol{\xi}^{u,1}./\boldsymbol{l}$, $\tilde{\boldsymbol{\xi}}^{v,1} := \boldsymbol{\xi}^{v,1}./\Big((1-\tau)w\,\boldsymbol{\xi}^{y}\circ\boldsymbol{y}^{1-\tau}\circ\boldsymbol{l}^{-\tau}\Big)$, $\tilde{\boldsymbol{\xi}}^{v,0} := \boldsymbol{\xi}^{v,0}./\Big((1-\tau)w\,\boldsymbol{\xi}^{y}\circ\boldsymbol{y}^{1-\tau}\circ\boldsymbol{l}^{-\tau}\Big)$. Note that $\boldsymbol{S}\circ\tilde{\boldsymbol{\lambda}}_c = \boldsymbol{\Pi}\,\overline{\boldsymbol{\lambda}}_c$.

The share-weighted *net* social value of liquidity (group level) solves:

$$\begin{aligned}\overline{\boldsymbol{\psi}} = \underbrace{\boldsymbol{D}_{\boldsymbol{\xi}^{u,0}\circ u'(\boldsymbol{c})}}_{\hat{\boldsymbol{M}}_0}\ \overline{\boldsymbol{\varpi}} - \underbrace{\boldsymbol{D}_{\boldsymbol{\xi}^{u,E}\circ u''(\boldsymbol{c})}}_{\hat{\boldsymbol{A}}_c}\Big(\overline{\boldsymbol{\lambda}}_c - (1+r)\,\boldsymbol{\Pi}\,\overline{\boldsymbol{\lambda}}_c\Big) \\ + \underbrace{(1-\tau)w\ \boldsymbol{D}_{\boldsymbol{\xi}^{y}\circ(\boldsymbol{y}\circ\boldsymbol{l})^{1-\tau}\circ\tilde{\boldsymbol{\xi}}^{u,1}\circ u''(\boldsymbol{c})}}_{\hat{\boldsymbol{A}}_\ell}\ \overline{\boldsymbol{\lambda}}_l\ +\ \underbrace{\boldsymbol{D}_{\boldsymbol{\Theta}\circ u''(\boldsymbol{c})}}_{\hat{\boldsymbol{A}}_{\mathrm{NK}}}\ \boldsymbol{S}\ -\ \mu\,\boldsymbol{S}.\end{aligned} \tag{98}$$

Unconstrained truncated histories satisfy the (grouped) implementability condition implied by (58). At the truncated level this yields an affine recursion:

$$\boldsymbol{P}\,\overline{\boldsymbol{\psi}} = \beta\,\boldsymbol{P}\Big((1+r)\,\overline{\boldsymbol{\Pi}}\,\overline{\boldsymbol{\psi}} + \ \boldsymbol{\Upsilon}\Big), \qquad (\boldsymbol{I}-\boldsymbol{P})\,\overline{\boldsymbol{\lambda}}_c = \boldsymbol{0}, \tag{99}$$

where the stacked vector $\boldsymbol{\Upsilon}$ captures the (known, steady-state) additive wedge driven by the Γ–Ψ block and nominal-bond return in (58).[9]

The wage, interest-rate, and progressivity FOCs become (stacked):

$$\Big(\boldsymbol{\xi}^{y}\circ(\boldsymbol{y}\circ\boldsymbol{l})^{1-\tau}\Big)^{\top}\overline{\boldsymbol{\psi}} = -(1-\tau)\Big(\boldsymbol{\xi}^{y}\circ(\boldsymbol{y}\circ\boldsymbol{l})^{1-\tau}\circ\tilde{\boldsymbol{\xi}}^{u,1}\circ u'(\boldsymbol{c})\Big)^{\top}\overline{\boldsymbol{\lambda}}_l, \tag{100}$$

$$\boldsymbol{a}^{\top}\overline{\boldsymbol{\psi}} = -\Big(\boldsymbol{\xi}^{u,E}\circ u'(\boldsymbol{c})\Big)^{\top}\boldsymbol{\Pi}\ \overline{\boldsymbol{\lambda}}_c, \tag{101}$$

$$\Big(\ln(\boldsymbol{y}\circ\boldsymbol{l})\circ\boldsymbol{\xi}^{y}\circ(\boldsymbol{y}\circ\boldsymbol{l})^{1-\tau}\Big)^{\top}\overline{\boldsymbol{\psi}} = -\Big((\boldsymbol{1}+(1-\tau)\ln(\boldsymbol{y}\circ\boldsymbol{l}))\circ\boldsymbol{\xi}^{y}\circ(\boldsymbol{y}\circ\boldsymbol{l})^{1-\tau}\circ\tilde{\boldsymbol{\xi}}^{u,1}\circ u'(\boldsymbol{c})\Big)^{\top}\overline{\boldsymbol{\lambda}}_l. \tag{102}$$

[9] Explicitly, $\boldsymbol{\Upsilon} := \mathbb{E}\big[\Gamma_{t+1}\big(r_{t+1} - (1-\tau^{k}_{t+1})(\tilde{R}^{n}_{t}/\Pi_{t+1} - 1)\big)\big]$ stacked by truncated history; at the steady state it is evaluated using the truncated transition $\boldsymbol{\Pi}$.

E.4.2 Solving the system

From the labor FOC block, solve for $\overline{\boldsymbol{\lambda}}_l$ as:

$$\boldsymbol{D}_{\tilde{\boldsymbol{\xi}}^{v,1}\circ v''(\boldsymbol{l})+\tau\,\tilde{\boldsymbol{\xi}}^{u,1}\circ u'(\boldsymbol{c})}\;\overline{\boldsymbol{\lambda}}_l=\overline{\boldsymbol{\psi}}-\boldsymbol{D}_{\tilde{\boldsymbol{\xi}}^{v,0}\circ v'(\boldsymbol{l})}\;\overline{\boldsymbol{\varpi}}+\mu\,\boldsymbol{\Phi},\qquad \overline{\boldsymbol{\lambda}}_l=\boldsymbol{M}_0\overline{\boldsymbol{\varpi}}+\boldsymbol{M}_1\overline{\boldsymbol{\psi}}+\mu\,\boldsymbol{V}_0,\tag{103}$$

where:

$$\boldsymbol{M}_1:=\Big(\boldsymbol{D}_{\tilde{\boldsymbol{\xi}}^{v,1}\circ v''(\boldsymbol{l})+\tau\,\tilde{\boldsymbol{\xi}}^{u,1}\circ u'(\boldsymbol{c})}\Big)^{-1},\quad \boldsymbol{M}_0:=-\boldsymbol{M}_1\,\boldsymbol{D}_{\tilde{\boldsymbol{\xi}}^{v,0}\circ v'(\boldsymbol{l})},$$

$$\boldsymbol{\Phi}:=F_L\,\boldsymbol{S}./\Big((1-\tau)w\,\boldsymbol{\xi}^y\circ\boldsymbol{y}^{-\tau}\circ\boldsymbol{l}^{-\tau}\Big),\qquad \boldsymbol{V}_0:=\boldsymbol{M}_1\,\boldsymbol{\Phi}.$$

Plug (103) into (98) to obtain:

$$\overline{\boldsymbol{\psi}}=\underbrace{\Big(\hat{\boldsymbol{M}}_0+\hat{\boldsymbol{A}}_\ell\boldsymbol{M}_0\Big)}_{\boldsymbol{N}_0}\;\overline{\boldsymbol{\varpi}}\;+\;\underbrace{\Big(-\hat{\boldsymbol{A}}_c+\hat{\boldsymbol{A}}_\ell\boldsymbol{M}_1\Big)}_{\boldsymbol{N}_1}\;\overline{\boldsymbol{\lambda}}_c\;+\;\underbrace{\Big(\hat{\boldsymbol{A}}_{\mathrm{NK}}\,\boldsymbol{S}-\mu\,\boldsymbol{S}+\hat{\boldsymbol{A}}_\ell\boldsymbol{V}_0\,\mu\Big)}_{\boldsymbol{n}_0+\mu\,\boldsymbol{n}_1}.\tag{104}$$

Use (99) to eliminate $\overline{\boldsymbol{\lambda}}_c$. Write $\boldsymbol{Q}:=\beta\,\boldsymbol{P}(1+r)\,\overline{\boldsymbol{\Pi}}$ and $\boldsymbol{q}:=\beta\,\boldsymbol{P}\,\boldsymbol{\Upsilon}$. Then:

$$\boldsymbol{P}\overline{\boldsymbol{\psi}}=\boldsymbol{Q}\,\overline{\boldsymbol{\psi}}+\boldsymbol{q},\qquad(\boldsymbol{I}-\boldsymbol{P})\overline{\boldsymbol{\lambda}}_c=\boldsymbol{0}.$$

Substituting (104) and solving for $\overline{\boldsymbol{\lambda}}_c$ gives:

$$\overline{\boldsymbol{\lambda}}_c=\boldsymbol{M}_5\,\overline{\boldsymbol{\varpi}}+\mu\,\boldsymbol{V}_2+\boldsymbol{V}_\Upsilon,\tag{105}$$

with:

$$\boldsymbol{M}_5:=-\Big((\boldsymbol{I}-\boldsymbol{P})+\boldsymbol{P}(\boldsymbol{I}-\boldsymbol{Q})\boldsymbol{N}_1\Big)^{-1}\boldsymbol{P}(\boldsymbol{I}-\boldsymbol{Q})\boldsymbol{N}_0,$$

$$\boldsymbol{V}_2:=-\Big((\boldsymbol{I}-\boldsymbol{P})+\boldsymbol{P}(\boldsymbol{I}-\boldsymbol{Q})\boldsymbol{N}_1\Big)^{-1}\boldsymbol{P}(\boldsymbol{I}-\boldsymbol{Q})\,\boldsymbol{n}_1,\quad \boldsymbol{V}_\Upsilon:=-\Big((\boldsymbol{I}-\boldsymbol{P})+\boldsymbol{P}(\boldsymbol{I}-\boldsymbol{Q})\boldsymbol{N}_1\Big)^{-1}\boldsymbol{q}.$$

Back-substitute (105) into (104):

$$\overline{\boldsymbol{\psi}}=\boldsymbol{M}_6\,\overline{\boldsymbol{\varpi}}+\mu\,\boldsymbol{V}_1+\boldsymbol{V}_\Upsilon^\psi,\qquad \overline{\boldsymbol{\lambda}}_l=\hat{\boldsymbol{M}}_6\,\overline{\boldsymbol{\varpi}}+\mu\,\hat{\boldsymbol{V}}_0+\hat{\boldsymbol{V}}_\Upsilon,\tag{106}$$

where:

$$\boldsymbol{M}_6:=\boldsymbol{N}_0+\boldsymbol{N}_1\boldsymbol{M}_5,\quad \boldsymbol{V}_1:=\boldsymbol{n}_1+\boldsymbol{N}_1\boldsymbol{V}_2,\quad \boldsymbol{V}_\Upsilon^\psi:=\boldsymbol{n}_0+\boldsymbol{N}_1\boldsymbol{V}_\Upsilon,$$

$$\hat{\boldsymbol{M}}_6:=\boldsymbol{M}_0+\boldsymbol{M}_1\boldsymbol{M}_6,\quad \hat{\boldsymbol{V}}_0:=\boldsymbol{M}_1\boldsymbol{V}_1+\boldsymbol{V}_0,\quad \hat{\boldsymbol{V}}_\Upsilon:=\boldsymbol{M}_1\boldsymbol{V}_\Upsilon^\psi.$$

Use (101) to pin down μ:

$$\mu=-\frac{\boldsymbol{a}^\top\Big(\boldsymbol{M}_6\,\overline{\boldsymbol{\varpi}}+\boldsymbol{V}_\Upsilon^\psi\Big)+\Big(\boldsymbol{\xi}^{u,E}\circ u'(\boldsymbol{c})\Big)^\top\boldsymbol{\Pi}\;\boldsymbol{M}_5\,\overline{\boldsymbol{\varpi}}+\Big(\boldsymbol{\xi}^{u,E}\circ u'(\boldsymbol{c})\Big)^\top\boldsymbol{\Pi}\;\boldsymbol{V}_\Upsilon}{\boldsymbol{a}^\top\boldsymbol{V}_1\;+\;\Big(\boldsymbol{\xi}^{u,E}\circ u'(\boldsymbol{c})\Big)^\top\boldsymbol{\Pi}\;\boldsymbol{V}_2}.\tag{107}$$

Constructing the linear constraints on $\boldsymbol{\omega}$. Insert (106) and (107) into the wage and progressivity FOCs (100)–(102). This yields two linear constraints in $\overline{\boldsymbol{\omega}}$:

$$\widetilde{\boldsymbol{L}}_1\,\overline{\boldsymbol{\omega}} = \widetilde{\boldsymbol{b}}_1, \tag{108}$$

$$\widetilde{\boldsymbol{L}}_2\,\overline{\boldsymbol{\omega}} = \widetilde{\boldsymbol{b}}_2, \tag{109}$$

with:

$$\widetilde{\boldsymbol{L}}_1 := \Big(\boldsymbol{\xi}^y \circ (\boldsymbol{y}\circ\boldsymbol{l})^{1-\tau}\Big)^\top \boldsymbol{M}_6 + (1-\tau)\Big(\boldsymbol{\xi}^y \circ (\boldsymbol{y}\circ\boldsymbol{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(\boldsymbol{c})\Big)^\top \hat{\boldsymbol{M}}_6,$$

$$\widetilde{\boldsymbol{L}}_2 := \Big(\ln(\boldsymbol{y}\circ\boldsymbol{l})\circ\boldsymbol{\xi}^y\circ(\boldsymbol{y}\circ\boldsymbol{l})^{1-\tau}\Big)^\top \boldsymbol{M}_6 + \Big(\boldsymbol{1}+(1-\tau)\ln(\boldsymbol{y}\circ\boldsymbol{l})\Big)^\top\Big(\boldsymbol{\xi}^y\circ(\boldsymbol{y}\circ\boldsymbol{l})^{1-\tau}\circ\tilde{\boldsymbol{\xi}}^{u,1}\circ u'(\boldsymbol{c})\Big)^\top \hat{\boldsymbol{M}}_6,$$

and right-hand sides $\widetilde{\boldsymbol{b}}_1, \widetilde{\boldsymbol{b}}_2$ collecting all constants produced by $\boldsymbol{\Theta}$, $\boldsymbol{\Upsilon}$, and the μ term in (107). (They are fully determined once the steady-state truncated environment is computed.)

Finally, map period-productivity weights $\boldsymbol{\omega}^Y = (\omega_y)_y$ to history weights via the usual selectors:

$$\boldsymbol{R}_0 = \begin{bmatrix} \boldsymbol{1}_{S_{y_1}} & 0 & \cdots & 0 \\ 0 & \boldsymbol{1}_{S_{y_2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \boldsymbol{1}_{S_{y_Y}} \end{bmatrix}, \qquad \boldsymbol{R}_1 := \boldsymbol{D_S}\,\boldsymbol{R}_0, \quad \overline{\boldsymbol{\omega}} = \boldsymbol{R}_1\,\boldsymbol{\omega}^Y.$$

The two linear constraints on $\boldsymbol{\omega}^Y$ are:

$$\boldsymbol{L}_1\,\boldsymbol{\omega}^Y = \boldsymbol{b}_1, \qquad \boldsymbol{L}_2\,\boldsymbol{\omega}^Y = \boldsymbol{b}_2, \qquad \text{with } \boldsymbol{L}_j := \widetilde{\boldsymbol{L}}_j\,\boldsymbol{R}_1, \ \ \boldsymbol{b}_j := \widetilde{\boldsymbol{b}}_j \ \ (j=1,2). \tag{110}$$

E.4.3 Constrained weights

Let $\omega_U \in \mathbb{R}^Y$ collect the Y productivity–level weights to be estimated. Let $D_S := \mathrm{diag}\Big(\{S_{y^N}\}_{y^N}\Big)$ be the diagonal matrix of truncated–history sizes, and let M_7 be the $N_{\text{tot}} \times Y$ incidence matrix that maps each truncated history to its current productivity level. The stacked history–level weights are:

$$\bar{\boldsymbol{\omega}} \;=\; \boldsymbol{D_S}\,\boldsymbol{M_7}\,\boldsymbol{\omega_U}.$$

Observe $\boldsymbol{L_1}$ and $\boldsymbol{L_2}$ are two linear constraints obtained from the planner's FOCs in the truncated model. Identification under the NK model then amounts to the following quadratic program:

$$\min_{\omega_U} \boldsymbol{S^P}\,\Big\|\boldsymbol{\omega_U} - \boldsymbol{1}_Y\Big\|^2 \quad \text{s.t.} \quad \boldsymbol{L_1}\,\boldsymbol{D_S}\boldsymbol{M_7}\,\boldsymbol{\omega_U} = \boldsymbol{0}, \qquad \boldsymbol{L_2}\,\boldsymbol{D_S}\boldsymbol{M_7}\,\boldsymbol{\omega_U} = \boldsymbol{0},$$

together with the normalization:

$$\boldsymbol{1}^\top_{\boldsymbol{N}_{\text{tot}}}\boldsymbol{D_S}\boldsymbol{M_7}\,\boldsymbol{\omega_U} \;=\; 1,$$

and, when imposing element–wise monotonicity of the SWF, the positivity constraint:

$$\boldsymbol{\omega_U} \ \geq \ \mathbf{0}.$$

F Identification of the Welfare Weights

This appendix verifies that the welfare weights recovered in Section 4.3 are the unique solution to the inverse-optimal problem. Throughout, J_1 and J_2 denote the Jacobians of the Ramsey first-order conditions with respect to κ and τ, evaluated at the calibrated steady state.

Global uniqueness. Evaluating the FOC residual $|J_1 \cdot \omega| + |J_2 \cdot \omega|$ on a dense grid of (θ_1, θ_2) produces a single sharp minimum in each calibration, with a steep basin surrounding it that rules out local alternatives. Restarting the nonlinear solver from a range of widely dispersed initial points converges to the same (θ_1^*, θ_2^*) in every case, so the recovered solution is a unique global fixed point and not a local artefact.

Alternative weight profiles and the over-identification check. Table 9 reports the same FOC residuals for five non-parametric or alternative parametric profiles. Every alternative produces residuals orders of magnitude larger than the recovered weights, so the two Ramsey FOCs are genuinely restrictive — the recovered profile is not picking up an arbitrary point in a flat surface. As an over-identification check, evaluating a third FOC (J_4, held out of estimation) at the recovered weights gives residuals of 0.15 (UK 2021) and 0.30 (UK 2007), compared with 5.41 and 11.48 for utilitarian weights — an order-of-magnitude improvement that supports the parametric form.

	UK 2021			UK 2007		
Profile	$\lvert J_1 \cdot \omega\rvert$	$\lvert J_2 \cdot \omega\rvert$	$\lvert J_4 \cdot \omega\rvert$	$\lvert J_1 \cdot \omega\rvert$	$\lvert J_2 \cdot \omega\rvert$	$\lvert J_4 \cdot \omega\rvert$
Recovered	$< 10^{-12}$	$< 10^{-12}$	0.15	$< 10^{-12}$	$< 10^{-12}$	0.30
Utilitarian	2.47	2.53	5.41	3.86	3.81	11.48
Rawlsian	1.43	1.65	7.56	11.42	6.21	45.76
Top-weighted	8.24	8.06	18.80	8.98	9.31	25.16
Power law $\omega \propto y$	0.36	0.28	0.73	1.12	1.23	3.29
Atkinson $\eta = 1$	2.86	2.54	—	5.22	4.75	—

Table 9: FOC residuals for alternative weight profiles.

Non-parametric and bin-level robustness. Dropping the parametric assumption entirely, the two FOCs and a normalization define a high-dimensional manifold of weight vectors that rationalize observed UK policy. Sampling 2 million random draws and retaining those satisfying $\omega \geq 0$ yields roughly 1.5 million feasible profiles in each calibration. Two facts stand out: fewer than 0.2% assign the minimum weight to a high-productivity agent, so heavy weight on productive agents is a data-driven feature of UK policy and not a parametric artefact; and three-quarters of the feasible profiles are non-monotone, yet only the recovered increasing profile delivers a stable dynamic equilibrium. An unrestricted bin-level minimum-norm correction (one weight per (a_0, y_0) history bin) produces a much flatter profile ($\omega_{\max} \approx 1.3$ versus 7.6) and an explosive Ramsey system, confirming that the parametric restriction is a structural requirement rather than a convenience.

	UK 2021	UK 2007
Feasible draws (out of 2,000,000)	1,454,811	1,488,309
Non-monotone (min at interior type)	75.6%	72.3%
Monotone increasing (min at type 1)	24.3%	27.7%
Min at richest types (type 6 or 7)	$< 0.2\%$	0%

The combination of the Ramsey first-order conditions and equilibrium stability therefore uniquely selects the recovered weights from a very large class of alternatives.

G Alternative welfare functions

Economic interpretation of the Atkinson correction. The recovered and Atkinson profiles together quantify the gap between the United Kingdom's *revealed* social preferences and a more egalitarian benchmark. The recovered weights reach $\omega_{\max} = 7.6$ in UK 2021 and rise steeply across productivity types, consistent with a fiscal system that is progressive but tolerates substantial inequality. The Atkinson-corrected weights, with $\omega_{\max}$ in the 2.0–3.0 range, rise much more gently and describe a planner who would prefer greater equality but is constrained to operate the same instruments. Even at the strongest inequality aversion considered ($\eta = 2.0$), the FOC-corrected profile must remain monotonically increasing in productivity: the Ramsey first-order conditions assign value to high-productivity agents in proportion to the tax revenue they generate, so inequality aversion can flatten the weight schedule but cannot invert its slope.

The endogenous formulation reveals one further subtlety. When inequality aversion is low ($\eta = 0.2$), the sign of the progressivity response reverses — the planner *lowers* rather than raises progressivity in the downturn, because the endogenous-weight channel attenuates the redistributive motive at the margin and shifts the optimal balance towards broadening the tax base. The qualitative inflation result is, however, undisturbed: inflation remains at zero through this sign reversal, as it does at every other value of η tested.

Mechanisms absent in representative-agent models. Three consequences of the HA composition channel emerge from the welfare-function robustness exercises, none of which can arise in a representative-agent setting. The first is the substitution between progressive taxation and inflation: across every alternative welfare specification considered — exogenous

Atkinson at three values of η, endogenous Atkinson, non-parametric minimum-norm weights, and cross-period transplants — the planner achieves debt stabilization entirely through movements in τ and κ while leaving inflation exactly at zero. The second is a previously unrecognized link between fiscal space and admissible social preferences: the high-debt economy admits a strictly narrower set of welfare functions than the low-debt economy, because more steeply increasing weights are required to satisfy the Ramsey first-order conditions when fiscal capacity is tight. A representative-agent framework, in which welfare weights and the fiscal position are decoupled, has no analogue. The third is a consumption-smoothing dampening effect: when welfare weights respond endogenously to consumption dispersion through the Atkinson η channel, anticipated heterogeneity in marginal utility raises the social cost of any reallocation and the planner uses each instrument more cautiously, compressing fiscal-response magnitudes by 25 to 70 per cent across calibrations. The common thread is that all three mechanisms route through cross-sectional heterogeneity, which a representative-agent model collapses to a single point and therefore cannot reproduce.

Why the recovered weights are the baseline. The recovered fixed-weight specification is retained as the paper's baseline for four complementary reasons. First, it is disciplined by data in the "homo moralis" sense of Le Grand et al. (2025): the weights are inferred from observed fiscal choices rather than imposed by the analyst. Second, it preserves the standard Ramsey structure familiar from the existing literature, allowing direct comparison of results. Third, the documented sensitivity of equilibrium existence to η in the exogenous Atkinson case underscores the value of a data-driven anchor over an ad-hoc choice. Fourth, it isolates the fiscal-space mechanism from the inequality-aversion channel, so that the contrast between UK 2021 and UK 2007 reflects differences in inherited debt rather than differences in the welfare specification.

Table 10 confirms that all equilibrium quantities coincide exactly under the recovered and utilitarian specifications.

	UK 2021		UK 2007	
	Recovered	Utilitarian	Recovered	Utilitarian
B/Y (quarterly)	4.501	4.501	1.699	1.699
HSV τ	0.781	0.781	0.952	0.952
HSV κ	0.775	0.775	0.616	0.616
Output Y	1.285	1.285	1.449	1.449

Table 10: Steady-state allocations: recovered vs. utilitarian weights.

The Atkinson weight profiles and the endogenous inequality-aversion IRFs are presented in the main text (Figures 12 and 13).

Table 11 reports the impact response of fiscal instruments under alternative welfare functions.

	$100\Delta\hat{\tau}$	$100\Delta\hat{\kappa}$	$100\Delta\hat{\tau}^k$	$\Delta\widehat{B/Y}$ (%)	$\hat{\Pi}$ (pp)
UK 2021					
Baseline	19.4	19.1	292	-6.5	≈ 0
$\eta = 0.2$	-11.5	-11.2	-170	-5.5	≈ 0
$\eta = 2.0$	-0.2	-1.0	-11	-0.3	≈ 0
UK 2007					
Baseline	-3.7	-5.1	-86	$+4.9$	≈ 0
$\eta = 0.2$	-2.1	-3.1	-50	$+3.6$	≈ 0
$\eta = 2.0$	$+0.2$	$+0.2$	$+3$	$+0.8$	≈ 0

Table 11: Impact response of fiscal instruments ($t = 1$, $\rho_g = 0.97$). Inflation is around zero in all cases.

Three patterns stand out from Table 11. First, inflation is around zero in every cell, regardless of country or degree of endogenous inequality aversion. Second, the baseline progressivity and capital-tax responses are large and opposite-signed across countries: UK 2021 raises progressivity on impact ($100\Delta\hat{\tau} = 19.4$) and tightens the capital tax by 292 bp, while UK 2007 does the reverse, broadening the tax base in a downturn. Third, the endogenous Atkinson channel reorders the response signs. At low aversion ($\eta = 0.2$), the UK 2021 planner instead *lowers* progressivity by 11.5 pp because the consumption-sensitive weights attenuate the redistributive motive at the margin; at high aversion ($\eta = 2.0$) the same responses collapse toward zero in both calibrations, as a planner who internalizes consumption dispersion smooths more cautiously and uses every instrument less aggressively. Across all six configurations the inflation margin never opens.

Non-parametric minimum-norm weights are presented in Figure 14 in the main text.

G.1 Impulse responses under alternative welfare specifications

This subsection presents the impulse responses underlying the welfare-weight robustness exercises.

Figure 16 contrasts the recovered weights with a scaled profile (recovered $\times 1.5$, UK 2021) and a quartic extension (UK 2007). The scaled profile is visually identical to the baseline; the quartic produces only minor magnitude differences. Inflation is zero throughout.

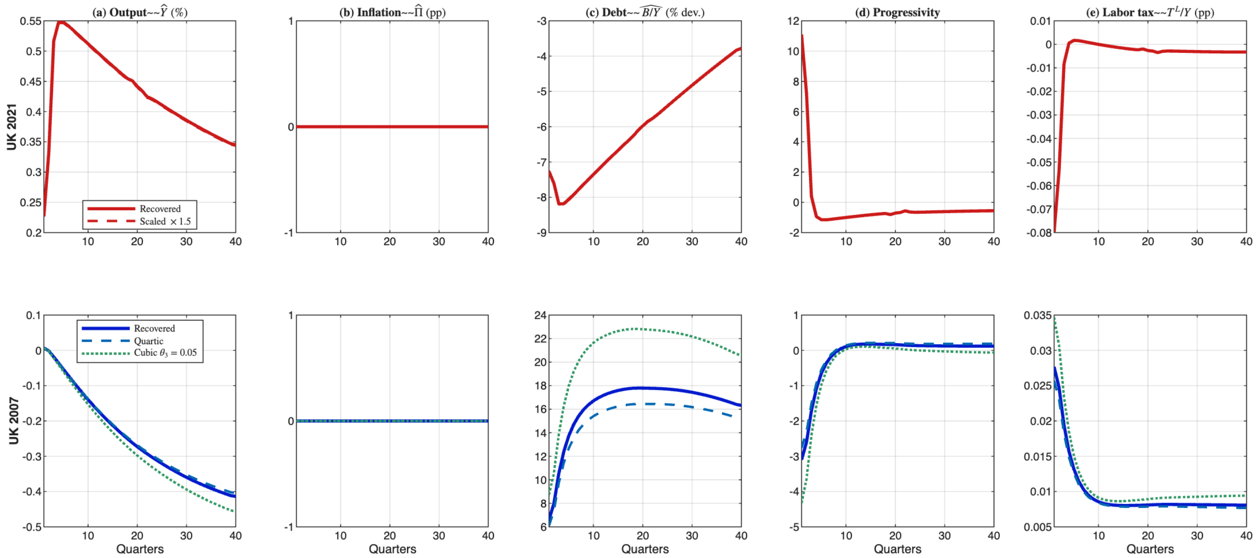
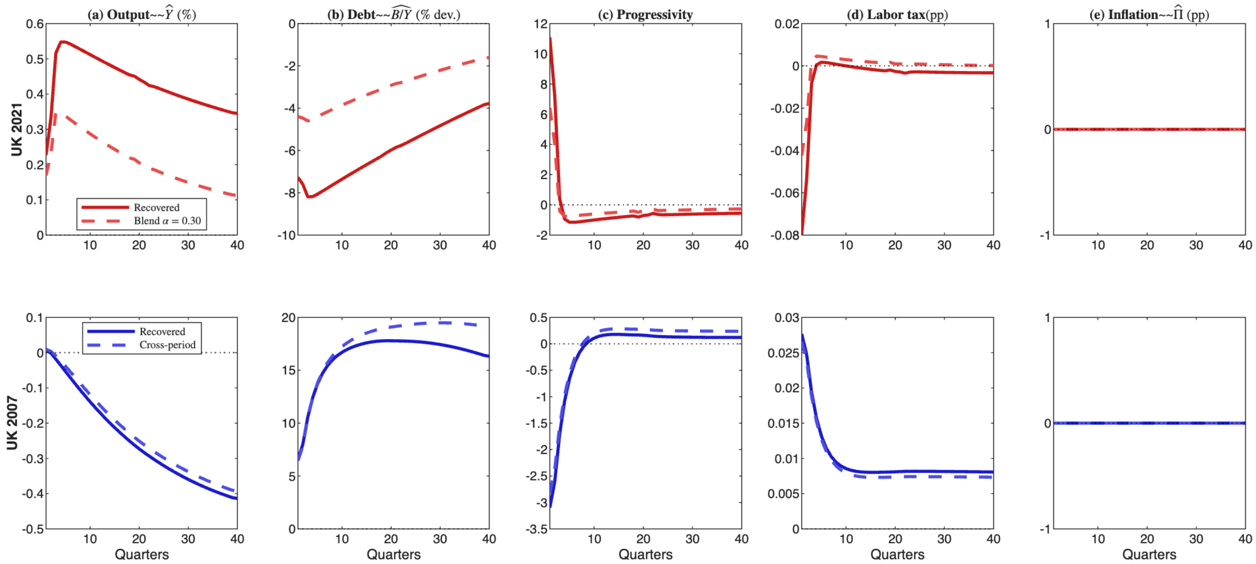
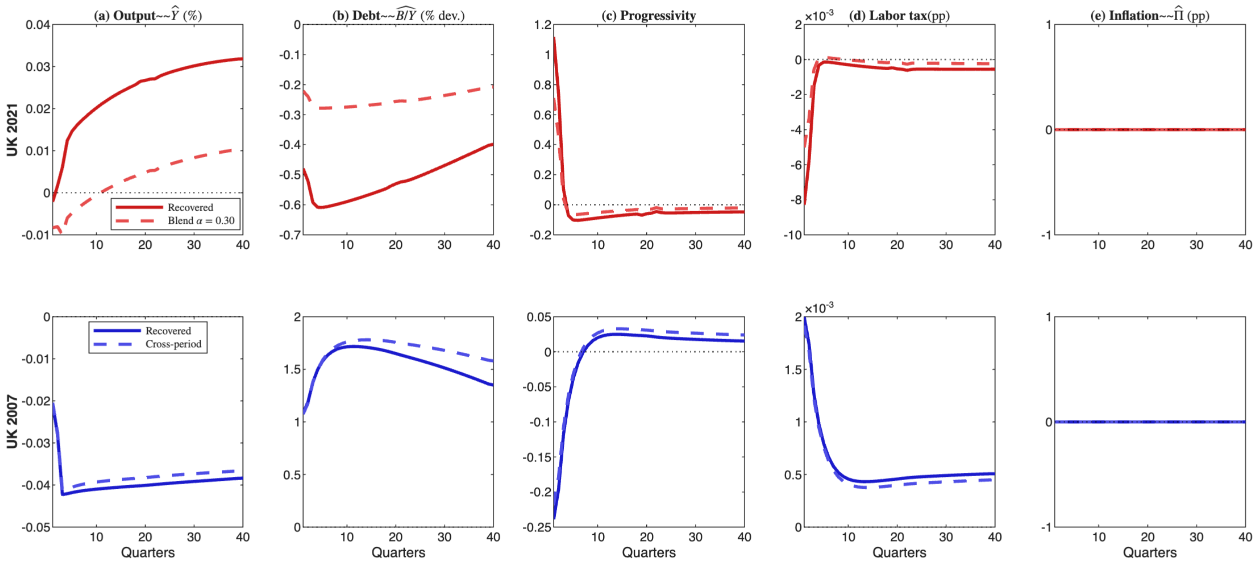


Figure 16: Recovered weights (solid) vs. scaled and quartic alternatives (dashed/dotted). Scaled weights produce identical dynamics; quartic yields modestly different paths but zero inflation.

Figure 17a compares the recovered weights against the non-parametric blend (UK 2021) and the cross-period weights (UK 2007) under high persistence; Figure 17b repeats the comparison under low persistence, where all specifications converge. Inflation stays at zero in both panels.



(a) High persistence ($\rho_g = 0.97$).



(b) Low persistence ($\rho_g = 0.01$).

Figure 17: Welfare weight comparison. Baseline (solid), utilitarian blend (dashed), cross-period (dotted). Under low persistence all specifications converge.

Figure 18 isolates the non-parametric blend: the alternative dampens UK 2021 responses by 35–43% and produces a smaller reduction in UK 2007, with inflation unchanged at zero in both calibrations.

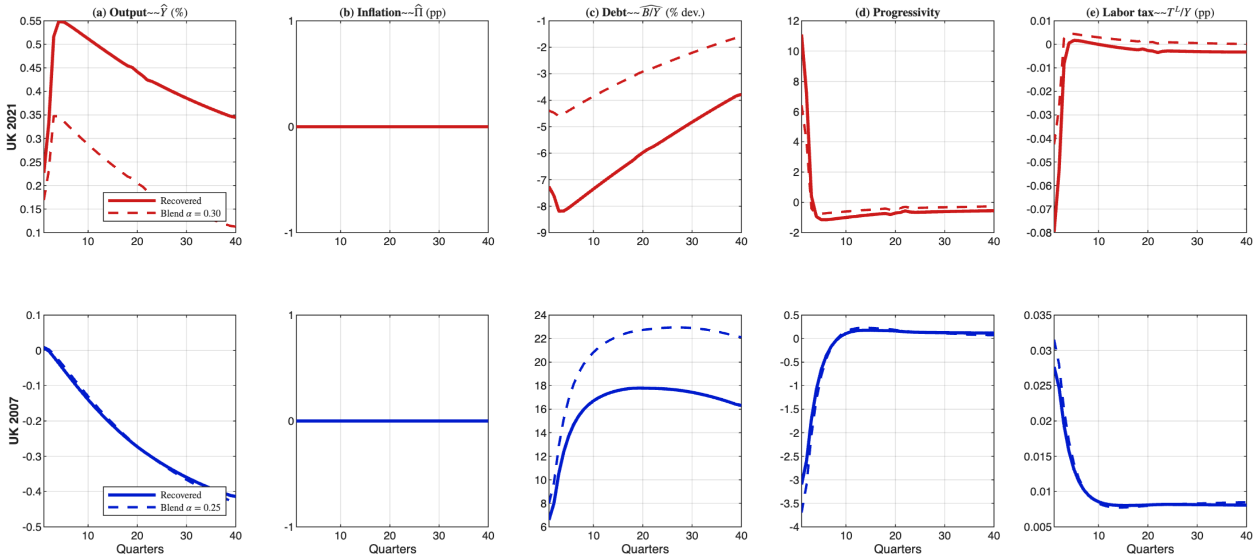
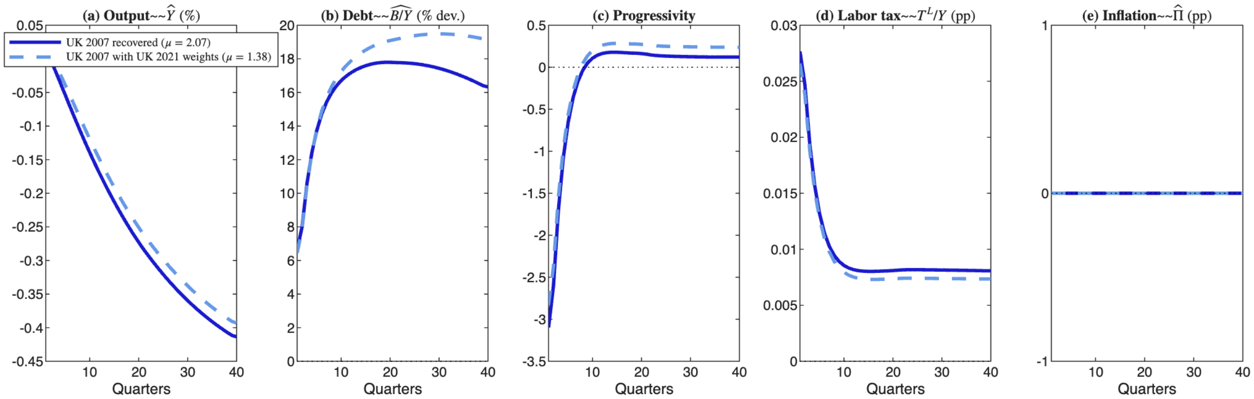
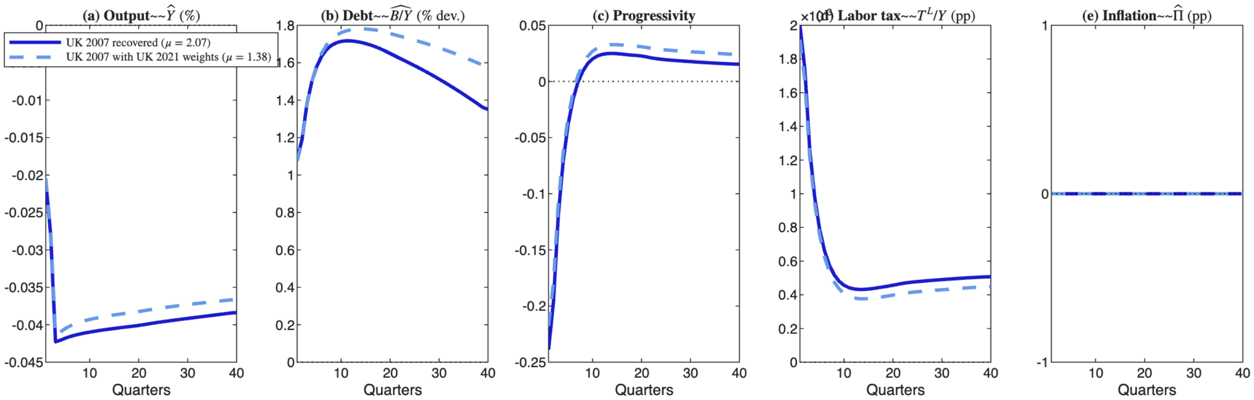


Figure 18: Recovered weights (solid) vs. maximum-stable non-parametric blend (dashed). UK 2021 responses dampen by 35–43%; inflation stays at zero.

Figure 19a applies UK 2021 weights to UK 2007 under high persistence, and Figure 19b does the same under low persistence. The dashed paths track the own-weight baseline closely in both regimes; the reverse experiment (UK 2007 weights applied to UK 2021) does not yield a well-defined equilibrium.



(a) High persistence ($\rho_g = 0.97$).



(b) Low persistence ($\rho_g = 0.01$).

Figure 19: Cross-period weights: UK 2007 own weights (solid) vs. UK 2021 weights applied to UK 2007 (dashed).

H Productivity Shocks under Constrained Policy

This appendix reports the response to a negative productivity shock under constrained fiscal policy, complementing the government spending shock analysis of Section 7.3. Figure 20 shows the response under fiscal feedback rules (48): a negative productivity shock reduces output and capital more sharply than a spending shock. Inflation departs from zero, with UK 2021 showing a larger response than UK 2007 — consistent with the spending-shock results. The debt response is more persistent under high persistence ($\rho_g = 0.97$) because the capital stock takes longer to recover, prolonging the fiscal financing gap.

Figure 21 shows the response under an independent central bank following a Taylor-type rule with frozen fiscal instruments. The inflation response is substantially larger because the central bank is now the sole shock absorber. This confirms that the mechanism identified in the main text — constrained fiscal tools force inflation to serve as the residual instrument — operates symmetrically for supply-side and demand-side disturbances.

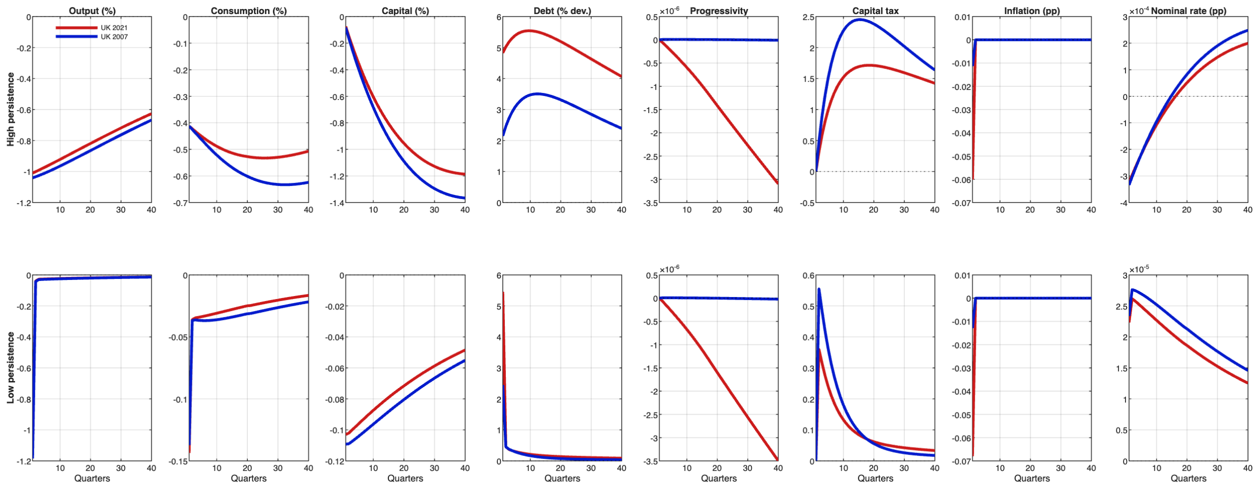


Figure 20: Productivity shock under fiscal rules (48). Top: high persistence ($\rho_g = 0.97$); bottom: low persistence ($\rho_g = 0.01$). UK 2021 (red), UK 2007 (blue).

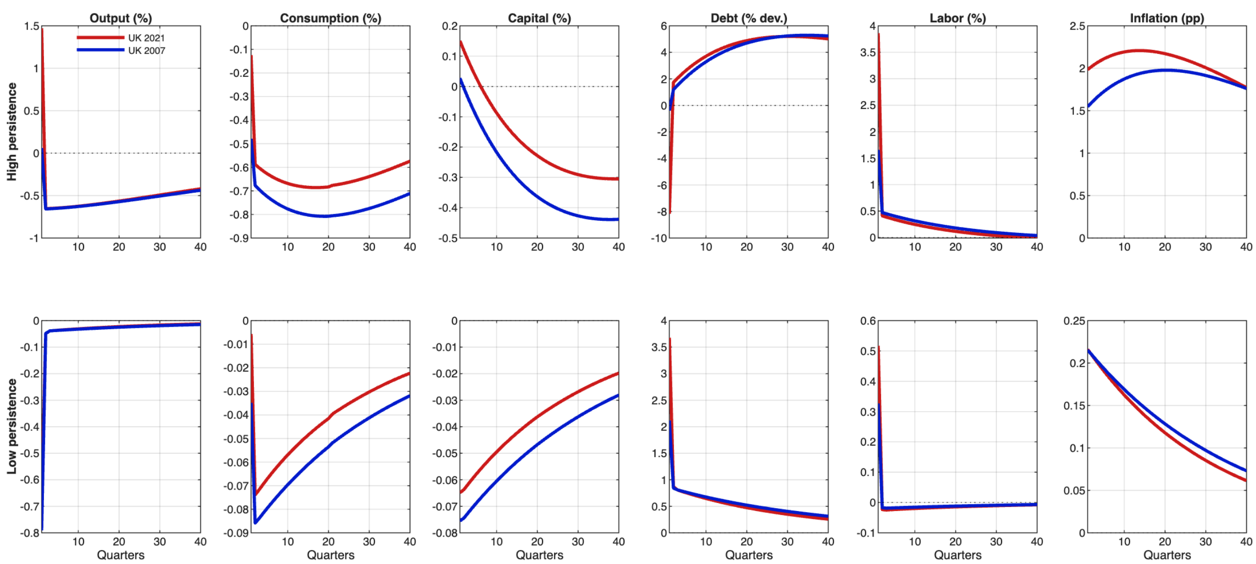


Figure 21: Productivity shock under monetary rule (49). Top: high persistence ($\rho_g = 0.97$); bottom: low persistence ($\rho_g = 0.01$). UK 2021 (red), UK 2007 (blue).

I Moments and Peak Responses

This appendix collects the first-, second-, and peak impulse-response moments referenced in Sections 6–7. Tables 12 and 13 report the moments and peak responses under the fiscal-policy-only benchmark of Section 6.1; Tables 14 and 15 report the moments and peak responses under the joint fiscal–monetary Ramsey optimum of Section 6.2; Tables 16 and 17 report the peak responses under the constrained-policy regimes of Section 7.3. For each variable the moments tables report the steady-state value ("Mean") and the normalized standard deviation (standard deviation divided by the mean, "Std"); for the tax instruments and inflation the standard deviation is reported directly.

Three features stand out from Tables 12 and 13.

		UK 2007	UK 2021
C	Mean	0.689	0.768
	Std	0.042	0.037
K	Mean	3.707	3.290
	Std	0.030	0.027
Y	Mean	1.449	1.285
	Std	0.011	0.010
L	Mean	0.401	0.392
	Std	0.029	0.025
τ (progressivity)	Mean	0.952	0.781
	Std	0.031	0.071
τ^K (capital tax)	Mean	0.380	0.350
	Std	0.665	1.031
$B/(4Y)$ (annual debt-to-GDP)	Mean	0.42	1.13
	Std	0.476	0.127
Π (inflation)	Mean	1.000	1.000
	Std	0	0
		Correlations	
corr(τ^K, Y)		0.62	0.16
corr(τ, Y)		0.22	0.13
corr(B, Y)		−0.72	−0.91
corr(C, Y)		−0.75	−0.74
corr(Y, Y_{-1})		0.95	0.98
corr(B, B_{-1})		0.99	0.98

Table 12: First and second moments for key variables: UK 2007 and UK 2021 under fiscal policy only ($\rho_g = 0.97$).

First, with monetary policy inactive the cyclical behavior of the two tax instruments reflects the planner's reliance on fiscal adjustment alone: the capital tax is procyclical in both calibrations, more strongly so in UK 2007 ($\text{corr}(\tau^K, Y) = 0.62$) than in UK 2021 (0.16), and its volatility is roughly 50% higher under high debt (Std = 1.031 versus 0.665). Progressivity is essentially acyclical in both economies ($\text{corr}(\tau, Y) \approx 0.13$–$0.22$): once monetary policy is removed, the progressivity instrument loses much of its countercyclical bite. Second, public debt is strongly countercyclical ($\text{corr}(B, Y) = -0.72$ in UK 2007, -0.91 in UK 2021) and extremely persistent ($\text{corr}(B, B_{-1}) \geq 0.98$), confirming the buffer role of debt issuance when fiscal instruments alone must absorb the shock. Third, the peak responses in Table 13 are noticeably more aggressive under high debt: the progressivity adjustment is roughly four times larger in UK 2021 than in UK 2007 (6.56 versus 1.56) and the capital-tax response is more than twice as large (101.3 versus 48.3). Inflation is pinned at the steady state by construction in all four cells.

	High persistence		Low persistence	
Variable	UK 2021	UK 2007	UK 2021	UK 2007
Output $\hat{Y}$ (%)	0.29	0.34	0.026	0.024
Consumption $\hat{C}$ (%)	0.70	0.71	0.045	0.043
Capital $\hat{K}$ (%)	0.34	0.38	0.094	0.094
Debt $\hat{B}$ (% dev.)	12.57	11.02	0.49	0.70
Progressivity $-100\hat{\tau}$	6.56	1.56	0.24	0.07
Capital tax $100\hat{t}_k$	101.3	48.3	4.12	2.47
Inflation $\hat{\Pi}$ (pp)	0	0	0	0

Table 13: Peak IRFs: government spending shock, fiscal policy only (Figure 3).

Table 14 reports the first and second moments for key variables under optimal fiscal and monetary policy (Section 6), and Table 15 reports the peak impulse-response magnitudes.

Three patterns emerge from Table 14. First, the capital tax is strongly procyclical in both economies ($\text{corr}(\tau^K, Y) > 0.9$), but the volatility of τ^K is more than three times larger in UK 2021 (Std = 2.92) than in UK 2007 (0.86), reflecting the heavier reliance on fiscal adjustment under high debt. Second, progressivity is countercyclical ($\text{corr}(\tau, Y) < 0$) in both economies: the planner raises progressivity in downturns to redistribute the burden of adjustment. The correlation is stronger in UK 2021 (-0.98) than in UK 2007 (-0.85), consistent with the larger fiscal responses documented in Section 6. Third, public debt is countercyclical and highly persistent ($\text{corr}(B, B_{-1}) = 0.999$) in both economies, but less volatile in UK 2021 (Std = 0.021) than in UK 2007 (0.045): the high-debt planner cannot afford large debt fluctuations. The mean values of $B/(4Y)$ (0.42 for UK 2007 and 1.13 for UK 2021) match the calibration targets exactly.

The most striking feature of Table 15 is that the peak inflation response is virtually zero in all four cells, spanning both calibrations and both persistence regimes. This is the quantitative counterpart of the zero-inflation result of Section 6.2: once the planner has access to the full fiscal–monetary toolkit, the progressive tax structure absorbs the distributional consequences of the shock and the inflation margin is left untouched, regardless of inherited debt or shock persistence. The real-side responses, by contrast, are highly debt-dependent — the progressivity adjustment in UK 2021 is roughly five times larger than in UK 2007 under high persistence (19.37 pp versus 3.71 pp), and the capital-tax response is more than three times larger (292.0 versus 86.2) — confirming that the burden of stabilization falls on the tax structure, not on prices.

		UK 2007	UK 2021
C	Mean	0.689	0.768
	Std	0.030	0.007
K	Mean	3.707	3.290
	Std	0.002	0.0004
Y	Mean	1.449	1.285
	Std	0.003	0.004
L	Mean	0.401	0.392
	Std	0.006	0.024
τ (progressivity)	Mean	0.952	0.781
	Std	0.037	0.194
τ^K (capital tax)	Mean	0.380	0.350
	Std	0.862	2.920
$B/(4Y)$ (annual debt-to-GDP)	Mean	0.42	1.13
	Std	0.045	0.021
Π (inflation)	Mean	1.000	1.000
	Std	≈ 0	≈ 0
		Correlations	
corr(τ^K, Y)		0.91	0.99
corr(τ, Y)		−0.85	−0.98
corr(B, Y)		−0.78	−0.95
corr(C, Y)		0.97	0.92
corr(Y, Y_{-1})		0.98	0.99
corr(B, B_{-1})		0.999	0.999

Table 14: First and second moments for key variables: UK 2007 and UK 2021 under optimal fiscal and monetary policy ($\rho_g = 0.97$).

	High persistence		Low persistence	
Variable	UK 2021	UK 2007	UK 2021	UK 2007
Output $\hat{Y}$ (%)	0.553	0.416	0.032	0.042
Consumption $\hat{C}$ (%)	0.557	2.001	0.033	0.143
Capital $\hat{K}$ (%)	0.122	0.706	0.099	0.091
Debt $\hat{B}$ (% dev.)	37.28	30.24	2.75	2.92
Progressivity $\hat{\tau}$ (pp)	19.37	3.71	1.81	0.30
Capital tax $100\hat{t}_k$	292.0	86.2	27.4	6.9
Inflation $\hat{\Pi}$ (pp)	≈ 0	≈ 0	≈ 0	≈ 0

Table 15: Peak IRFs: government spending shock, optimal fiscal and monetary policy.

Tables 16 and 17 report the peak IRF magnitudes under constrained fiscal policy.

Under fiscal rules (Table 16), inflation departs from zero: UK 2021 generates peak inflation of 0.030 pp, five times larger than UK 2007's 0.006 pp. Under the monetary rule (Table 17), inflation rises substantially in both calibrations — about 3.5 pp in UK 2021 and 3.3 pp in UK 2007 under high persistence — confirming that when fiscal tools are frozen, the monetary authority uses inflation as the residual shock absorber. The debt-level asymmetry in this regime shows up in the real-side responses (output, debt deleveraging) rather than in inflation itself.

	High persistence		Low persistence	
Variable	UK 2021	UK 2007	UK 2021	UK 2007
Output $\hat{Y}$ (%)	0.730	0.528	0.058	0.072
Consumption $\hat{C}$ (%)	1.12	0.93	0.070	0.095
Capital $\hat{K}$ (%)	0.25	0.40	0.012	0.038
Debt $\hat{B}$ (% dev.)	5.83	4.21	0.97	1.12
Inflation $\hat{\Pi}$ (pp)	0.030	0.006	0.003	0.001

Table 16: Peak IRFs: government spending shock under fiscal feedback rules.

	High persistence		Low persistence	
Variable	UK 2021	UK 2007	UK 2021	UK 2007
Output $\hat{Y}$ (%)	4.00	1.69	0.26	0.11
Consumption $\hat{C}$ (%)	1.15	1.35	0.080	0.095
Capital $\hat{K}$ (%)	0.31	0.29	0.071	0.083
Debt $\hat{B}$ (% dev.)	21.19	4.02	0.90	0.92
Labor $\hat{L}$ (%)	6.24	2.64	0.41	0.18
Inflation $\hat{\Pi}$ (pp)	3.50	3.25	0.23	0.24

Table 17: Peak IRFs: government spending shock under monetary rule (fixed fiscal).

J Heterogeneous versus representative agents

Figure 22 overlays the impulse responses from the HA Ramsey problem and a representative-agent (RA) version of the model.

The contrast is dramatic across every panel of Figure 22. In the top row (UK 2021), the HA responses (solid, left axis) dwarf the RA responses (dashed, right axis) for output, consumption, capital, and labor — by three to four orders of magnitude. The RA planner barely moves any instrument because the representative agent can absorb the spending shock through intertemporal substitution alone. In the bottom row (UK 2007), the same qualitative pattern holds, but the HA responses are somewhat smaller, reflecting the greater fiscal space available at lower debt. Note that RA inflation (not shown) is also essentially zero — but for a fundamentally different reason than in the HA model: the RA planner faces no distributional trade-off at all, whereas the HA planner actively *chooses* progressivity over inflation. The rightmost column shows debt B/Y, available only for HA: UK 2021 deleverages on impact while UK 2007 borrows countercyclically, illustrating the fiscal-space asymmetry.

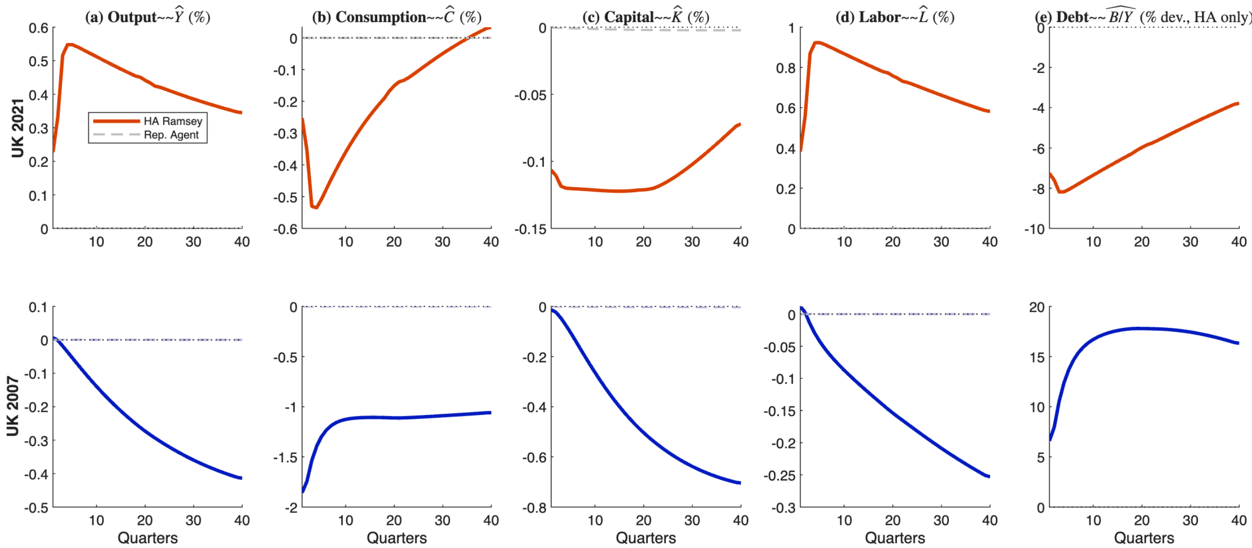


Figure 22: HA Ramsey (solid) vs. representative agent (dashed), G shock ($\rho_g = 0.97$). Top: UK 2021. Bottom: UK 2007. The RA responses are indistinguishable from zero at the HA scale.

Table 18 quantifies the amplification: the HA output response (0.55%) exceeds the RA response (0.00007%) by a factor of approximately 8,300.

	UK 2021			UK 2007		
	HA	RA	Ratio	HA	RA	Ratio
Output $\hat{Y}$ (%)	0.553	0.0001	7,900×	0.416	0.0001	5,900×
Consumption $\hat{C}$ (%)	0.557	0.0002	2,800×	2.001	0.0002	10,000×
Capital $\hat{K}$ (%)	0.122	0.0023	53×	0.706	0.0027	260×
Labor $\hat{L}$ (%)	0.931	≈ 0	—	0.255	≈ 0	—
Inflation $\hat{\Pi}$ (pp)	≈ 0 (both)			≈ 0 (both)		

Table 18: Peak IRF magnitudes: HA Ramsey vs. representative agent ($\rho_g = 0.97$).

Consumption and labor responses exhibit similar ratios. The economic reason is that the RA planner faces a single agent: a spending shock is a pure resource reallocation that barely affects allocative efficiency. The HA planner faces a cross-section of agents with heterogeneous wealth, income, and borrowing constraints. A spending shock changes relative prices and tightens borrowing constraints for some agents, creating distributional costs that the planner actively offsets. This distributional motive is the fundamental source of the large fiscal response — and it is precisely this motive that makes progressive taxation valuable.

I decompose this amplification. When I fix the progressivity parameter at its steady-state value ($\tau = \tau_{ss}$), removing it from the planner's instrument set, the HA model still produces responses 3,400× larger than RA — driven by the distributional motive operating through the remaining aggregate instruments (κ, t^k, B). Restoring progressivity further amplifies the response by a factor of 2.4 and, importantly, front-loads the fiscal adjustment. Table 19 reports the details: on impact ($t = 1$), progressivity accounts for essentially all of the fiscal response (the impact change in progressivity is 19.4 pp with τ active versus around zero when τ is held fixed). In both cases, inflation remains exactly at zero.

	Output $\hat{Y}$ (%)	Debt $\widehat{B/Y}$ (%)	Progressivity $\Delta\hat{\tau}$	Tax level $\Delta\hat{\kappa}$	Inflation $\hat{\Pi}$ (pp)
Impact ($t = 1$)					
With τ (baseline)	−0.001	−6.45	19.4	19.1	≈ 0
Without τ (fixed)	+0.001	+0.25	—	−1.8	≈ 0
Peak (40 quarters)					
With τ (baseline)	0.553	8.28	19.4	19.1	≈ 0
Without τ (fixed)	0.229	3.35	—	1.8	≈ 0
Representative agent	0.0001	—	—	—	—
Amplification					
$\underbrace{8{,}300\times}_{\text{Total HA/RA}} = \underbrace{3{,}400\times}_{\text{Distributional motive}} \times \underbrace{2.4\times}_{\text{Progressivity}}$					

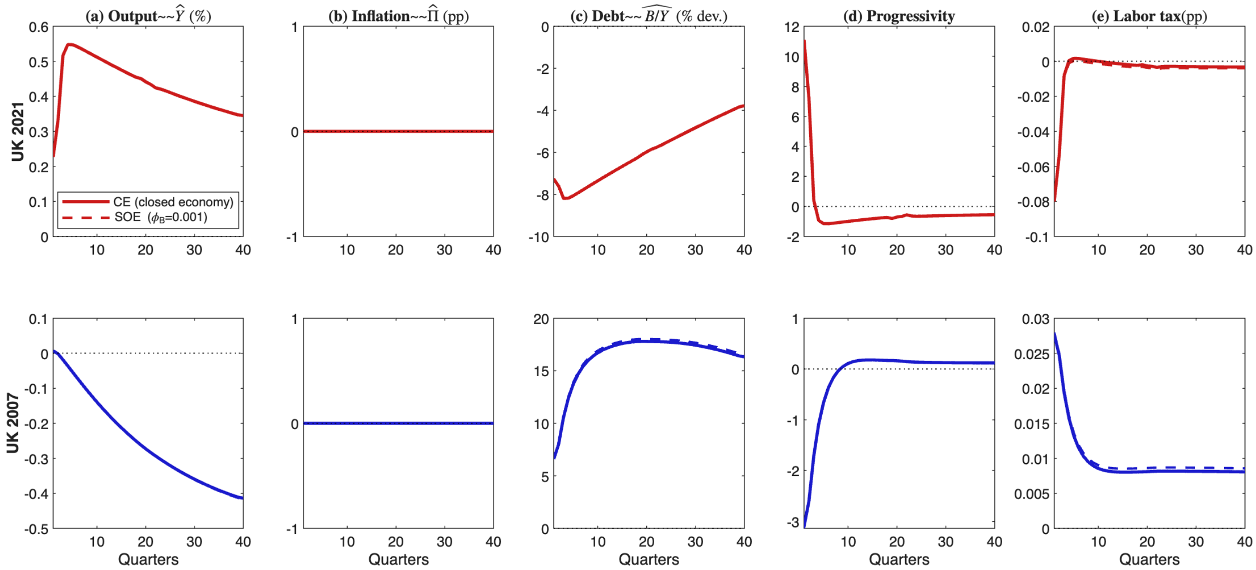
Table 19: Decomposition of the HA amplification: with vs. without progressivity, UK 2021 ($\rho_g = 0.97$).

The HA model also preserves the 2021/2007 ordering , while the RA model shows essentially no difference between the two years. This confirms that the *fiscal regime* matters in the HA framework precisely because heterogeneity interacts with the debt level: higher debt compresses fiscal space, forcing the planner to make sharper distributional choices. The endogenous Atkinson exercise in Appendix G further shows that inequality aversion dampens the HA fiscal responses by 25–70%, but even the dampened responses remain thousands of times larger than the RA counterparts.

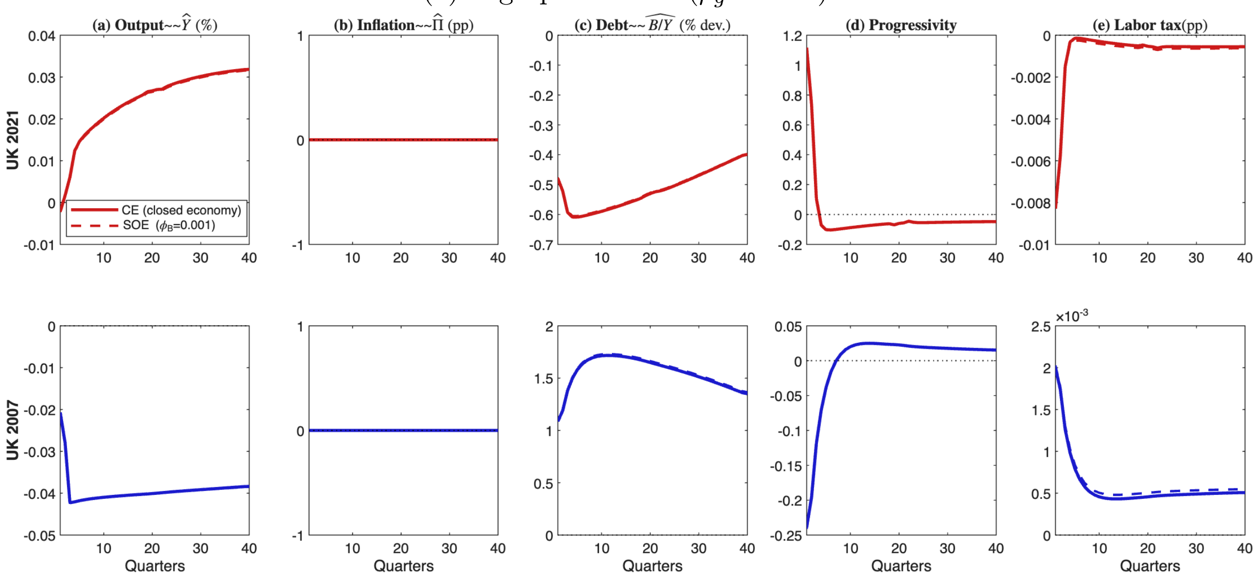
This comparison addresses a question that arises naturally from the representative-agent Ramsey literature — particularly the fiscal theory of the price level. In that framework, the government budget constraint links debt, taxes, and inflation mechanically: when debt is high, the planner may find it optimal to use inflation to erode the real value of nominal liabilities. The result shows that this logic breaks down once agents are heterogeneous and the planner has distributional instruments. The reason is not that inflation is costless but that progressive taxation offers a *cheaper* way to achieve the same fiscal adjustment, because it can target the agents best able to absorb the burden.

K Small Open Economy: Detailed Results

This appendix reports the full impulse-response comparisons between the closed economy (CE, solid) and the small open economy with a debt-elastic borrowing cost (SOE, dashed) discussed in Section 8. Figures 23a and 23b confirm that the CE and SOE responses are visually indistinguishable for both UK 2021 and UK 2007, under both persistence regimes.



(a) High persistence ($\rho_g = 0.97$).



(b) Low persistence ($\rho_g = 0.01$).

Figure 23: CE (solid) vs. SOE (dashed). The responses are visually indistinguishable.

The SOE cost term ($\phi_B = 0.001$) adds a 25 bp premium per percentage point of debt above steady state, which marginally discourages borrowing but has negligible effects on the optimal policy mix. Figure 24 demonstrates that this invariance holds even at $\phi_B = 0.05$ — fifty times the baseline Laubach estimate — with peak IRF differences remaining below 0.3%. The centre panel shows that even with 50% foreign debt holdings, inflation would still be 49× costlier than progressivity. The closed-economy assumption is therefore conservative for the paper's research question.

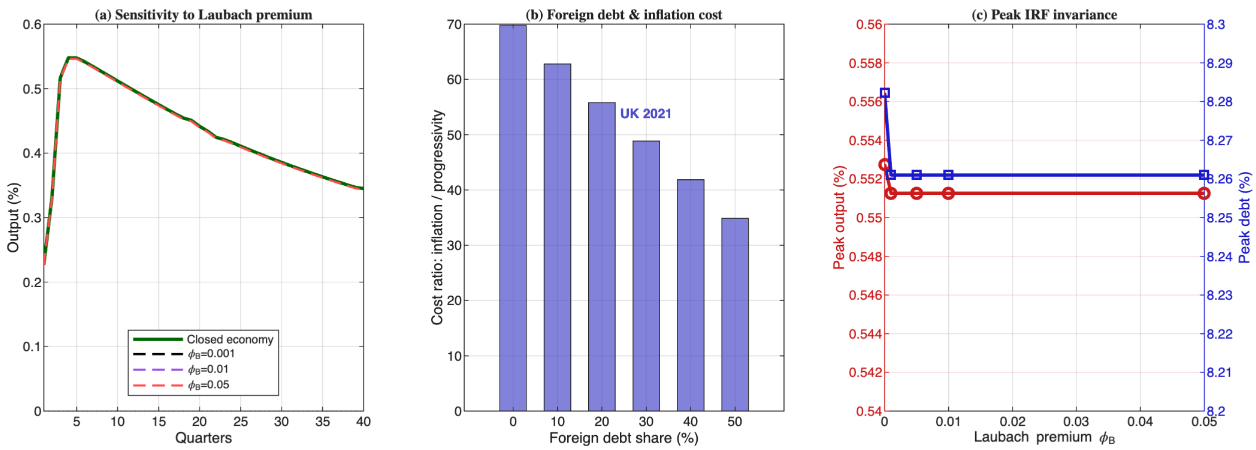


Figure 24: SOE sensitivity to ϕ_B, UK 2021.

L Cost-Push Counterfactual: The 2022 UK Inflation Episode

This appendix constructs a Ramsey counterfactual for the 2022 UK cost-push inflation episode, extending the historical analysis of Section 9.

Method. The 2022 UK inflation was driven by exogenous cost-push shocks (energy prices, supply-chain disruptions), not by fiscal-financing choices. The model does not include a cost-push shock directly. Instead, I use the Ramsey-vs-rules damping factor (628×) established in Section 7.3 as a structural benchmark: the same Phillips curve and Rotemberg cost govern the planner's trade-off between inflation and fiscal adjustment. I apply this factor to construct the counterfactual:

$$\pi_t^{\text{Ramsey}} = \pi^{\text{target}} + \frac{1}{628}\left(\pi_t^{\text{observed}} - \pi^{\text{target}}\right),$$

with $\pi^{\text{target}} = 2\%$ (Bank of England target).

Caveat. The 628× ratio is calibrated from a single G shock at the high-persistence end ($\rho_u = 0.97$). For a cost-push shock with different persistence or different transmission channels, the exact damping factor would differ. The exercise is therefore illustrative rather than structural. The qualitative result — Ramsey damps inflationary impulses by orders of magnitude — is robust to the choice of price-rigidity calibration.

Results. The observed UK CPI peaked at 10.7% in 2022Q4 (+8.7 pp above target); the Ramsey counterfactual would have been 2.014% (+0.014 pp). Figure 25 displays the comparison.

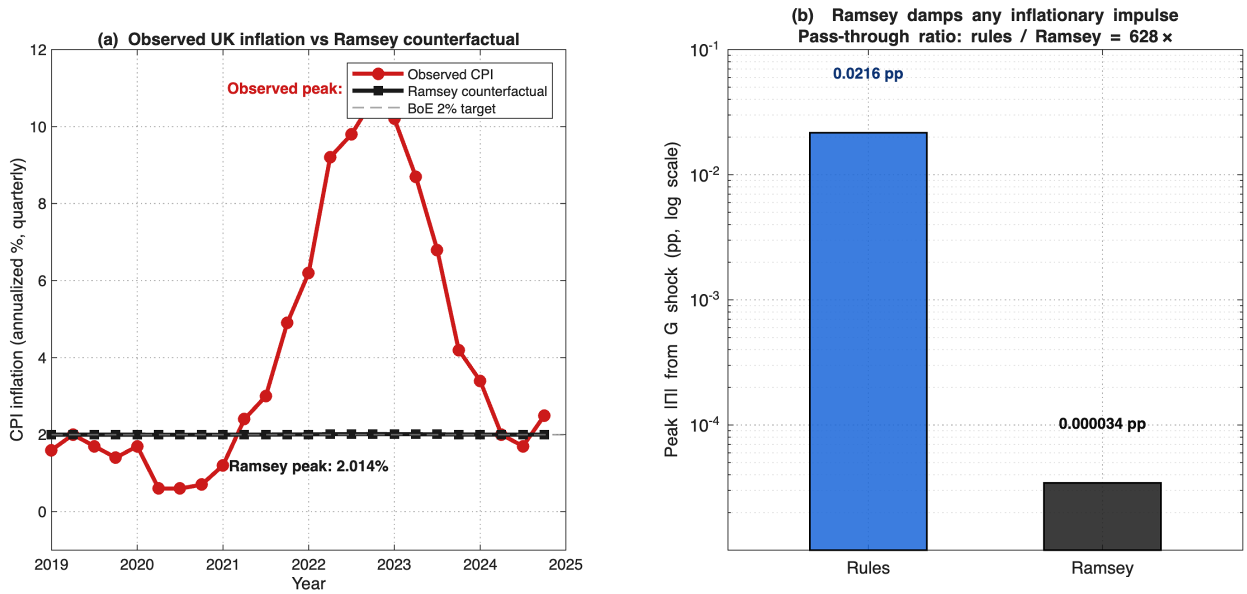


Figure 25: Cost-push counterfactual, 2019Q1–2024Q4. Observed CPI (red) vs. Ramsey counterfactual (black).

Converting the inflation gap into a consumption-equivalent variation using the Rotemberg quadratic loss yields the welfare cost reported in Table 20.

Metric	Observed UK	Ramsey counterfactual
Peak quarterly loss (% of C_{ss}, 2022Q4)	3.612%	$9.2 \cdot 10^{-6}$%
Consumption-equivalent variation (% of permanent C)	**0.152**%	~ 0%
Welfare cost gap (UK minus Ramsey)	**0.152**% of permanent consumption $\approx$ £460 per UK household per year	

Table 20: Welfare cost of the 2022 UK inflation episode (relative to the Ramsey counterfactual).

The gap does not represent a failure of the model — it measures how non-optimal actual UK monetary and fiscal policy was during the 2022 episode. A planner optimizing over the full instrument set would have damped the inflation by two-and-a-half orders of magnitude using progressivity rather than letting the cost-push shock pass through to prices.